DISSERTATION

DEVELOPMENT AND VALIDATION OF ADJUSTED REDUNDANCY INDEX FOR MECHANICS-BASED SCREENING OF DISRUPTION SCENARIOS BEFORE GRAPH NEURAL NETWORK PREDICTION

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LIST OF ABBREVIATIONS AND SYMBOLS

Symbol	Meaning
ARI	*Adjusted Redundancy Index*
CI	*Criticality Index* atau indeks kekritisan
GNN	*Graph Neural Network*
CRC	Indeks kekritisan acuan konvensional
$\mathcal{E}_v$, $\Delta\mathcal{E}$, $r_{\mathcal{E}}$	Energi listrik, selisih energi, dan fraksi penghematan energi
ι_j, $\widehat{M}_{\mathrm{CO_2e},v}$	Intensitas emisi listrik dan estimasi emisi operasional
G^M	Graf berbobot mekanis dari model utuh
e, $s = (e, \lambda)$	Elemen dan skenario gangguan elemen
$\mathcal{S}$	Himpunan skenario gangguan; tidak mencakup kontrol $\lambda = 1$
L_e, E_e, A_e, I_e	Panjang, modulus elastisitas, luas, dan momen inersia penampang
k_e, k_{sys}	Atribut kekakuan elemen dan kekakuan global ekuivalen
R_e, D_e, q_e	Kapasitas, tuntutan model utuh, dan faktor kondisi awal
K, γ, ω_r	Batas jumlah lintasan, faktor penurunan, dan bobot urutan lintasan
$S(G)$	Skor graf; berbeda dari himpunan skenario $\mathcal{S}$
U_e^*, η	Utilisasi awal dibatasi satu dan koefisien penyesuaiannya
RR_s, ARI_s, CI_s	Rasio skor jalur, ARI, dan indeks kekritisan penyaringan
$\mathcal{F}$, $\mathcal{A}$	Himpunan diteruskan dan himpunan abstensi ARI
C	Himpunan kasus kritis menurut acuan evaluasi

Symbol	Meaning
ϵ_w, ϵ_d	Regularisasi bobot dan biaya lintasan yang tak berdimensi
$\epsilon_{R,e}$, ϵ_u, ϵ_k	Regularisasi dengan satuan kapasitas, perpindahan, dan kekakuan yang sesuai

Chapter I

INTRODUCTION

1.1 Background

Structural assessment concerns not only a system's ability to carry loads in its intact state, but also the consequences when one of its elements loses some or all of its mechanical capability. In this study, such conditions are represented as element-weakening or element-loss scenarios. Element-level disruptions must be examined at the system level because a change in one part can affect force distribution, displacement, and stability. Assessment of the intact state is therefore a starting point, not a substitute for examining post-disruption consequences.

Assessing the consequences of element loss or weakening requires attention to alternative paths and changes in structural response. Research on redundancy and structural resilience provides a basis for distinguishing the existence of alternative paths from a structure's ability to maintain its function after a disruption (Frangopol dan Curley, 1987; Baker et al., 2008; Starossek dan Haberland, 2011). In this dissertation, neither concept is equated with graph connectivity alone.

This distinction matters because redundancy can refer to several different concepts. Topological redundancy indicates the availability of alternative connections, whereas static redundancy concerns indeterminacy and the distribution of element contributions. Mechanical redundancy requires the remaining paths to be capable of transferring loads in the physical model. Structural resilience, meanwhile, concerns the consequences of a disruption for system function. These four concepts are related, but they answer different questions. Accordingly, the number of connections or paths cannot be interpreted directly as a measure of a structure's ability to maintain its performance.

An alternative path may exist topologically, but its contribution to the response depends on the properties of its elements and their arrangement in the

system. A highly flexible path cannot automatically be treated as equivalent to a stiffer one; limited element capacity may also constrain its ability to accept force redistribution. In addition, member orientation and support conditions determine whether a particular displacement is actually restrained. In structural mechanics formulations, this information is represented through compatibility and equilibrium relations and stiffness assembly, not merely through a list of connections between nodes (Zienkiewicz et al., 2013). The existence of a path is therefore relevant information, but its sufficiency as a basis for disruption prioritization must still be tested.

The fundamental problem addressed in this study is information sufficiency: does an alternative path summarized by a scalar score also reflect the constraints needed to maintain mechanical stability? Developing an index provides a means to test this relationship explicitly. Research outcomes are therefore assessed not only by screening capability, but also by the ability to identify the boundary between information represented by the score and mechanical consequences that require separate examination.

The need to establish priorities arises as the set of candidate disruptions grows. In this research design, a scenario is defined by the disrupted element and its degree of weakening. Increasing the number of elements, weakening levels, and parent models increases the number of scenarios to be evaluated. Examining each candidate requires repeated model preparation, computation, and output assessment. This raises an operational question: can information available before post-disruption analysis be used to prioritize scenarios without overlooking cases that are important according to the mechanical reference?

For models with many candidate disruptions, this study tests whether a screening method based on the intact model can reduce the number of scenarios that must be predicted by a GNN. This benefit is not assumed in advance: the screening computation may cost more than the inference it avoids. The value of ARI must therefore be demonstrated through critical-case coverage and total time savings, not merely through the percentage reduction in candidates.

ARI is proposed as a prioritization index that combines mechanical attributes, alternative paths, and initial utilization. A graph represents relationships among components (Newman, 2010), while the reference response is still obtained from verified structural analysis. The GNN is treated as a fixed predictor, not as a component modified by the screening method. This study does not examine training-data selection, label savings, or improvements to the GNN architecture.

Conceptually, the path-score ratio in ARI is intended to summarize changes in path support between a disrupted graph and the intact graph. The initial utilization adjustment incorporates information about demand relative to capacity in the intact state, without treating it as a calculation of post-disruption force redistribution. This combination is a representational hypothesis to be tested, not a guarantee that every aspect of mechanical redundancy is captured. Assigning stiffness or capacity weights to graph edges does not, by itself, make a path score equivalent to solving the structural equations. ARI and the criticality index CI are therefore limited to relative-priority measures; they are not measured residual capacity, failure probabilities, or safety decisions.

The boundary on input information is an important part of this design. The screening method uses only the intact model and the disruption definition, including initial-response information specified in the formulation. Post-disruption responses are kept separate as evaluation references and are not used to construct the score. If the response being prioritized had to be calculated first to form the index, the rationale for screening would be inconsistent. This separation ensures that the evaluation assesses whether initial information can prioritize consequences that are unknown at the screening stage, rather than merely summarizing outputs already available.

The research gap addressed here is therefore not simply the need for another index. The study must explain which mechanical information is retained in the score, which consequences it can distinguish, and under what conditions the score is inadequate. The question also concerns added value: does combining attributes and path search provide more useful priorities than simpler pre-analysis baselines? Answering this requires an evaluation with equivalent input access and decision rules.

Formulation complexity or agreement on a few examples is not sufficient to establish the index's superiority.

Accordingly, computational verification is distinguished from validation of interpretation. Verification assesses whether the formulas and algorithms produce outputs consistent with their definitions, including whether path search matches its objective function and whether uncertified searches are handled through abstention. Validation of interpretation assesses whether a correctly computed score is adequately related to mechanical consequences in the tested domain. A correct algorithm can still produce scores with limited discriminatory power if its input representation lacks necessary information. Findings that limit interpretation of the index are therefore research results, not merely implementation failures.

Canonical verification in this study reveals scenarios in which the graph remains connected and CI is close to zero, while stiffness rank decreases. This example makes the representational sufficiency boundary a central finding: correctly computing a score is not equivalent to having sufficient information to assess stability. The demonstrated limitation is that the score cannot guarantee stability for the configurations examined; it is not a theorem that all graph methods are incapable of representing structural mechanics. The usefulness of ARI for ranking priorities remains a separate question: a counterexample does not, by itself, prove that all of its rankings are unhelpful.

This finding motivates a separate check of kinematic information through an intact-model-based safeguard, without changing CI. In the reported regression suite and development diagnostics, the safeguard forwarded every disruption with a rank deficiency. However, capacity exceedance or increased displacement may still be missed. Thus, successfully detecting mechanisms does not establish coverage of all critical consequences. Enriched information must be assessed according to the types of consequences it can detect, and success on development models requires independent evaluation before the claim is extended.

At the decision level, ranking must also be distinguished from screening. Ranking orders candidates, whereas screening determines which scenarios are

forwarded and which are not. Candidate reduction does not indicate success if critical cases are missed. Conversely, forwarding nearly every candidate may preserve coverage but does not demonstrate the value of reduction. The feasibility of ARI must therefore be assessed through the relationship between critical-case coverage and scenario reduction, while accounting for abstentions and failures over the same candidate set. A scenario that is not forwarded is not evaluated by the GNN; it is not thereby declared safe.

The benefit of integration requires additional evidence because weight calculation, path search, safeguard checks, and forwarding decisions also take time. Evaluation cannot simply count avoided inferences; it must compare the total cost of screening and prediction with the cost of predicting every candidate. The reference GNN and its trained parameters must remain the same so that changes in outcomes can be attributed to the forwarding policy, not to a different predictor. Total time savings and final coverage are treated as hypotheses to be tested, not as properties assumed to follow from using ARI.

On this basis, the study aims to produce an auditable formulation and implementation, identify the limits of ARI's representational sufficiency, and establish a basis for evaluating screening before GNN prediction. The available evidence comes from synthetic canonical models and development diagnostics of the safeguard; cross-family validation, calibration of a final policy, and measurement of integration benefits remain future work. Separating index validity, screening feasibility, and integration benefit provides the basis for the research questions that follow. Within this framework, the contribution includes both score development and an account of the limits on decisions supported by its information.

1.2 Problem Identification

The research problems are distinguished as problems of construct, measurement, and application. At the construct level, redundancy may refer to static indeterminacy, the distribution of element contributions, alternative paths, or the

ability to maintain function. Summing or multiplying indicators from different concepts without operational definitions can produce a number that is easy to calculate but difficult to interpret. ARI must therefore be defined as a scenario-priority score, not as a substitute for every concept of structural redundancy.

At the measurement level, a graph stores connectivity, but a scalar representation does not automatically contain force direction, end conditions, displacement compatibility, or stability. The central problem is to determine the limits of the information retained by the ARI representation: the existence of a path in a graph must be distinguished from the sufficiency of constraints in the mechanical model. Weights that combine stiffness and capacity require normalization and tests of compensation among attributes. A disruption to graph weights must also be distinguished from a change in cross-section or material in the physical model. If this distinction is overlooked, validation may unknowingly test two different definitions of disruption.

At the application level, a screening method may capture many critical cases but still be unhelpful if score-computation costs exceed the inference costs avoided. Conversely, substantial reduction may occur because difficult cases are excluded. The study must assess coverage, abstentions, failures, and total time over the same candidate set.

The research problem is specified in the following six topics. Their order is used consistently in the research questions, objectives, and conclusions so that each topic has a traceable question, objective, and evidence status.

1. **ARI formulation.** The definition of ARI's construct, inputs, outputs, domain, units, normalization, regularization, disruption operation, and mathematical properties must be made explicit. Without these specifications, a priority score may be misinterpreted as residual capacity or a failure probability.
2. **Algorithm verification.** Path search must conform to the defined objective function. A plausible-looking greedy solution is not enough; agreement with an exact reference, output certification, and handling of

search limits through abstention must be examined.

3. **Mechanical adequacy and added value.** Mathematical consistency and mechanical representational adequacy are distinct issues. Conditions in which a path score fails to distinguish mechanical consequences must be identified as limits on index interpretation. ARI's added value over simpler pre-analysis baselines must be tested with equivalent input access and evaluation rules.
4. **Sensitivity and scope of applicability.** The effects of index components, parameters, boundary cases, and differences among structural families are not yet sufficiently known. Data partitioning and uncertainty estimation must follow parent models so that related scenarios are not treated as independent.
5. **Screening feasibility.** Scenario reduction does not indicate success if critical cases are missed or difficult cases are excluded. Coverage, reduction, abstentions, and failures must be evaluated over the same candidate set and at a forwarding budget equivalent to that of the baseline.
6. **ARI–GNN integration benefit.** The compatibility of the reference GNN with the structural family and target must be confirmed; available code alone does not mean that a model is ready to use. Screening costs may exceed the inference costs avoided, so benefits must be demonstrated through total time and final coverage.

1.3 Research Questions

The following six research questions follow the topics identified above. The first two are addressed through formulation and verification; the third through fifth are limited to development diagnostics. The sixth concerns the design of an integration evaluation, not GNN results.

1. **ARI formulation.** How can ARI be consistently defined, including its construct, inputs, outputs, domain, units, normalization, regularization,

disruption operation, and mathematical properties?

2. **Algorithm verification.** How can the ARI algorithm be implemented and verified so that path search conforms to its objective function, outputs are certified, and uncertified searches result in abstention?
3. **Mechanical adequacy and added value.** In the canonical cases examined, when does the ARI score fail to assign high priority to a loss of stability, and how does coverage change after adding a kinematic safeguard without changing the score?
4. **Sensitivity and scope of applicability.** What limits on interpretation can be established from variations in weakening and parametric diagnostics, and which aspects of sensitivity and generalization remain unanswered by those development cases?
5. **Screening feasibility.** At the specified diagnostic threshold, how do the ARI, safeguard, and combined policies differ in the number of scenarios forwarded and in missed mechanism, capacity, or displacement cases, and what evidence is needed before feasibility can be claimed?
6. **ARI–GNN integration benefit.** How should an evaluation separate screening cost, inference cost, and final coverage so that future integration benefits can be tested without equating them with scenario reduction alone?

1.4 Research Objectives

This study aims to identify the limits of ARI's representational sufficiency through an auditable definition and algorithm and examination of canonical models. Mapping these limits provides a basis for distinguishing index validity, screening feasibility, and integration benefit. Validation for each structural family is a subsequent objective; it is not assumed to have been achieved through canonical verification. Specifically, the following six objectives correspond one-to-one with the research questions above.

1. **ARI formulation.** Formulate a consistent operational definition of ARI, including its construct, inputs, outputs, domain, units, normalization, regularization, disruption operation, mathematical properties, and limits on score interpretation.
2. **Algorithm verification.** Produce an auditable ARI implementation and verify path-search results against an exact reference, output certification, and abstention rules when a search is unresolved or uncertified.
3. **Mechanical adequacy and added value.** Identify canonical counterexamples in which ARI priority does not reflect loss of stability, and measure diagnostic coverage changes after adding a kinematic safeguard without changing the score.
4. **Sensitivity and scope of applicability.** Establish limits on interpretation supported by variations in weakening and parametric diagnostics, distinguishing these from parameter sensitivity and cross-family generalization that still require independent evaluation.
5. **Screening feasibility.** Compare forwarding counts and missed critical cases for the ARI, safeguard, and combined policies in the development package, and specify the evidence needed for feasibility decisions beyond that package.
6. **ARI–GNN integration benefit.** Develop a protocol for evaluating total time and final coverage with a compatible reference GNN, using the same model and trained parameters, without treating the protocol itself as evidence of integration benefit.

Each objective is paired with an output and decision criterion, not merely a computational activity. The first and second objectives are addressed through the reported formulation and verification. For the third and fourth, the canonical cases provide counterexamples to the sufficiency of path scores to guarantee stability, while parametric safeguard diagnostics clarify the role of kinematic information. Comprehensive comparison and generalization still require further validation. The fifth

objective has been examined diagnostically to distinguish mechanism detection from general mechanical screening feasibility. The sixth objective produces a protocol; its implementation through GNN integration is not yet available. The protocol for further evaluation is kept separate from the development results.

The title retains the programmatic direction of developing and validating ARI before GNN prediction. At the reported stage, empirical outputs are limited to verification and diagnosis, while the integration output is a protocol. Comparison at an equal budget, independent cross-family validation, and measurement of GNN benefits remain future objectives of that program. Aligning the objectives with the available evidence does not imply that the comprehensive validation indicated by the title has been completed.

1.5 Scope and Limits of Claims

1. The primary formulation uses line elements, linear-elastic response, small deformations, static loading, and deterministic attributes.
2. The intact model must be feasible and stable. Disruption scenarios that create mechanisms are not excluded merely because they lack an ordinary elastic solution; they are handled through a separate mechanical classification.
3. The primary disruptions are the loss of one element or partial weakening of an element. The undisturbed control is not included in the denominator for scenario reduction.
4. ARI inputs come from the intact model and the disruption definition. Post-disruption responses are used only in separately conducted development and evaluation.
5. Numerical evidence includes four canonical 1D/2D axial-bar models and 16 additional parametric models for developing the kinematic safeguard. These variants are not independent population samples. Nonlinear analysis and external models belong to the subsequent protocol and are

not yet available as confirmatory results. Nonlinear comparison is an additional examination of ARI's scope of applicability with its formula and score held fixed, not a replacement of the concept or the baseline linear reference, as discussed in Section 3.15.1.

6. The integration protocol is limited to one reference GNN compatible with planar trusses. A trained model and inference experiments are not yet available; the dissertation does not infer integration benefits from canonical verification.

1.6 Research Contributions and Status of Evidence

The research contribution centers on the relationship between the definition of ARI, its interpretive failure in particular cases, and policy changes after kinematic information is added. The claimed novelty is not the general assertion that a graph differs from a mechanical model, but a diagnosis that can be traced to a formula, path configuration, and specific scenario. Historical priority or superiority over all previous indices is not inferred from this verification package.

Three outcomes are supported by the available evidence: a verifiable ARI formulation and implementation; a counterexample to interpreting the score as a guarantee of stability; and a diagnostic evaluation of a policy with a kinematic safeguard that preserves the original score. Proposed capacity and displacement bounds and the integration protocol are treated separately as analytical outputs and research design, not as empirical validation results.

The position relative to prior research follows the synthesis in Table 2.11. Baselines based on static indeterminacy and redundancy matrices must not be replaced solely by weaker topological measures. The relationship between the safeguard and mechanical redundancy information is discussed in Section 4.2; the contribution here is the adaptation and evaluation of a policy, not a claim to a new foundational mechanical method.

The decisive added-value question is whether ARI further improves the

Table 1.1: Contribution map, supporting evidence, and claims not yet established.

Component	Contribution and available evidence	Claim boundary
ARI formulation	Score definition and disruption operation; implementation and 342 canonical checks	Implementation agreement, not ranking superiority
Counterexample	Two element-loss scenarios in a one-diagonal square have CI near zero alongside a reduction in rank	Tested configuration, not all parameters or graph representations
Kinematic safeguard	Adaptation of an intact-model check; all 40 rank-deficient cases forwarded in the development package	Not a discovery of the redundancy matrix or coverage of all critical cases
Consequence enrichment	Formulation of capacity and displacement bounds with explicit domains and input audits	Analytical proposal; costs and empirical coverage have not been tested
GNN integration	Protocol for comparing total cost and final coverage	No inference results or measured savings yet

selection of critical cases once this mechanical information is available. Answering it requires an ablation with and without ARI using equivalent inputs, test models, and evaluation budgets. If adding ARI only increases cost or the number of forwarded cases without improving outcomes under equivalent conditions, the appropriate conclusion is that it adds no value in that domain. Formula complexity is not evidence of benefit.

1.7 Research Benefits

1.7.1 Theoretical Benefits

This study clarifies the limits of the relationship between path scores, initial utilization, and disruption priority through definitions and verifiable counterexamples. The canonical findings show that alternative paths in the ARI representation do not, by themselves, guarantee mechanical stability. The theoretical benefit is to establish limits on score interpretation and a basis for examining kinematic information

separately, rather than equating path redundancy with every aspect of a structure's ability to maintain its function.

1.7.2 Methodological Benefits

The verification package makes it possible to examine discrepancies in mechanical interpretation separately from implementation accuracy. The protocol separates formula development, threshold calibration, blind validation, and final testing. Sampling units, labels, decision criteria, and the handling of unresolved cases are specified so that other researchers can assess whether results truly arise from the index or from evaluation choices. The documentation format supports replication with different algorithms and software without changing the experimental definition.

1.7.3 Practical Benefits and Their Limits

The findings provide a basis for distinguishing score-based priorities, kinematic triggers, and the need for capacity and displacement evaluation. The intended operational output is a list of priority scenarios together with forwarding status, reasons for abstention, and estimated computational benefits in the tested domain. This output may help organize the order of evaluation, but it does not remove the responsibility to assess the structure. Use beyond the tested families, parameter ranges, or disruption types requires additional examination. No safety or efficiency benefit is claimed to have been achieved merely because a protocol has been prepared.

Chapter II

THEORETICAL BACKGROUND AND ARI FORMULATION

2.1 Mechanical Foundations of Structural Systems

Structures are viewed as systems of elements connected through equilibrium, compatibility, constitutive relations, and boundary conditions. In the linear elastic regime, after constrained degrees of freedom have been eliminated, the global relationship is written as

$$\mathbf{K}_{ff}\mathbf{u}_f = \mathbf{p}_f. \tag{2.1}$$

The matrix $\mathbf{K}_{ff}$ is the stiffness matrix for the free degrees of freedom, not the path-count limit K or the scalar stiffness k_{sys}. Element-matrix assembly and coordinate transformations must be consistent with the model used (Zienkiewicz et al., 2013). In this dissertation, this equation provides a reference for distinguishing changes in physical response from changes in the graph score.

For an axially loaded truss without initial strain, define the compatibility matrix $\mathbf{B}$ and the diagonal element-stiffness matrix $\mathbf{D}_k = \text{diag}(E_e A_e / L_e)$. Then

$$\boldsymbol{\Delta} = \mathbf{B}\mathbf{u}_f, \qquad \mathbf{n} = \mathbf{D}_k\boldsymbol{\Delta}, \qquad \mathbf{K}_{ff} = \mathbf{B}^{\mathsf{T}}\mathbf{D}_k\mathbf{B}. \tag{2.2}$$

The rows of $\mathbf{B}$ depend on bar orientation and the constrained degrees of freedom. Replacing $\mathbf{B}$ with an ordinary adjacency matrix would remove this information. This is why stiffness information on graph edges is not equivalent to a complete mechanical representation.

Pellegrino dan Calladine (1986) relate the rank and vector spaces of the equilibrium matrix to states of self-stress and deformation modes without bar extension. This foundation treats static indeterminacy and kinematic mechanisms as distinct properties. In the unstressed domain considered here, the existence of a nonzero displacement mode satisfying $\mathbf{B}\mathbf{u}_f = \mathbf{0}$ explains the loss of constraint in

a linear model. This discussion does not claim that prestress effects or nonlinear geometric stability are captured by ARI.

In frames, translational and rotational displacements are coupled with axial forces, shear forces, and moments. The attribute EI/L^3 is only one indicator used to compare elements under a consistent idealization. Coefficients such as end conditions or moment releases must not be assumed to be represented unless they are included in the model and tested.

2.2 Redundancy, Resilience, and Alternative Paths

In this dissertation, topological redundancy means the availability of alternative connections; static redundancy concerns indeterminacy and its distribution; and mechanical redundancy requires the remaining paths to be capable of transferring loads in the physical model. Structural resilience assesses the consequences of disruption for system function. This distinction reflects the interpretive needs for resilience measures discussed in the cited literature (Frangopol dan Curley, 1987; Baker et al., 2008; Starossek dan Haberland, 2011).

Many paths are not necessarily effective if their members are highly flexible, have low capacity, or have support conditions that do not enable the transfer direction. Conversely, an element with a small initial force may become important when other paths are lost. ARI tests the usefulness of initial information for prioritization; it does not assume that initial forces explain all post-disruption redistribution.

Table 2.1: Distinctions among constructs and their interpretive limits.

Construct	Key question	Does not automatically establish
Connectivity	Are the nodes still connected?	Stability and capacity
Static indeterminacy	How are constraints and element contributions distributed?	Response under all loading patterns
Alternative paths	How is load transferred after disruption?	Adequacy for all nonlinear mechanisms
ARI score	Which scenarios are prioritized using initial information?	Failure probability or a safety decision

2.3 Role of Alternative-Path Analysis

The FHWA study by Agrawal et al. (2020) uses finite-element simulations of a truss-bridge example to examine alternative paths, with demand-to-capacity measures in elastic analysis and strain measures in nonlinear dynamic analysis. In the ARI study, this source informs confirmatory questions about changes in mechanism; it is not evidence that ARI has already been validated for bridges or dynamic collapse.

Operationally, mechanical confirmation asks whether the priority ranking obtained from the elastic model remains useful when the idealization is extended. A substantial change in ranking narrows the index's domain. Model extensions must not be applied only to cases that initially favor ARI; their selection is determined by structural strata and criticality.

2.4 GNN Foundations and Representation Limits

The *message-passing* framework separates message construction between nodes, aggregation, and representation updates (Gilmer et al., 2017). Using node representations $\mathbf{h}_i^{(\ell)}$ and edge attributes $\mathbf{z}_{ij}$, a generic form for describing the model interface is

$$\mathbf{m}_i^{(\ell)} = \mathrm{AGG}_{j\in\mathcal{N}(i)}\, M_\ell(\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}, \mathbf{z}_{ij}), \tag{2.3}$$

$$\mathbf{h}_i^{(\ell+1)} = U_\ell(\mathbf{h}_i^{(\ell)}, \mathbf{m}_i^{(\ell)}). \tag{2.4}$$

These equations do not propose a new architecture or specify the final reference model. The functions, depth, and outputs are selected according to the implementation that passes the compatibility audit.

Sanchez-Gonzalez et al. (2020) developed *Graph Network-based Simulators*, which represent physical systems using particles as nodes and learn dynamics through *message passing*. Their experiments include fluids, rigid bodies, and deformable materials. Here, this source broadens the foundation for graph-based physical representations; it does not provide a reference truss GNN that is automatically

compatible. Particle-dynamics simulation and prediction of static structural response have different interfaces and targets. Transferring a model requires checking its features, boundary conditions, outputs, and trained parameters before the benefits of ARI integration can be assessed.

Xu et al. (2019) discuss the discriminative power of graph structures in neighborhood-aggregation GNNs. This foundation helps distinguish graph representation capacity from a predictor's suitability for a mechanics problem. In the ARI design, the GNN prediction function is frozen; the filter only changes the list of input scenarios. Therefore, a change in error on the selected list must not be described as an improvement in the network's representational capacity.

A GNN that produces one response does not necessarily provide every CRC component. For example, displacement outputs must be accompanied by definitions for reconstructing internal forces and capacities if the target also uses utilization. The cost of reconstruction and validity checks is part of the interface, not a free step outside the experiment.

2.5 Constructs and Representation

Conventional structural analysis provides the response reference through checks of equilibrium, compatibility, and boundary conditions (Zienkiewicz et al., 2013). The ARI graph representation does not replace the stiffness matrix and does not automatically enforce equilibrium, moment transfer, or stability of the degrees of freedom. Two graphs that appear topologically equivalent may require different mechanical assessments.

Each element retains its own identity. When multiple elements connect the same pair of nodes, their edge identities are preserved as a multigraph rather than merged in a way that removes parallel paths. The sets of loaded and supported nodes are denoted by V_L and V_T.

Figure 2.1 shows the mapping of a truss to a graph and two edge-disjoint paths for one load–support pair. Element identities remain distinct; the highlighted

paths are not the result of a force-redistribution calculation.

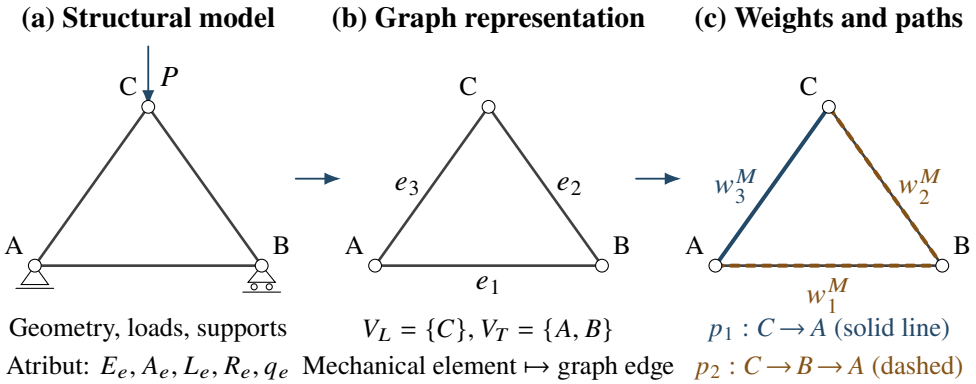


Figure 2.1: Schematic mapping of a truss to a mechanically weighted graph. For the C–A pair, paths p_1 and p_2 are edge-disjoint although they share an endpoint. Path labels do not indicate cost order; ordering is performed after costs are calculated. Panel (c) shows only one load–support pair, not the full graph-score aggregation. Colored lines show graph paths, not the magnitude or direction of forces from structural analysis.

2.6 Attributes and Normalization

The primary stiffness indicator is defined by element family:

$$k_e^{\text{aksial}} = \frac{E_e A_e}{L_e}, \qquad k_e^{\text{lentur}} = \frac{E_e I_e}{L_e^3}. \tag{2.5}$$

L_e is the element length, and L_e^3 is its cube. The bending indicator is not intended to represent all coefficients of the stiffness matrix; the effects of end conditions must be documented in the idealization and tested in validation.

For physically comparable positive attributes, use

$$\widehat{x_e} = \frac{x_e}{x_e + s_x}, \qquad x_e > 0, \quad s_x > 0. \tag{2.6}$$

The scale s_x is the median positive attribute in the development data within a group sharing the same units and meaning. The scale is frozen before validation and converted when units change. Axial and moment capacities are not pooled into one group. Missing, negative, or non-finite inputs are rejected; a zero attribute on an

active element requires model inspection. Values outside the development range are flagged as extrapolations.

The candidate weight is defined as

$$w_e^M = q_e\left(\alpha_k \widehat{k}_e + \alpha_R \widehat{R}_e\right), \quad \alpha_k + \alpha_R = 1, \quad \alpha_k, \alpha_R \geq 0, \quad 0 < q_e \leq 1. \tag{2.7}$$

The factor q_e represents the initial condition, not a hypothetical disruption. A value of one indicates only that no initial degradation is modeled; it does not establish that the physical condition has been inspected. Length is not added again to the primary weight because it is already included in the stiffness indicator. The additive form must be compared with geometric, harmonic, and minimum forms to examine compensation between attributes. Intact-model demand enters only through the utilization adjustment.

2.7 Alternative-Path Score

The edge cost and cost of a simple path p are

$$c_e^M = \frac{1}{w_e^M + \epsilon_w}, \qquad d^M(p) = \sum_{e \in p} c_e^M, \quad \epsilon_w > 0. \tag{2.8}$$

Let $\mathfrak{P}_{K,ij}(G)$ denote the set of all choices of at most K simple edge-disjoint paths between i and j, including the empty choice. Paths within each choice are ordered by nondecreasing cost. The pair score is defined by the optimization

$$\Phi_{ij}^{(K)}(G) = \max_{\mathcal{P} \in \mathfrak{P}_{K,ij}(G)} \sum_{r=1}^{|\mathcal{P}|} \frac{\omega_r}{d^M(p_{(r)}) + \epsilon_d}, \tag{2.9}$$

$$\omega_r = \frac{\gamma^{r-1}}{\sum_{t=1}^{K} \gamma^{t-1}}, \qquad 0 < \gamma \leq 1, \quad \epsilon_d > 0. \tag{2.10}$$

Unfilled path positions contribute zero; rank weights are not renormalized when the number of paths decreases. The initial candidate uses $K = 2$ and $\gamma = 1$. The variations $K \in \{2, 3\}$ and $\gamma \in \{0.5, 1\}$ are selected only during development. $K = 1$

is used as an ablation baseline.

For each active load component $c \in \mathcal{D}$, node and component weights are derived from the intact model:

$$F_c = \sum_{i \in V_L} |f_{ic}|, \qquad a_{ic} = \frac{|f_{ic}|}{F_c}, \qquad \pi_c = \frac{F_c / f_{c,\mathrm{ref}}}{\sum_{h \in \mathcal{D}} F_h / f_{h,\mathrm{ref}}}, \tag{2.11}$$

$$B_c = \sum_{j \in V_T} |r_{jc}^{(0)}|, \quad b_{jc} = \begin{cases} \dfrac{\beta}{|V_T|} + (1-\beta)\dfrac{|r_{jc}^{(0)}|}{B_c}, & B_c > 0, \\ \dfrac{1}{|V_T|}, & B_c = 0, \end{cases} \tag{2.12}$$

where $0 < \beta \leq 1$. Only components with $F_c > 0$ are included. The scale $f_{c,\mathrm{ref}} > 0$ has appropriate units, so forces and moments are not summed directly. The initial candidate uses $\beta = 0.10$, and the uniform baseline uses $\beta = 1$. All weights remain fixed across disruptions of the same model.

The graph score is

$$S(G) = \sum_{c \in \mathcal{D}} \pi_c \sum_{i \in V_L} a_{ic} \sum_{j \in V_T} b_{jc} \Phi_{ij}^{(K)}(G). \tag{2.13}$$

Models without active loads or supports are not scored. A node that is both loaded and supported uses one empty path of zero cost, contributing ω_1 / ϵ_d. This contribution is constant across disruptions, may dominate the score, and must be tested for sensitivity. A model dominated by this contribution is not considered validated without a dedicated check.

Exhaustive enumeration of all paths and edge-disjoint combinations is used as the reference for small graphs. Finding the shortest path and then greedily deleting its edges is not considered equivalent to Equation (2.9). Agreement on small cases verifies the implementation; it does not prove universal optimality. Any replacement algorithm must have an equivalence basis or an optimality certificate for the output used. If the time limit is reached without certification, the scenario receives abstention

status and must not be discarded by the filter.

2.8 Disruption, Score Ratio, and ARI

For scenario $s = (e, \lambda)$, complete loss at $\lambda = 0$ is implemented by deleting the edge. For $0 < \lambda \leq 1$, the primary candidate uses

$$w^M_{e,\lambda} = \lambda w^M_e. \tag{2.14}$$

The normalization and weights of other elements are not recomputed. This operation defines disruption in the score; it does not assert that renormalizing λk_e and λR_e would yield the same result. The stiffness- and capacity-weakening rules in the physical reference model are specified separately, consistently with the scenario definition, and recorded before evaluation.

Figure 2.2 distinguishes a change in the weight of an active edge from edge deletion. In the example, weakening does not change the number of available paths, but changes the cost of paths using the target edge. Edge loss removes the paths that traverse it.

For $S(G^M) > \epsilon_S$, define

$$\mathrm{RR}_e(\lambda) = \frac{S(G^M_{e,\lambda})}{S(G^M)}, \tag{2.15}$$

$$U^*_e = \min\left\{\frac{D_e}{R_e + \epsilon_{R,e}}, 1\right\}, \tag{2.16}$$

$$\mathrm{ARI}_e(\lambda) = \mathrm{RR}_e(\lambda)(1 - \eta U^*_e), \quad 0 \leq \eta \leq 1, \tag{2.17}$$

$$\mathrm{CI}_e(\lambda) = 1 - \mathrm{ARI}_e(\lambda). \tag{2.18}$$

Demand $D_e \geq 0$ and capacity $R_e > 0$ must refer to the same force mode. For elements with multiple modes, the capacity and utilization-aggregation rules must be specified by family before computation. The regularizer $\epsilon_{R,e}$ has the units of the relevant capacity. No regularizer is added to the denominator of RR, so that the control $\mathrm{RR}_e(1) = 1$ remains exact. A baseline score that is too small or non-finite

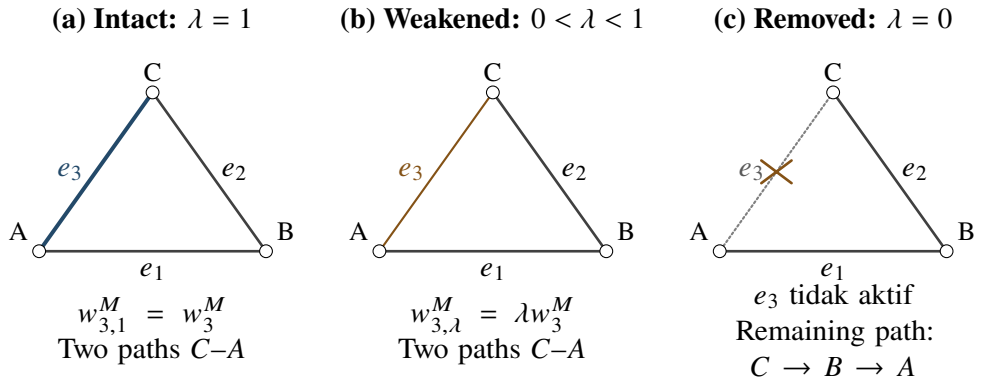


Figure 2.2: Disruption operations on the example graph in Figure 2.1. Weakening changes the edge weight without removing paths, whereas loss deletes the edge and the paths that use it. The dotted line with a cross marks only the location of the deleted edge, not an active edge. Line thickness is not to scale. Path availability does not establish mechanical stability; response and physical weakening rules are examined separately.

causes all scenarios for that model to abstain rather than receive an imputed zero.

Low ARI or high CI indicates analysis priority, not the magnitude of failure probability. Under the no-disruption control,

$$\mathrm{ARI}_e(1) = 1 - \eta U^*_e, \qquad \mathrm{CI}_e(1) = \eta U^*_e. \tag{2.19}$$

Thus, CI includes baseline priority, not only disruption consequences. $1 - \mathrm{RR}_e(\lambda)$ is reported separately as the relative loss in path score. The control $\lambda = 1$ is not included in the disruption set $\mathcal{S}$.

Figure 2.3 summarizes the relationship between path score and initial utilization. The disruption operation changes only the specified edge; normalization, pair weights, and the score denominator remain based on the intact model.

2.9 Mathematical Properties and Boundary Cases

With positive attributes, fixed weights, and exact optimization, weakening does not decrease the cost of any path. Edge deletion reduces the set of feasible paths.

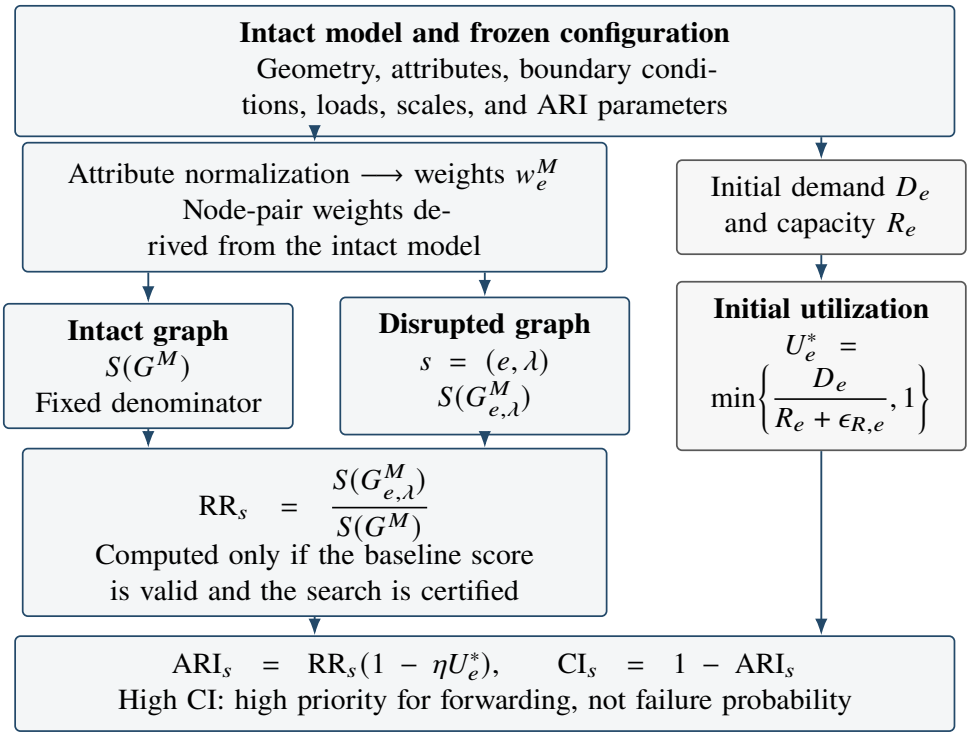


Figure 2.3: ARI construction from two intact-model components: path score and utilization. In the disrupted graph, $\lambda = 0$ deletes the edge and $0 < \lambda \leq 1$ multiplies its edge weight by λ, without renormalization. An invalid score or uncertified search results in abstention.

For each choice that remains feasible, the objective contribution does not increase; therefore, the post-disruption maximum cannot exceed the pre-disruption maximum. It follows that

$$0 \le S(G^M_{e,\lambda}) \le S(G^M), \qquad 0 \le \mathrm{RR}_e(\lambda) \le 1, \qquad 0 \le \mathrm{ARI}_e(\lambda) \le 1. \tag{2.20}$$

This proof does not establish correspondence with the stiffness or capacity of a real structure. Under these assumptions, an increase in score after weakening indicates a violation in the implementation or assumptions.

Testing must include the following cases.

1. Two identical parallel paths: for $K = 2$ and $\gamma = 1$, losing one path halves the pair score.
2. The graph remains connected while the structure becomes a mechanism: compare the graph result with the mechanical-stability classification.
3. Stiffness approaches zero while capacity remains high: test whether the additive form assigns inappropriate priority.
4. $\eta = 0$: ARI is identical to RR. For $\eta = 1$ and $U^*_e = 1$, ARI is zero at every disruption level for that element, eliminating the ability to distinguish disruptions through the path score. Report this condition as saturation, not as evidence that all scenarios have the same physical consequences.
5. Dominance by empty paths, unit changes, node-label permutations, bounded path costs, and sensitivity to regularization.

All parameters are selected during development. Formula revisions are not justified solely to improve final-test results.

2.10 Green AI and Computational Efficiency

The study's connection to green technology is limited to sustainable computing, particularly *Green AI*. Schwartz et al. (2019) propose treating efficiency as a criterion for evaluating AI research, alongside accuracy and other measures of

quality. This perspective is relevant to testing whether the additional cost of ARI screening can be offset by lower inference costs without compromising the specified coverage of critical cases.

Execution time, electrical energy, and greenhouse-gas emissions are distinct quantities. Reporting energy use and estimating emissions require a clear system boundary, hardware information, and electricity-supply characteristics (Henderson et al., 2020). Therefore, a reduced number of scenarios or shorter execution time alone is not evidence of ARI's environmental benefit.

In this dissertation, the connection to green technology is an operational benefit hypothesis that must be measured. Claims about material savings, extended structural service life, reduced embodied carbon, and hardware lifecycle impacts lie outside the available evidence. The protocol in Section 3.18.7 separates energy measurements, emission estimates, and output quality; this perspective does not change the ARI formula or establish screening feasibility.

2.11 Literature Search Results and Documentation

This section documents the search supporting the theoretical background and synthesis of previous research. Screening records, thematic summaries, and the annotated catalog are presented together so that the use of each source and the limits of its review can be traced.

2.11.1 Catalog of Discovered Sources and Methodological Connections

The focused search on September 8, 2026 identified 80 additional sources beyond the 23 sources in the initial inventory, for a total of 103 unique works. Eight additional sources have been integrated into the Chapter 2 synthesis, bringing the integrated bibliography to 31 sources; the other 72 findings are recorded as discovered sources, not as cited works. These figures are actual inventory results, not target counts or evidence that all methods have been audited. The catalog's core and supporting categories are proposed dominant roles in relation to ARI.

Table 2.2: Additional-source search results by theme.

Theme	Core	Supporting	Total
Mechanics and redundancy	9	6	15
Graphs and algorithms	4	6	10
GNNs and physical representations	8	14	22
Reliability and sensitivity	3	11	14
Validation and uncertainty	5	8	13
Energy and computational cost	0	5	5
Damage monitoring	0	1	1
Total	29	51	80

The full catalog contains titles, identified authors, years, publication details, recorded DOIs, primary-source locations, groups, connections to ARI, usage limits, reviewed material, and integration status. It is available in `results/literature/discovered_sources.csv` and as a readable version in `results/literature/source_catalog.md`. The following summary identifies sources that are especially close starting points for method comparison; the ordering reflects the author's judgment, not a bibliometric ranking or a guarantee of numerical compatibility.

Table 2.3: Selected catalog sources and their comparison roles.

Discovered source	Connection and limits
Tian, Y., Guan, X., Sun, H., and Bao, Y. (2024). *An adaptive structural dominant failure modes searching method based on graph neural network*. Primary source.	A close comparator for structural failure-mode search through adaptive graph classification. Sample selection and failure-mode targets differ from deterministic screening based on the intact model.
Parisi, F., Ruggieri, S., Lovreglio, R., Fanti, M. P., and Uva, G. (2024). *On the use of mechanics-informed models to structural engineering systems: Application of graph neural networks for structural analysis*. Primary source.	A candidate mechanics-informed GNN comparator predicting 2D and 3D truss deformations. Compatibility of constraint features and trained parameters must still be checked for integration.
Nourian, N., El-Badry, M., and Jamshidi, M. (2023). *Design Optimization of Truss Structures Using a Graph Neural Network-Based Surrogate Model*. Primary source.	Comparator for truss-displacement prediction and total cost of reducing FEM evaluations. Sizing optimization is not disruption screening, and fewer FEM evaluations do not always mean faster execution.
von Scheven, M., Ramm, E., and Bischoff, M. (2021). *Quantification of the Redundancy Distribution in Truss and Beam Structures*. Primary source.	Comparator of static-indeterminacy distribution with ARI element priorities. A mechanical-redundancy measure is not identical to a path score.

Discovered source	Connection and limits
Krake, T., et al. (2022). *Efficient Update of Redundancy Matrices for Truss and Frame Structures*. Primary source.	Comparator for updating redundancy information when elements change or are removed. Update costs and domains must be aligned with ARI scenarios.
Tkachuk, A., Krake, T., Gade, J., and von Scheven, M. (2023). *Efficient Computation of Redundancy Matrices for Moderately Redundant Truss and Frame Structures*. Primary source.	Alternative computation of redundancy matrices for comparison with the safeguard. Efficiency on moderately redundant structures does not guarantee performance on graphs of every size.
Pellegrino, S. (1993). *Structural computations with the singular value decomposition of the equilibrium matrix*. Primary source.	Reference for diagnosing rank and mechanism spaces through the equilibrium matrix. Numerical tolerances and boundary conditions require separate auditing.
Connelly, R., and Whiteley, W. (1996). *Second-Order Rigidity and Prestress Stability for Tensegrity Frameworks*. Primary source.	Basis for distinguishing first-order rigidity and connectivity from prestress stability. Tensegrity and second-order assumptions are not automatically covered by ARI's linear axial solver.
Hager, W. W. (1989). *Updating the Inverse of a Matrix*. Primary source.	Basis for low-rank matrix updates as an alternative to full reanalysis. The inverse formula requires invertibility, so mechanisms need special treatment.

Discovered source	Connection and limits
Yen, J. Y. (1971). *Finding the K Shortest Loopless Paths in a Network*. Primary source.	Reference for finding a set of simple paths. Loopless paths are not necessarily edge-disjoint or optimal for ARI.
Moustapha, M., Marelli, S., and Sudret, B. (2022). *Active learning for structural reliability: Survey, general framework and benchmark*. Primary source.	Comparator framework for reducing costly evaluations through active learning. Access to performance-function evaluations and probability targets differ from ARI.
Cawley, G. C., and Talbot, N. L. C. (2010). *On Over-fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation*. Primary source.	Basis for preventing bias when choosing ARI thresholds and configurations. Selection-bias literature does not replace independent evaluation on structures.

Review was conducted on the identity, abstract, or available source sections as recorded for each entry. Not every full text has been audited. Entries marked as having authors not yet extracted still include a title and publisher location but are not ready for final bibliographic citation. An empty DOI means it has not been recorded, not that none exists. Preprint versions are distinguished from journal-volume years and are not counted again as separate works. This search is not claimed to be systematic or exhaustive of all available publications.

Supporting sources need not test ARI directly. Their relevance may concern graph-flow assumptions, boundary-condition information, update algorithms, sample selection, or evaluation measures. However, shared terminology is not sufficient to equate method inputs, outputs, and domains. The matrix of already cited sources remains separate in `results/literature/reference_matrix.csv`. Adding sources does not change ARI experimental results or turn other research

findings into evidence that this index passes validation.

2.11.2 Annotated Catalog of Discovered Sources

The following catalog contains 80 additional sources identified by the search. DOI or primary-source links are included to make publications traceable. Core and supporting labels indicate dominant function, not source quality. Detailed review notes are available in the accompanying CSV. Sources marked as integrated have been cited in the Chapter 2 synthesis. Unavailable author metadata are marked rather than fabricated. Tables are grouped by theme; source codes are retained so that identities, annotations, and statuses can be matched to the accompanying catalog.

Mechanics and Redundancy

Table 2.4: Annotated source catalog: Mechanics and redundancy.

Code	Identity and source link	Annotation and status
S01	Krake, T., dkk. (2022). *Efficient Update of Redundancy Matrices for Truss and Frame Structures*. Journal of Theoretical, Computational and Applied Mechanics. DOI: `10.46298/jtcam.9615`	**Role:** Core. **Relevance:** Comparator for updating redundancy information when elements change or are removed. **Limitations:** Update costs and domains must be aligned with ARI scenarios. **Status:** not used in the Q1-journal synthesis or integrated bibliography; retained only as a discovered-source record.

Continued on the next page

Tabel 2.4 (lanjutan)

Code	Identity and source link	Annotation and status
S02	Tkachuk, A., Krake, T., Gade, J., dan von Scheven, M. (2023). *Efficient Computation of Redundancy Matrices for Moderately Redundant Truss and Frame Structures*. Journal of Theoretical, Computational and Applied Mechanics. DOI: 10.46298/jtcam.11056	**Role:** Core. **Relevance:** Alternative computation of redundancy matrices for comparison with the safeguard. **Limitations:** Efficiency on moderately redundant structures does not guarantee performance on graphs of every size. **Status:** not used in the Q1-journal synthesis; retained only as a discovered-source record.
S03	von Scheven, M., Ramm, E., dan Bischoff, M. (2021). *Quantification of the Redundancy Distribution in Truss and Beam Structures*. International Journal of Solids and Structures 213, 41–49. DOI: 10.1016/j.ijsolstr.2020.11.002	**Role:** Core. **Relevance:** Comparator of static-indeterminacy distribution with ARI element priorities. **Limitations:** A mechanical-redundancy measure is not identical to a path score. **Status:** integrated into the synthesis and bibliography.
S04	Gade, J., Tkachuk, A., von Scheven, M., dan Bischoff, M. (2021). *A continuum-mechanical theory of redundancy in elastostatic structures*. International Journal of Solids and Structures 226–227. DOI: 10.1016/j.ijsolstr.2021.01.022	**Role:** Core. **Relevance:** Foundation for comparing discrete and continuum redundancy constructs. **Limitations:** A continuum operator does not directly yield the ARI formula. **Status:** integrated into the synthesis and bibliography on October 10, 2026; assessment remains based on metadata and abstracts, not a full-text audit.

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Tabel 2.4 (lanjutan)

Code	Identity and source link	Annotation and status
S05	Spyridis, P., dan Strauss, A. (2020). *Robustness Assessment of Redundant Structural Systems Based on Design Provisions and Probabilistic Damage Analyses*. Buildings 10(12), 213. DOI: 10.3390/buildings10120213	**Role:** Core. **Relevance:** Comparator for assessing redundant-system resilience under damage analysis. **Limitations:** Probabilistic analysis requires a probability model that ARI does not provide. **Status:** discovered source, not yet cited in the argument.
S06	Laman, G. (1970). *On graphs and rigidity of plane skeletal structures*. Journal of Engineering Mathematics 4, 331–340. DOI: 10.1007/BF01534980	**Role:** Core. **Relevance:** Basis for distinguishing connectivity from rigidity in planar frameworks. **Limitations:** Check generic-rigidity assumptions and geometry before applying the criteria. **Status:** discovered source, not yet cited in the argument.
S07	Pellegrino, S. (1993). *Structural computations with the singular value decomposition of the equilibrium matrix*. International Journal of Solids and Structures 30(21), 3025–3035. DOI: 10.1016/0020-7683(93)90210-X	**Role:** Core. **Relevance:** Reference for diagnosing rank and mechanism spaces through the equilibrium matrix. **Limitations:** Numerical tolerances and boundary conditions require separate auditing. **Status:** integrated into the synthesis and bibliography.
S08	Starossek, U. (2007). *Typology of progressive collapse*. Engineering Structures 29(9), 2302–2307. DOI: 10.1016/j.engstruct.2006.11.025	**Role:** Supporting. **Relevance:** Context for collapse mechanisms not captured by a single score. **Limitations:** Collapse classification is not numerical validation of elastic ARI. **Status:** discovered source, not yet cited in the argument.

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Tabel 2.4 (lanjutan)

Code	Identity and source link	Annotation and status
S09	Izzuddin, B. A., Vlassis, A. G., Elghazouli, A. Y., dan Nethercot, D. A. (2008). *Progressive collapse of multi-storey buildings due to sudden column loss — Part I: Simplified assessment framework*. Engineering Structures 30(5), 1308–1318. DOI: 10.1016/j.engstruct.2007.07.011	**Role:** Supporting. **Relevance:** Framework for assessing column-loss consequences to delimit the domain. **Limitations:** Dynamic effects and ductility are not represented by the linear axial solver. **Status:** discovered source, not yet cited in the argument.
S10	Vlassis, A. G., Izzuddin, B. A., Elghazouli, A. Y., dan Nethercot, D. A. (2008). *Progressive collapse of multi-storey buildings due to sudden column loss — Part II: Application*. Engineering Structures 30(5), 1424–1438. DOI: 10.1016/j.engstruct.2007.08.011	**Role:** Supporting. **Relevance:** Application example for column-loss assessment as context for comparator cases. **Limitations:** The application section differs from Part I but is not independent evidence for ARI. **Status:** discovered source, not yet cited in the argument.
S11	Ellingwood, B. R., dkk. (2007). *Best Practices for Reducing the Potential for Progressive Collapse in Buildings*. NISTIR 7396. DOI: 10.6028/NIST.IR.7396	**Role:** Supporting. **Relevance:** Context for alternative load paths and progressive-collapse assessment. **Limitations:** A historical report is not a statement of current design provisions. **Status:** discovered source, not yet cited in the argument.
S59	Connelly, R., dan Whiteley, W. (1996). *Second-Order Rigidity and Prestress Stability for Tensegrity Frameworks*. SIAM Journal on Discrete Mathematics 9(3), 453–491. DOI: 10.1137/S0895480192229236	**Role:** Core. **Relevance:** Basis for distinguishing first-order rigidity/connectivity from prestress stability. **Limitations:** Tensegrity and second-order assumptions are not automatically covered by ARI's linear axial solver. **Status:** discovered source, not yet cited in the argument.

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Tabel 2.4 (lanjutan)

Code	Identity and source link	Annotation and status
S60	Connelly, R., dan Whiteley, W. (1992). *The Stability of Tensegrity Frameworks*. International Journal of Space Structures 7(2). DOI: 10.1177/026635119200700208	**Role:** Supporting. **Relevance:** Context for the hierarchy of stability criteria in cable-and-strut systems. **Limitations:** This is not evidence that a path score can replace kinematic constraints. **Status:** discovered source, not yet cited in the argument.
S61	Power, S. C. (2024). *Equilibrium stresses and rigidity for infinite tensegrities and frameworks*. Journal of Mathematical Analysis and Applications 529(1), 127655. DOI: 10.1016/j.jmaa.2023.127655	**Role:** Supporting. **Relevance:** Conceptual extension of the relationship among equilibrium stress, prestress, and stiffness. **Limitations:** An infinite framework is outside the finite canonical-model domain of this ARI study. **Status:** discovered source, not yet cited in the argument.
S62	Hager, W. W. (1989). *Updating the Inverse of a Matrix*. SIAM Review 31(2), 221–239. DOI: 10.1137/1031049	**Role:** Core. **Relevance:** Basis for low-rank matrix updates as an alternative to full reanalysis. **Limitations:** The inverse formula requires invertibility, so mechanism cases need special handling. **Status:** discovered source, not yet cited in the argument.

Graphs and Algorithms

Table 2.5: Annotated source catalog: Graphs and algorithms.

Code	Identity and source link	Annotation and status
S12	Dijkstra, E. W. (1959). *A note on two problems in connexion with graphs*. Numerische Mathematik 1, 269–271. DOI: 10.1007/BF01386390	**Role:** Core. **Relevance:** Foundation for path search as an algorithmic comparator. **Limitations:** The shortest-path objective differs from ARI path-combination optimization. **Status:** discovered source, not yet cited in the argument.
S13	Yen, J. Y. (1971). *Finding the K Shortest Loopless Paths in a Network*. Management Science 17(11), 712–716. DOI: 10.1287/mnsc.17.11.712	**Role:** Core. **Relevance:** Reference for finding a specified number of simple paths. **Limitations:** Loopless paths are not necessarily edge-disjoint or optimal for ARI. **Status:** integrated into the synthesis and bibliography.
S14	Eppstein, D. (1998). *Finding the k Shortest Paths*. SIAM Journal on Computing 28(2), 652–673. DOI: 10.1137/S0097539795290477	**Role:** Core. **Relevance:** Comparator for the complexity of finding multiple paths. **Limitations:** The formulation permits repeated nodes or edges and is therefore not a direct substitute. **Status:** discovered source, not yet cited in the argument.
S15	Freeman, L. C. (1977). *A Set of Measures of Centrality Based on Betweenness*. Sociometry 40(1), 35–41. DOI: 10.2307/3033543	**Role:** Supporting. **Relevance:** Basis for ranking topological importance as a simple comparator. **Limitations:** Centrality does not encode truss compatibility or member capacity. **Status:** discovered source, not yet cited in the argument.

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Tabel 2.5 (lanjutan)

Code	Identity and source link	Annotation and status
S16	Brandes, U. (2001). *A faster algorithm for betweenness centrality*. The Journal of Mathematical Sociology 25(2), 163–177. DOI: 10.1080/0022250X.2001.9990249	**Role:** Core. **Relevance:** Basis for implementing betweenness as a comparator and measuring its ranking cost. **Limitations:** Centrality results are not a measure of mechanical safety. **Status:** discovered source, not yet cited in the argument.
S17	Spielman, D. A., dan Srivastava, N. (2011). *Graph Sparsification by Effective Resistances*. SIAM Journal on Computing 40(6), 1913–1926. DOI: 10.1137/080734029	**Role:** Supporting. **Relevance:** Alternative graph information through effective resistance and quadratic forms. **Limitations:** A scalar-Laplacian guarantee does not directly guarantee framework stiffness. **Status:** discovered source, not yet cited in the argument.
S18	Benczúr, A. A., dan Karger, D. R. (1996). *Approximating s-t minimum cuts in $\tilde{O}(n^2)$ time*. Proceedings of STOC 1996, 47–55. DOI: 10.1145/237814.237827	**Role:** Supporting. **Relevance:** Cut-based alternative comparator for connectivity loss. **Limitations:** A graph cut does not detect every loss of kinematic constraint. **Status:** discovered source, not yet cited in the argument.
S19	Batson, J., Spielman, D. A., dan Srivastava, N. (2014). *Twice-Ramanujan Sparsifiers*. SIAM Review 56(2), 315–334. DOI: 10.1137/130949117	**Role:** Supporting. **Relevance:** Context for preserving spectral information when simplifying graphs. **Limitations:** The SIAM Review version is counted once and not duplicated with the earlier version. **Status:** discovered source, not yet cited in the argument.

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Tabel 2.5 (lanjutan)

Code	Identity and source link	Annotation and status
S66	Newman, M. E. J. (2005). *A measure of betweenness centrality based on random walks*. Social Networks 27(1), 39–54. DOI: 10.1016/j.socnet.2004.11.009	**Role:** Supporting. **Relevance:** Ranking comparator that does not rely only on shortest paths. **Limitations:** A random-walk flow model is not truss equilibrium or compatibility. **Status:** discovered source, not yet cited in the argument.
S67	Borgatti, S. P. (2005). *Centrality and network flow*. Social Networks 27(1), 55–71. DOI: 10.1016/j.socnet.2004.11.008	**Role:** Supporting. **Relevance:** Basis for checking the assumed flow type behind centrality measures. **Limitations:** Social-network interpretations are not transferred as force-redistribution laws. **Status:** discovered source, not yet cited in the argument.

GNNs and Physical Representations

Table 2.6: Annotated source catalog: GNNs and physical representations.

Code	Identity and source link	Annotation and status
S20	Kipf, T. N., dan Welling, M. (2017). *Semi-Supervised Classification with Graph Convolutional Networks*. ICLR 2017. Sumber primer: openreview.net	**Role:** Core. **Relevance:** Dasar arsitektur konvolusi graf sebagai pembanding representasi. **Limitations:** Klasifikasi simpul tidak identik dengan prediksi respons struktur. **Status:** discovered source, not yet cited in the argument.

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Tabel 2.6 (lanjutan)

Code	Identity and source link	Annotation and status
S21	Veličković, P., dkk. (2018). *Graph Attention Networks*. ICLR 2018. Sumber primer: openreview.net	**Role:** Core. **Relevance:** Alternatif pembobotan interaksi tetangga dengan attention. **Limitations:** Bobot attention bukan kekakuan atau kontribusi mekanis yang terukur. **Status:** discovered source, not yet cited in the argument.
S22	Hamilton, W. L., Ying, R., dan Leskovec, J. (2017). *Inductive Representation Learning on Large Graphs*. Advances in Neural Information Processing Systems 30. Sumber primer: papers.nips.cc	**Role:** Core. **Relevance:** Konteks pembelajaran representasi induktif untuk graf yang belum dilihat. **Limitations:** Generalisasi graf harus dibuktikan lagi pada keluarga struktur sasaran. **Status:** discovered source, not yet cited in the argument.
S23	Pfaff, T., Fortunato, M., Sanchez-Gonzalez, A., dan Battaglia, P. W. (2021). *Learning Mesh-Based Simulation with Graph Networks*. ICLR 2021. Sumber primer: openreview.net	**Role:** Supporting. **Relevance:** Contoh representasi mesh dan pembelajaran simulasi fisik berbasis graf. **Limitations:** Simulasi mesh bukan model statik rangka yang langsung kompatibel dengan ARI. **Status:** integrated into the synthesis and bibliography.
S24	Battaglia, P. W., dkk. (2018). *Relational inductive biases, deep learning, and graph networks*. Pracetak arXiv:1806.01261. Sumber primer: arxiv.org	**Role:** Core. **Relevance:** Kerangka bias relasional untuk menjelaskan pilihan representasi graf. **Limitations:** Pracetak konseptual bukan hasil validasi ARI atau jaminan semua GNN. **Status:** discovered source, not yet cited in the argument.

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Tabel 2.6 (lanjutan)

Code	Identity and source link	Annotation and status
S25	Scarselli, F., Gori, M., Tsoi, A. C., Hagenbuchner, M., dan Monfardini, G. (2009). *The Graph Neural Network Model*. IEEE Transactions on Neural Networks 20(1), 61–80. DOI: 10.1109/TNN.2008.2005605	**Role:** Core. **Relevance:** Landasan model neural yang bekerja pada struktur graf. **Limitations:** Formulasi generik memerlukan pemetaan fitur mekanis dan target. **Status:** discovered source, not yet cited in the argument.
S26	Raissi, M., Perdikaris, P., dan Karniadakis, G. E. (2019). *Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations*. Journal of Computational Physics 378, 686–707. DOI: 10.1016/j.jcp.2018.10.045	**Role:** Supporting. **Relevance:** Alternatif pemanfaatan persamaan fisika dalam model prediksi. **Limitations:** PINN bukan GNN dan biaya optimasi solusi tidak sama dengan inferensi tetap. **Status:** discovered source, not yet cited in the argument.
S27	Karniadakis, G. E., dkk. (2021). *Physics-informed machine learning*. Nature Reviews Physics 3, 422–440. DOI: 10.1038/s42254-021-00314-5	**Role:** Supporting. **Relevance:** Sintesis cara memasukkan pengetahuan fisika ke pembelajaran mesin. **Limitations:** Artikel tinjauan bukan pembandingan eksperimen ARI. **Status:** discovered source, not yet cited in the argument.
S28	Li, Z., dkk. (2021). *Fourier Neural Operator for Parametric Partial Differential Equations*. ICLR 2021. Sumber primer: openreview.net	**Role:** Supporting. **Relevance:** Alternatif model pengganti pemetaan parameter ke solusi. **Limitations:** Representasi operator Fourier berbeda dari graf skenario ARI. **Status:** discovered source, not yet cited in the argument.

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Tabel 2.6 (lanjutan)

Code	Identity and source link	Annotation and status
S29	Lu, L., Jin, P., Pang, G., Zhang, Z., dan Karniadakis, G. E. (2021). *Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators*. Nature Machine Intelligence 3, 218–229. DOI: 10.1038/s42256-021-00302-5	**Role:** Supporting. **Relevance:** Alternatif pembelajaran operator untuk prediksi respons parametrik. **Limitations:** Teorema aproksimasi tidak menetapkan akurasi pada data struktur sasaran. **Status:** discovered source, not yet cited in the argument.
S30	Bronstein, M. M., Bruna, J., LeCun, Y., Szlam, A., dan Vandergheynst, P. (2017). *Geometric deep learning: going beyond Euclidean data*. IEEE Signal Processing Magazine 34(4), 18–42. DOI: 10.1109/MSP.2017.2693418	**Role:** Supporting. **Relevance:** Konteks pembelajaran pada domain non-Euclidean dan representasi geometri. **Limitations:** Geometri dalam representasi tidak otomatis menjamin keseimbangan mekanis. **Status:** discovered source, not yet cited in the argument.
S31	Brody, S., Alon, U., dan Yahav, E. (2022). *How Attentive are Graph Attention Networks?*. ICLR 2022. Sumber primer: openreview.net	**Role:** Supporting. **Relevance:** Kritik daya pembeda attention untuk pemilihan arsitektur pembanding. **Limitations:** Batas keluarga attention tertentu bukan teorema kegagalan seluruh GNN mekanis. **Status:** discovered source, not yet cited in the argument.

Continued on the next page

Tabel 2.6 (lanjutan)

Code	Identity and source link	Annotation and status
S51	Tian, Y., Guan, X., Sun, H., dan Bao, Y. (2024). *An adaptive structural dominant failure modes searching method based on graph neural network*. Reliability Engineering & System Safety 243, 109841. DOI: 10.1016/j.ress.2023.109841	**Role:** Core. **Relevance:** Pembanding dekat untuk pencarian mode kegagalan struktural melalui klasifikasi graf adaptif. **Limitations:** Pemilihan sampel dan target mode kegagalan berbeda dari penyaringan deterministik berbasis model utuh. **Status:** discovered source, not yet cited in the argument.
S52	Nourian, N., El-Badry, M., dan Jamshidi, M. (2023). *Design Optimization of Truss Structures Using a Graph Neural Network-Based Surrogate Model*. Algorithms 16(8), 380. DOI: 10.3390/a16080380	**Role:** Core. **Relevance:** Pembanding prediksi perpindahan truss dan biaya total pengurangan evaluasi FEM. **Limitations:** Optimasi ukuran bukan penyaringan gangguan dan lebih sedikit evaluasi FEM tidak selalu lebih cepat. **Status:** discovered source, not yet cited in the argument.
S53	Tang, A., Li, C., Yang, J., Zhang, H., Zheng, Q., dan Zhang, J. (2025). *Training and application of graph neural networks for predicting structural responses targeted at tall building structures*. Journal of Building Engineering 103, 112131. DOI: 10.1016/j.jobe.2025.112131	**Role:** Supporting. **Relevance:** Konteks representasi bangunan tinggi dan prediksi perpindahan drift serta periode. **Limitations:** Domain beban angin dan bangunan beton tidak langsung setara dengan kehilangan batang aksial. **Status:** discovered source, not yet cited in the argument.

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Tabel 2.6 (lanjutan)

Code	Identity and source link	Annotation and status
S54	Penulis belum diekstrak (2025). *A novel scalable graph neural network for response prediction of structures with different topologies*. Engineering Structures 343, 121146. DOI: 10.1016/j.engstruct.2025.121146	**Role:** Supporting. **Relevance:** Kandidat pengayaan kondisi batas dan evaluasi generalisasi lintas topologi truss. **Limitations:** Penulis belum diekstrak dan klaim lintas topologi sumber bukan bukti generalisasi ARI. **Status:** discovered source, not yet cited in the argument.
S55	Zheng, S., Qiu, L., dan Lan, F. (2023). *TSO-GCN: A Graph Convolutional Network approach for real-time and generalizable truss structural optimization*. Applied Soft Computing 134, 110015. DOI: 10.1016/j.asoc.2023.110015	**Role:** Supporting. **Relevance:** Alternatif generasi topologi rangka tiga dimensi berbasis GCN. **Limitations:** Keluaran rancangan optimal berbeda dari label konsekuensi kehilangan elemen. **Status:** discovered source, not yet cited in the argument.
S56	Ariyasinghe, N., Wickremasinghe, T., Weeratunge, H., Mallikarachchi, C., dan Herath, S. (2026). *Topology and size optimization of trusses using graph neural networks: Towards efficient surrogate modeling*. Structures 88, 111934. DOI: 10.1016/j.istruc.2026.111934	**Role:** Supporting. **Relevance:** Perbandingan konvolusi GNN dan fitur beban serta tumpuan pada truss. **Limitations:** Optimasi topologi dan ukuran bukan penilaian keselamatan pascagangguan. **Status:** discovered source, not yet cited in the argument.
S57	Wang, Z.-Q., Zheng, Z., MengXiang, J.-W., dan Jiang, W.-Q. (2023). *Inversion Method of Uncertain Parameters for Truss Structures Based on Graph Neural Networks*. Advanced Steel Construction 19(4), 366–374. DOI: 10.18057/IJASC.2023.19.4.5	**Role:** Supporting. **Relevance:** Konteks ketidakpastian kekakuan sambungan dan identifikasi parameter truss. **Limitations:** Masalah invers memerlukan informasi respons sehingga bukan masukan operasional ARI yang sama. **Status:** discovered source, not yet cited in the argument.

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Tabel 2.6 (lanjutan)

Code	Identity and source link	Annotation and status
S58	Parisi, F., Ruggieri, S., Lovreglio, R., Fanti, M. P., dan Uva, G. (2024). *On the use of mechanics-informed models to structural engineering systems: Application of graph neural networks for structural analysis*. Structures 59, 105712. DOI: 10.1016/j.istruc.2023.105712	**Role:** Core. **Relevance:** Kandidat GNN pembanding dengan informasi mekanis dan prediksi deformasi truss 2D serta 3D. **Limitations:** Kecocokan fitur kendala dan parameter terlatih masih harus diperiksa untuk integrasi. **Status:** discovered source, not yet cited in the argument.
S63	Alon, U., dan Yahav, E. (2021). *On the Bottleneck of Graph Neural Networks and its Practical Implications*. International Conference on Learning Representations. Sumber primer: openreview.net	**Role:** Supporting. **Relevance:** Konteks kehilangan informasi interaksi jarak jauh pada message passing. **Limitations:** Over-squashing tidak identik dengan rank kurang matriks kekakuan. **Status:** discovered source, not yet cited in the argument.
S64	Topping, J., Di Giovanni, F., Chamberlain, B. P., Dong, X., dan Bronstein, M. M. (2022). *Understanding over-squashing and bottlenecks on graphs via curvature*. International Conference on Learning Representations, pracetak arXiv:2111.14522. Sumber primer: arxiv.org	**Role:** Supporting. **Relevance:** Alternatif menilai hambatan aliran informasi dan pengayaan topologi GNN. **Limitations:** Rewiring untuk komputasi pesan tidak boleh disamakan dengan penambahan batang fisik. **Status:** discovered source, not yet cited in the argument.

Reliability and Sensitivity

Table 2.7: Annotated source catalog: Reliability and sensitivity.

Code	Identity and source link	Annotation and status
S32	Moustapha, M., Marelli, S., dan Sudret, B. (2022). *Active learning for structural reliability: Survey, general framework and benchmark*. Structural Safety 96, 102174. DOI: 10.1016/j.strusafe.2021.102174	**Role:** Supporting. **Relevance:** Kerangka pembanding pengurangan evaluasi mahal melalui pembelajaran aktif. **Limitations:** Akses evaluasi fungsi kinerja dan target probabilitas berbeda dari ARI. **Status:** integrated into the synthesis and bibliography.
S33	Au, S.-K., dan Beck, J. L. (2001). *Estimation of small failure probabilities in high dimensions by subset simulation*. Probabilistic Engineering Mechanics 16(4), 263–277. DOI: 10.1016/S0266-8920(01)00019-4	**Role:** Supporting. **Relevance:** Alternatif pemilihan evaluasi untuk kejadian kegagalan langka. **Limitations:** Probabilitas kegagalan memerlukan distribusi masukan dan bukan keluaran langsung ARI. **Status:** discovered source, not yet cited in the argument.
S34	Cheng, K., Lu, Z., Xiao, S., dan Lei, J. (2022). *Estimation of small failure probability using generalized subset simulation*. Mechanical Systems and Signal Processing 163, 108114. DOI: 10.1016/j.ymssp.2021.108114	**Role:** Supporting. **Relevance:** Pengembangan estimasi kejadian langka sebagai pembanding konseptual penyaringan. **Limitations:** Kebijakan sampling probabilistik tidak sama dengan penerusan skenario deterministik. **Status:** discovered source, not yet cited in the argument.
S35	Saltelli, A., Annoni, P., Azzini, I., Campolongo, F., Ratto, M., dan Tarantola, S. (2010). *Variance based sensitivity analysis of model output. Design and estimator for the total sensitivity index*. Computer Physics Communications 181(2), 259–270. DOI: 10.1016/j.cpc.2009.09.018	**Role:** Core. **Relevance:** Dasar evaluasi pengaruh parameter dan interaksi pada keluaran ARI. **Limitations:** Distribusi masukan dan definisi keluaran harus ditetapkan sebelum mengestimasi sensitivitas. **Status:** discovered source, not yet cited in the argument.

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Tabel 2.7 (lanjutan)

Code	Identity and source link	Annotation and status
S36	Bhattacharyya, B. (2020). *Global sensitivity analysis: A Bayesian learning based polynomial chaos approach*. Journal of Computational Physics 415, 109539. DOI: 10.1016/j.jcp.2020.109539	**Role:** Supporting. **Relevance:** Alternatif model pengganti untuk analisis sensitivitas parameter. **Limitations:** Ketidakpastian model pengganti perlu dibedakan dari ketidakpastian indeks. **Status:** discovered source, not yet cited in the argument.
S68	Schöbi, R., Sudret, B., dan Wiart, J. (2015). *Polynomial-Chaos-based Kriging*. Pracetak arXiv:1502.03939. Sumber primer: arxiv.org	**Role:** Supporting. **Relevance:** Alternatif model pengganti untuk mengurangi evaluasi numerik mahal. **Limitations:** Pembangunan surrogate membutuhkan sampel respons yang tidak tersedia sebagai masukan ARI operasional. **Status:** discovered source, not yet cited in the argument.
S69	Sudret, B. (2008). *Global sensitivity analysis using polynomial chaos expansions*. Reliability Engineering & System Safety 93(7), 964–979. DOI: 10.1016/j.ress.2007.04.002	**Role:** Core. **Relevance:** Dasar evaluasi sensitivitas global dan pembanding metode estimasi Sobol. **Limitations:** Akurasi surrogate dan distribusi masukan harus diperiksa sebelum menghitung indeks. **Status:** discovered source, not yet cited in the argument.
S72	Novák, L. (2022). *On distribution-based global sensitivity analysis by polynomial chaos expansion*. Computers & Structures 267, 106808. DOI: 10.1016/j.compstruc.2022.106808	**Role:** Supporting. **Relevance:** Alternatif sensitivitas berbasis distribusi di luar dekomposisi varians. **Limitations:** Ukuran berbasis distribusi dan Sobol tidak boleh diperlakukan identik. **Status:** discovered source, not yet cited in the argument.

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Tabel 2.7 (lanjutan)

Code	Identity and source link	Annotation and status
S73	Shang, X., Wang, L., Fang, H., Lu, L., dan Zhang, Z. (2024). *Active Learning of Ensemble Polynomial Chaos Expansion Method for Global Sensitivity Analysis*. Reliability Engineering & System Safety 249, 110226. DOI: 10.1016/j.ress.2024.110226	**Role:** Supporting. **Relevance:** Alternatif pemilihan sampel untuk sensitivitas global dengan anggaran terbatas. **Limitations:** Memerlukan pelatihan surrogate dan evaluasi model tambahan. **Status:** discovered source, not yet cited in the argument.
S74	Garcia-Cabrejo, O., dan Valocchi, A. (2014). *Global Sensitivity Analysis for multivariate output using Polynomial Chaos Expansion*. Reliability Engineering & System Safety. DOI: 10.1016/j.ress.2014.01.005	**Role:** Supporting. **Relevance:** Konteks sensitivitas bersama beberapa keluaran seperti gaya dan perpindahan. **Limitations:** Struktur korelasi keluaran dan galat surrogate perlu dinilai tersendiri. **Status:** discovered source, not yet cited in the argument.
S75	Bhattacharyya, B. (2023). *On the use of sparse Bayesian learning-based polynomial chaos expansion for global reliability sensitivity analysis*. Journal of Computational and Applied Mathematics 420, 114819. DOI: 10.1016/j.cam.2022.114819	**Role:** Supporting. **Relevance:** Alternatif sensitivitas probabilitas kegagalan dengan PCE sparse. **Limitations:** Target probabilitas kegagalan berbeda dari prioritas deterministik ARI. **Status:** discovered source, not yet cited in the argument.
S76	Penulis belum diekstrak (2026). *Uncertainty quantification and global sensitivity analysis of RC slab under near-field explosion using the generalized Polynomial Chaos Expansion*. Structures 87, 111618. DOI: 10.1016/j.istruc.2026.111618	**Role:** Supporting. **Relevance:** Contoh perluasan ketidakpastian material desain dan gangguan pada struktur beton. **Limitations:** Penulis belum diekstrak dan domain ledakan nonlinier berbeda dari solver ARI. **Status:** discovered source, not yet cited in the argument.

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Tabel 2.7 (lanjutan)

Code	Identity and source link	Annotation and status
S79	Chaudhuri, A., Marques, A. N., dan Willcox, K. E. (2019). *mfEGRA: Multifidelity Efficient Global Reliability Analysis through Active Learning for Failure Boundary Location*. Pracetak arXiv:1910.02497. Sumber primer: arxiv.org	**Role:** Supporting. **Relevance:** Alternatif pengurangan biaya melalui beberapa fidelitas dan pencarian batas kegagalan. **Limitations:** Versi yang dicatat adalah pracetak 2019 dan versi jurnal tidak dihitung sebagai sumber tambahan. **Status:** discovered source, not yet cited in the argument.
S80	Sobol', I. M. (2001). *Global sensitivity indices for nonlinear mathematical models and their Monte Carlo estimates*. Mathematics and Computers in Simulation 55(1–3), 271–280. DOI: 10.1016/S0378-4754(00)00270-6	**Role:** Core. **Relevance:** Dasar menilai pengaruh parameter dan interaksinya terhadap keluaran indeks. **Limitations:** Sensitivitas matematis tidak membuktikan kesesuaian mekanis atau recall penyaringan. **Status:** discovered source, not yet cited in the argument.

Validation and Uncertainty

Table 2.8: Annotated source catalog: Validation and uncertainty.

Code	Identity and source link	Annotation and status
S37	Varma, S., dan Simon, R. (2006). *Bias in error estimation when using cross-validation for model selection*. BMC Bioinformatics 7, 91. DOI: 10.1186/1471-2105-7-91	**Role:** Core. **Relevance:** Dasar memisahkan pemilihan konfigurasi dan estimasi kinerja. **Limitations:** Validasi bersarang tetap harus menjaga pengelompokan model induk. **Status:** discovered source, not yet cited in the argument.

Continued on the next page

Tabel 2.8 (lanjutan)

Code	Identity and source link	Annotation and status
S38	Cawley, G. C., dan Talbot, N. L. C. (2010). *On Over-fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation*. Journal of Machine Learning Research 11, 2079–2107. Sumber primer: www.jmlr.org	**Role:** Core. **Relevance:** Dasar mencegah bias saat memilih ambang dan konfigurasi ARI. **Limitations:** Literatur bias seleksi tidak menggantikan evaluasi independen pada struktur. **Status:** integrated into the synthesis and bibliography.
S39	Saito, T., dan Rehmsmeier, M. (2015). *The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets*. PLOS ONE 10(3), e0118432. DOI: `10.1371/journal.pone.0118432`	**Role:** Core. **Relevance:** Dasar memilih pelaporan precision dan recall ketika kasus kritis tidak seimbang. **Limitations:** Kurva precision–recall tidak sama dengan recall–reduksi atau jaminan keselamatan. **Status:** integrated into the synthesis and bibliography.
S40	Davis, J., dan Goadrich, M. (2006). *The Relationship Between Precision-Recall and ROC Curves*. Proceedings of ICML 2006, 233–240. DOI: `10.1145/1143844.1143874`	**Role:** Core. **Relevance:** Dasar interpretasi kurva pemeringkatan dan ketidakseimbangan kelas. **Limitations:** Interpolasi serta sasaran optimasi PR perlu dibedakan dari ROC. **Status:** discovered source, not yet cited in the argument.
S41	Geifman, Y., dan El-Yaniv, R. (2017). *Selective Classification for Deep Neural Networks*. Advances in Neural Information Processing Systems 30. Sumber primer: proceedings.neurips.cc	**Role:** Supporting. **Relevance:** Pembanding konseptual abstensi dan pertukaran risiko dengan cakupan prediksi. **Limitations:** Coverage prediksi bukan recall kasus kritis dan penolakan bukan keputusan aman. **Status:** discovered source, not yet cited in the argument.

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Tabel 2.8 (lanjutan)

Code	Identity and source link	Annotation and status
S42	Guo, C., Pleiss, G., Sun, Y., dan Weinberger, K. Q. (2017). *On Calibration of Modern Neural Networks*. Proceedings of Machine Learning Research 70, 1321–1330. Sumber primer: proceedings.mlr.press	**Role:** Supporting. **Relevance:** Dasar kalibrasi probabilitas bila keluaran GNN berupa peluang. **Limitations:** CI deterministik tidak menjadi probabilitas hanya karena dinormalisasi. **Status:** discovered source, not yet cited in the argument.
S43	Lakshminarayanan, B., Pritzel, A., dan Blundell, C. (2017). *Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles*. Advances in Neural Information Processing Systems 30. Sumber primer: proceedings.neurips.cc	**Role:** Supporting. **Relevance:** Alternatif estimasi ketidakpastian pada tahap prediksi. **Limitations:** Ensemble menambah biaya dan tidak menjamin ketercakupan penyaring ARI. **Status:** discovered source, not yet cited in the argument.
S44	Ovadia, Y., dkk. (2019). *Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift*. Advances in Neural Information Processing Systems 32. Sumber primer: proceedings.neurips.cc	**Role:** Supporting. **Relevance:** Dasar menguji ketidakpastian ketika keluarga struktur atau distribusi berubah. **Limitations:** Hasil benchmark klasifikasi tidak dipindahkan menjadi kinerja GNN struktur. **Status:** discovered source, not yet cited in the argument.
S45	Cawley, G. C., dan Talbot, N. L. C. (2007). *Preventing Over-Fitting during Model Selection via Bayesian Regularisation of the Hyper-Parameters*. Journal of Machine Learning Research 8, 841–861. Sumber primer: www.jmlr.org	**Role:** Supporting. **Relevance:** Alternatif pengendalian overfitting dalam pemilihan konfigurasi. **Limitations:** Metode untuk kernel machine tidak langsung menjadi prosedur kalibrasi ARI. **Status:** discovered source, not yet cited in the argument.

Continued on the next page

Tabel 2.8 (lanjutan)

Code	Identity and source link	Annotation and status
S46	Guyon, I., Saffari, A., Dror, G., dan Cawley, G. (2010). *Model Selection: Beyond the Bayesian/Frequentist Divide*. Journal of Machine Learning Research 11, 61–87. Sumber primer: jmlr.org	**Role:** Supporting. **Relevance:** Konteks prinsip parsimoni dan pilihan metodologis evaluasi model. **Limitations:** Editorial sintesis bukan eksperimen mekanis atau validasi indeks. **Status:** discovered source, not yet cited in the argument.
S70	Demšar, J. (2006). *Statistical Comparisons of Classifiers over Multiple Data Sets*. Journal of Machine Learning Research 7, 1–30. Sumber primer: www.jmlr.org	**Role:** Core. **Relevance:** Dasar pembandingan metode pada beberapa dataset secara statistik. **Limitations:** Skenario dari model induk sama bukan dataset independen untuk pengujian. **Status:** discovered source, not yet cited in the argument.
S71	García, S., dan Herrera, F. (2008). *An Extension on Statistical Comparisons of Classifiers over Multiple Data Sets for all Pairwise Comparisons*. Journal of Machine Learning Research 9, 2677–2694. Sumber primer: www.jmlr.org	**Role:** Supporting. **Relevance:** Pendukung koreksi pembandingan berpasangan antar banyak metode. **Limitations:** Tidak menggantikan rancangan partisi independen dan ukuran efek. **Status:** discovered source, not yet cited in the argument.
S77	Geifman, Y., dan El-Yaniv, R. (2019). *SelectiveNet: A Deep Neural Network with an Integrated Reject Option*. Proceedings of Machine Learning Research 97, 2151–2159. Sumber primer: proceedings.mlr.press	**Role:** Supporting. **Relevance:** Alternatif konseptual keputusan abstensi dan kurva risiko terhadap cakupan. **Limitations:** Penolakan prediksi terlatih berbeda dari abstensi sertifikasi pencarian ARI. **Status:** discovered source, not yet cited in the argument.

Energy and Computational Cost

Table 2.9: Annotated source catalog: Energy and computational cost.

Code	Identity and source link	Annotation and status
S47	Strubell, E., Ganesh, A., dan McCallum, A. (2019). *Energy and Policy Considerations for Deep Learning in NLP*. Proceedings of ACL 2019, 3645–3650. DOI: 10.18653/v1/P19-1355	**Role:** Supporting. **Relevance:** Konteks biaya pengembangan dan pelatihan dalam evaluasi efisiensi AI. **Limitations:** Biaya NLP tidak mewakili konsumsi GNN struktur tanpa pengukuran. **Status:** discovered source, not yet cited in the argument.
S48	Lacoste, A., Luccioni, A., Schmidt, V., dan Dandres, T. (2019). *Quantifying the Carbon Emissions of Machine Learning*. Pracetak arXiv:1910.09700. Sumber primer: arxiv.org	**Role:** Supporting. **Relevance:** Konteks pengaruh lokasi perangkat dan pasokan listrik terhadap estimasi emisi. **Limitations:** Estimasi operasional bukan analisis daur hidup atau pengukuran langsung. **Status:** discovered source, not yet cited in the argument.
S49	Lannelongue, L., Grealey, J., dan Inouye, M. (2021). *Green Algorithms: Quantifying the Carbon Footprint of Computation*. Advanced Science 8, 2100707. DOI: 10.1002/advs.202100707	**Role:** Supporting. **Relevance:** Metode estimasi emisi komputasi sebagai pendukung pelaporan manfaat. **Limitations:** Batas sistem operasional tidak mencakup keseluruhan daur hidup perangkat. **Status:** discovered source, not yet cited in the argument.

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Tabel 2.9 (lanjutan)

Code	Identity and source link	Annotation and status
S50	Anthony, L. F. W., Kanding, B., dan Selvan, R. (2020). *Carbontracker: Tracking and Predicting the Carbon Footprint of Training Deep Learning Models*. ICML Workshop on Challenges in Deploying and Monitoring Machine Learning Systems, arXiv:2007.03051. Sumber primer: arxiv.org	**Role:** Supporting. **Relevance:** Perangkat pelacakan energi dan estimasi emisi untuk eksperimen tambahan. **Limitations:** Dukungan perangkat dan batas pengukuran harus diperiksa pada mesin eksperimen. **Status:** discovered source, not yet cited in the argument.
S78	Luccioni, A. S., Jernite, Y., dan Strubell, E. (2023). *Power Hungry Processing: Watts Driving the Cost of AI Deployment?*. Pracetak arXiv:2311.16863. Sumber primer: arxiv.org	**Role:** Supporting. **Relevance:** Konteks pengukuran biaya energi inferensi bukan hanya pelatihan. **Limitations:** Angka model bahasa atau generatif tidak ditransfer ke GNN struktur. **Status:** discovered source, not yet cited in the argument.

Damage Monitoring

Table 2.10: Annotated source catalog: Damage monitoring.

Code	Identity and source link	Annotation and status
S65	Jia, J., dan Li, Y. (2023). *Deep Learning for Structural Health Monitoring: Data, Algorithms, Applications, Challenges, and Trends*. Sensors 23(21), 8824. DOI: `10.3390/s23218824`	**Role:** Supporting. **Relevance:** Tinjauan sumber data dan deteksi kerusakan untuk konteks aplikasi gangguan. **Limitations:** SHM berbasis pengukuran tidak sama dengan penyaringan praanalisis ARI. **Status:** discovered source, not yet cited in the argument.

2.11.3 Deduplication, Extraction, and Updates

Deduplication uses the DOI when available, followed by title, authors, and year. The year of initial online publication is distinguished from the volume year. For example, the Song et al. article is listed under its 2023 volume; the initial online date is not used to create two entries for the same work.

Each source record includes its inclusion rationale, structural domain, disruption definition, input requirements, target, algorithm, and the availability of models and code. Decision changes resulting from new literature are recorded with the protocol version. Updates are made before the method is frozen, before the final test is unblinded, and before submission of the dissertation. New publications are not used to change hypotheses retroactively without explanation.

2.12 Ranking and Screening Decisions

Ranking orders scenarios, whereas screening adds a decision about how many scenarios to forward or which threshold to apply. Two indices may have similar rank correlations but different coverage in the most critical tail. Conversely, high recall can be achieved by forwarding every scenario. This study therefore uses paired coverage and reduction measures rather than a single measure that could conceal failure on either objective.

When higher CI indicates higher priority, lowering the threshold expands the forwarded set. Under a fixed score configuration, this nested property provides an implementation check: a lower threshold must not reduce the number of forwarded cases. Abstention and tie-handling rules must remain fixed so that changes in selection policy do not violate this property.

2.13 Previous Research and Synthesis

The reviewed literature shows that mechanical redundancy does not have a single meaning. Gade et al. (2021) formulate elastostatic redundancy using a continuum operator, whereas von Scheven et al. (2021) measure the distribution of static indeterminacy in trusses and beams using redundancy matrices. For predominantly axial structures, Okasha et al. (2023) use the distribution of static indeterminacy to construct a member-redundancy index for material distribution optimization. All three provide mechanical comparisons for ARI, but their constructs, units of analysis, and optimization objectives differ from disruption-scenario prioritization. A valid comparison must align the model, disruption, and output definition; a path score cannot be treated as a direct substitute for a redundancy operator or matrix.

Damage studies illustrate why this equivalence matters. Caredda et al. (2022) assess local-failure scenarios in a validated steel truss bridge and report activation of alternative load paths that also depends on the capacity of other elements and the flexural action of the superstructure and connections. Abbracciavento dan De Biagi (2025) show that damage can change system behavior from parallel to series. These findings position redistribution, connection conditions, and behavioral transitions as consequences that cannot be inferred merely from the presence of graph paths. Setiawan et al. (2024) provide an approach to quantifying the robustness of concrete frames; the measures and scenarios from that domain must be mapped before they can serve as ARI reference labels.

For prediction, Song et al. (2023) develop physics-informed GNN-based elastic analysis, whereas Chou et al. (2024) present a GNN framework for static structural analysis. Both concern response prediction, not the pre-analysis decision about which scenarios to forward. Therefore, feature, boundary condition, output, trained-parameter, and inference-cost compatibility must be audited before either is used as the reference GNN. Using ARI as a filter does not by itself show that the prediction model becomes more accurate.

Echard et al. (2011) develop AK-MCS to estimate failure probability using

Kriging, Monte Carlo simulation, and iterative selection of performance function evaluations. Its connection to ARI is the reduction of costly evaluations, not a shared target. AK-MCS uses performance-function evaluations to update a surrogate model; operational ARI uses the intact model and disruption definition. AK-MCS is therefore treated as a conceptual comparison. Numerical comparison requires aligning data access, objectives, and costs rather than directly transferring reliability results as evidence of ARI coverage.

Accordingly, the narrowly defined gap is whether the reviewed sources have answered the following question: can a weighted graph score using only intact-model information screen scenarios before prediction without missing critical cases and while reducing total cost? This is not a claim that no similar method exists anywhere in the literature. Testing it requires three levels of comparison: agreement with mechanical references, paired recall–reduction including abstentions, and screening plus inference costs using the same prediction model.

Table 2.11: Focused synthesis matrix and research decisions.

Reference	Focus considered	Implication for ARI
Pellegrino dan Calladine (1986)	Rank, self-stress, and truss mechanisms	Distinguish kinematic constraints from graph connectivity.
Frangopol dan Curley (1987)	Damage, redundancy, and reliability	Do not equate a deterministic score with reliability.
Baker et al. (2008); Starossek dan Haberland (2011)	Structural-resilience measures	Define disruption, response, and interpretive limits.
Agrawal et al. (2020)	Alternative paths in truss bridges	Design confirmation tests for changes in mechanism.

Tabel 2.11 (lanjutan)

Rujukan	Fokus yang digunakan	Implikasi untuk ARI
Gade et al. (2021); von Scheven et al. (2021)	Mechanical-redundancy theory and distribution	Distinguish redundancy operators or matrices from graph path scores.
Okasha et al. (2023)	Member redundancy based on static indeterminacy	Verify the comparison in the axial domain.
Caredda et al. (2022); Abbracciavento dan De Biagi (2025)	Local damage and load-path changes	Test whether pre-analysis priorities capture relevant redistribution consequences.
Setiawan et al. (2024)	Robustness quantification for concrete frames	Map scenario definitions and consequences before establishing a reference.
Echard et al. (2011)	Kriging active learning for reliability	Separate probability estimation from deterministic scenario priority.
Sanchez-Gonzalez et al. (2020)	Particle-based physical simulation with graph networks	Check representation compatibility before transfer to structures.
Song et al. (2023)	Physics-informed GNN elastic analysis	Distinguish solution development from fixed-model inference.
Chou et al. (2024)	Hierarchical structural prediction with virtual nodes	Audit domain and outputs before use in S2.

Tabel 2.11 (lanjutan)

Rujukan	Fokus yang digunakan	Implikasi untuk ARI
Roberts et al. (2017)	Validation for structured data	Keep parent models as the split groups.
Cawley dan Talbot (2010)	Model-selection and evaluation bias	Separate configuration selection from final evaluation.

Table 2.11 is a focused synthesis, not an exhaustive list of all research. Each row must be followed by checks of assumptions, equations, data, and reproducibility when a comparison implementation is developed. Accuracy percentages from published articles are not transferred as expected performance on ARI data.

2.13.1 Methodological Connections to Additional Sources

Sources that are only partly related to ARI remain useful when the component being compared is specified. The following eight additional sources broaden the comparison of constructs, algorithms, physical representations, and evaluation. Their use is conceptual; it does not mean that their algorithms have been reproduced on the ARI dataset.

Foundations of mechanical redundancy. Gade et al. (2021) derive a continuum theory of redundancy for elastostatic structures, while von Scheven et al. (2021) develop matrices describing the distribution of static indeterminacy in discrete structures. These works show that mechanical-redundancy information contains system properties that cannot be reduced to a graph path count. ARI should be tested against these mechanical references using comparable domains, disruptions, and input information; computational advantage is not assumed without measuring total cost.

Pellegrino (1993) use singular-value decomposition of the equilibrium matrix for structural computations. Its connection is the diagnosis of mechanism

spaces and rank, not an equivalence between the ARI score and a singular value. Mechanism findings in ARI verification are situated within existing mechanics foundations; this dissertation does not claim to discover the concept of a mechanism.

Path search and the definition of an optimum. Yen (1971) discuss finding the K shortest loopless paths. This provides a more specific algorithmic comparison than a generic reference to path search. However, loopless paths are not necessarily edge-disjoint. The shortest-path cost objective also differs from the ARI path-score combination objective. Replacing the algorithm requires checking the objective function, constraints, and candidate completeness; the number of paths found is not an optimality certificate for the ARI formulation.

Physical prediction and reduced evaluation. Pfaff et al. (2021) demonstrate the use of graph networks for mesh-based simulation. This source broadens the context of physical representation but does not provide direct evidence of compatibility with static truss prediction under the ARI protocol. Features, boundary conditions, targets, and solution-generation procedures still need to be matched before transfer.

Moustapha et al. (2022) provide a review, framework, and benchmark for active learning in structural reliability. Its connection to ARI is the effort to reduce costly evaluations, while the difference lies in the reliability target and the use of additional evaluations to learn the performance function. A fair comparison must include the cost of building the surrogate, not only the final evaluation. The source benchmark is not used as an ARI performance figure.

Configuration selection and error reporting. Cawley dan Talbot (2010) demonstrate the risk of overfitting during model selection and bias in performance estimates. For this study, this implies separating parameter and threshold selection from final evaluation. Parent models remain the grouping unit so that related scenarios are not spread across development and test partitions.

Saito dan Rehmsmeier (2015) discuss the value of precision–recall curves when classes are imbalanced. This source supports metric checks for rare critical cases, but precision–recall is not equivalent to recall–reduction. The ARI study must still report the number of missed critical cases, the fraction of scenarios forwarded, and their mechanical consequences. Performance on one measure does not replace another.

The catalog in Section 2.11.2 contains the identities and connections of 80 additional sources, including the eight synthesized here. It distinguishes sources cited in the argument from those recorded for future review, without counting the same work twice.

2.14 Synthesis Strategy and Evidence Quality

The search is grouped into redundancy mechanics, graphs and ranking, GNN response prediction, and statistical evaluation. The unit of record is a traceable primary source. A preprint and its journal version are linked so that they are not counted as two independent pieces of evidence. Assessment considers not only journal reputation but also question fit, methodological clarity, model completeness, and reproducibility. For journal articles used as prior evidence and comparisons, this synthesis is restricted to active Scopus journals in Q1 of CiteScore 2025 in relevant ASJC categories; books, proceedings, and software documentation are identified separately as foundations or implementation references.

For this protocol, class A means the method and data are sufficiently clear to plan a reproduction; class B means the source is relevant but requires additional information; and class C means it provides contextual support only. These classes are internal assessments, not universal rankings of scientific quality. Each classification must include a rationale and reviewer. The focused review on September 6, 2026 did not produce a systematic inclusion–exclusion count; search totals are not reported without an actual log.

2.14.1 Core and Supporting References

The literature is grouped by its role in the argument, not merely by title or keyword similarity. **Core references** directly ground the construct, formulation, algorithmic and mechanical references, comparator candidates, or primary validation design. **Supporting references** provide context, alternatives not yet adopted, and extensions to the domain or benefit measures. A supporting source may play a direct role in an additional analysis, such as energy measurement, without grounding ARI's mechanical validity.

This grouping does not indicate source quality or audit completeness and is distinct from the A–C reproducibility-readiness classes described earlier. Software documentation may be a core reference for an implementation contract, but it is not evidence of scientific novelty. A GNN candidate may be core to the comparison design even when a compatible implementation and trained parameters are unavailable. A source with multiple roles is counted once under its stated dominant function.

The bibliography remains a single alphabetical list; books, proceedings, and software documentation are distinguished by function from journal articles. For journal articles used in the synthesis, inclusion requires an active Scopus journal in Q1 according to CiteScore 2025 in a relevant ASJC category. Metadata were checked on October 10, 2026. Quartile is a property of a journal and category, not a guarantee of the quality or applicability of every article. Table 2.12 contains 43 of the 44 records in `prism-uploads/export.ris` that met these criteria. The record from *Next Sustainability* is excluded because the journal was not Q1 in the examined CiteScore update. The table summarizes metadata and initial roles; inclusion does not mean that every article was audited in full or used as validation evidence. The catalog in Section 2.11.2 remains a search inventory, not a list of sources all adopted in the Q1 synthesis.

Table 2.12: Q1 journal articles in the RIS file and their relevance to the ARI review.

RIS source	Theme and comparison limits
Buitrago et al. (2023). *Robustness of a full-scale precast building structure subjected to corner-column failure*. Structures, 52, 824–841. DOI: 10.1016/j.istruc.2023.03.146	Bukti eksperimental skala penuh; cocokkan jenis kolom, kondisi batas, dan ukuran konsekuensi.
Caredda et al. (2022). *Analysing local failure scenarios to assess the robustness of steel truss-type bridges*. Engineering Structures, 262. DOI: 10.1016/j.engstruct.2022.114341	Jalur alternatif pada jembatan; audit validasi model dan perbedaan aksi lentur terhadap model ARI.
Li et al. (2023). *Machine learning prediction of structural dynamic responses using graph neural networks*. Computers and Structures, 289. DOI: 10.1016/j.compstruc.2023.107188	Prediksi respons dinamik GNN; bedakan target dinamik dari penyaringan skenario statik.
Setiawan et al. (2024). *Practical methodology for quantifying the structural robustness of RC building structures*. Structures, 70. DOI: 10.1016/j.istruc.2024.107898	Pengukuran ketahanan; petakan definisi skenario dan konsekuensinya sebelum membandingkan indeks.
Zhao et al. (2024). *Collapse-resistant mechanisms induced by various perimeter column damage scenarios in RC flat plate structures*. Structures, 59. DOI: 10.1016/j.istruc.2023.105716	Mekanisme pascakerusakan; respons nonlinier dan jalur gaya tidak direpresentasikan penuh oleh skor lintasan.
Buitrago et al. (2025). *Robustness of a full-scale precast building structure after edge column failure*. Engineering Structures, 326. DOI: 10.1016/j.engstruct.2024.119495	Uji kolom tepi skala penuh; periksa kesetaraan kondisi uji dan kasus kanonik.

Tabel 2.12 (lanjutan)

RIS source	Theme and comparison limits
Ribeiro et al. (2025). *Optimal risk-based design of planar RC frames under progressive collapse: Influence of frame aspect ratio and column cross-section*. Engineering Structures, 342. DOI: 10.1016/j.engstruct.2025.120905	Desain berbasis risiko; probabilitas dan optimasi desain berbeda dari prioritas deterministik ARI.
Soleimani et al. (2026). *Progressive collapse of post-and-beam mass-timber buildings: Quantifying robustness and dynamic increase factors*. Journal of Building Engineering, 130, 117116. DOI: 10.1016/j.jobe.2026.117116	Ketahanan dan faktor dinamik; transfer ke material dan sistem struktur lain perlu diuji.
Wijethunga et al. (2025). *Robust and efficient dual-graph neural networks for structural damage detection and localization*. Engineering Structures, 343. DOI: 10.1016/j.engstruct.2025.121265	Deteksi dan lokalisasi kerusakan; bedakan tugas diagnosis dari prediksi konsekuensi gangguan.
Lee et al. (2026). *Seismic response prediction of reinforced concrete frames using graph neural networks with physical interpretability*. Structures, 90. DOI: 10.1016/j.istruc.2026.112487	Prediksi seismik berbasis GNN; audit kesesuaian beban, fitur, dan target keluaran.
Makoond et al. (2024). *The effect of continuity reinforcement on the progression of collapse in reinforced concrete buildings*. Structures, 61. DOI: 10.1016/j.istruc.2024.105981	Pengaruh detailing terhadap progresi runtuh; tidak setara dengan perubahan skor graf saja.

Tabel 2.12 (lanjutan)

RIS source	Theme and comparison limits
Parisi et al. (2024). *On the use of mechanics-informed models to structural engineering systems: Application of graph neural networks for structural analysis*. Structures, 59. DOI: 10.1016/j.istruc.2023.105712	GNN dengan informasi mekanis; bandingkan fitur, keluaran, dan parameter model sebelum integrasi.
Caredda et al. (2025). *Analysing the cost-effectiveness of improving building robustness through segmentation*. Engineering Structures, 343. DOI: 10.1016/j.engstruct.2025.121165	Efektivitas biaya intervensi; jangan disamakan dengan biaya komputasi penyaringan.
Cai dan Jelovica (2026). *A hybrid global–local computational framework for ship hull structural analysis using homogenized model and graph neural network*. Marine Structures, 109. DOI: 10.1016/j.marstruc.2026.104077	GNN struktur kapal; transfer domain dan pemetaan model lokal-global perlu diperiksa.
Martins dan Thrall (2026). *Investigating modular truss bridges with hybrid system and internal redundancy*. Journal of Constructional Steel Research, 245. DOI: 10.1016/j.jcsr.2026.110513	Redundansi jembatan modular; cocokkan mekanisme sistem dan definisi gangguan.
López et al. (2025). *Reliability-based structural assessment of steel truss bridges subjected to failure scenarios*. Engineering Structures, 341. DOI: 10.1016/j.engstruct.2025.120850	Reliabilitas jembatan; pisahkan peluang kegagalan dari skor prioritas ARI.

Tabel 2.12 (lanjutan)

RIS source	Theme and comparison limits
Huber et al. (2021). *Finite element analysis of alternative load paths to prevent disproportionate collapse in platform-type CLT floor systems*. Engineering Structures, 240. DOI: 10.1016/j.engstruct.2021.112362	Jalur alternatif pada lantai kayu; batas material dan model elemen perlu dicatat.
Esparza et al. (2026). *A graph neural network and decision tree modeling approach for predicting wildfire-induced building damage*. Computer-Aided Civil and Infrastructure Engineering, 47. DOI: 10.1016/j.cacaie.2026.100085	Prediksi kerusakan akibat bahaya; berbeda dari analisis konsekuensi pascagangguan struktur.
Li et al. (2025). *Dynamic graph-based approach for prediction of spatiotemporal response of RC structure to impact loads*. Computers and Structures, 316. DOI: 10.1016/j.compstruc.2025.107861	Respons dinamik tumbukan; uji transfer dari prediksi temporal ke penyaringan praanalisis.
Buitrago et al. (2024). *Failure analysis after the progressive collapse of a precast building*. Engineering Structures, 321. DOI: 10.1016/j.engstruct.2024.118893	Analisis kegagalan pascaruntuh; gunakan untuk konteks mekanisme, bukan validasi pemeringkatan ARI.
Song et al. (2026). *Efficient Full-Range Nonlinear Analyses of Structural Systems Based on Heterogeneous Graph Learning*. Engineering, 59, 320–336. DOI: 10.1016/j.eng.2025.07.001	Analisis nonlinier GNN; bandingkan domain, biaya, dan sasaran keluaran secara terpisah.

Tabel 2.12 (lanjutan)

RIS source	Theme and comparison limits
Zeng et al. (2026). *Prediction model for progressive collapse resistance of concrete beam-column subassemblies considering tensile catenary action using machine learning algorithms*. Results in Engineering, 29. DOI: 10.1016/j.rineng.2025.108944	Prediksi kapasitas subrakitan; target terlatih bukan ukuran langsung prioritas elemen.
Elkady et al. (2024). *Progressive collapse: Past, present, future and beyond*. Structures, 62. DOI: 10.1016/j.istruc.2024.106131	Tinjauan progresi runtuh; sumber konteks, bukan bukti validasi ARI.
Makoond et al. (2023). *Corner-column failure scenarios in building structures: Current knowledge and future prospects*. Structures, 49, 958–982. DOI: 10.1016/j.istruc.2023.01.121	Tinjauan skenario kolom sudut; gunakan untuk cakupan skenario dan batas transfer.
Ghasemi et al. (2024). *A new generation of DCR by introducing reliability-survivability safety measure*. Structures, 62. DOI: 10.1016/j.istruc.2024.106154	Ukuran keselamatan berbasis reliabilitas; definisi dan targetnya tidak otomatis sebanding dengan ARI.
Zhou et al. (2026). *Physics-informed graph neural networks for minimal mass design of cable-strut structures*. Computer-Aided Civil and Infrastructure Engineering, 50. DOI: 10.1016/j.cacaie.2026.100210	Desain optimum berbasis GNN; bedakan tugas optimasi dari prediksi respons dan penyaringan.

Tabel 2.12 (lanjutan)

RIS source	Theme and comparison limits
Arias-Gracey et al. (2026). *Effectiveness of roof-level retrofit beams in improving the robustness of existing steel building structures*. Journal of Constructional Steel Research, 246. DOI: 10.1016/j.jcsr.2026.110574	Efek retrofit; intervensi struktural berbeda dari prioritas skenario gangguan.
Sio et al. (2026). *Simplified tying-bending model for robustness assessment of floor systems under corner column loss*. Engineering Structures, 366. DOI: 10.1016/j.engstruct.2026.123486	Model mekanis kehilangan kolom; periksa asumsi penyederhanaan dan respons yang diukur.
Tumbeva et al. (2022). *Investigating the redundancy of steel truss bridges composed of modular joints*. Journal of Constructional Steel Research, 188. DOI: 10.1016/j.jcsr.2021.107038	Kehilangan batang diagonal; periksa redistribusi, aksi lentur sambungan, dan efek dinamik.
Mirzahosseini et al. (2023). *Robustness assessment of RC frame buildings with HPFRCC subjected to progressive collapse*. Results in Engineering, 17. DOI: 10.1016/j.rineng.2022.100809	Ketahanan rangka beton; cocokkan material dan model perilaku pascakerusakan.
Francioli et al. (2023). *Structural robustness analysis of RC frames under seismic and blast chained loads scenarios*. Journal of Building Engineering, 67. DOI: 10.1016/j.jobe.2023.105970	Bahaya berantai; cakupan beban multi-bahaya belum diwakili ARI saat ini.

Tabel 2.12 (lanjutan)

RIS source	Theme and comparison limits
Martinelli et al. (2024). *Robustness assessment of half-joint RC girder bridges*. Engineering Structures, 306. DOI: 10.1016/j.engstruct.2024.117712	Ketahanan jembatan girder; periksa jenis kegagalan dan geometri terhadap domain model.
Kiakojouri et al. (2023). *Experimental studies on the progressive collapse of building structures: A review and discussion on dynamic column removal techniques*. Structures, 57. DOI: 10.1016/j.istruc.2023.105059	Tinjauan uji pelepasan kolom; gunakan untuk protokol respons dan validasi eksperimen.
Ravasini et al. (2024). *A robustness-oriented procedure for the design of tying reinforcement in precast concrete hollow-core floors*. Structures, 63. DOI: 10.1016/j.istruc.2024.106427	Detail tulangan pengikat; desain ketahanan bukan skor prioritas praanalisis.
Chou et al. (2024). *StructGNN: An efficient graph neural network framework for static structural analysis*. Computers and Structures, 299. DOI: 10.1016/j.compstruc.2024.107385	Kandidat prediktor GNN statik; audit fitur, keluaran, parameter terlatih, dan waktu integrasi.
Lim et al. (2022). *System-reliability-based disaster resilience analysis: Framework and applications to structural systems*. Structural Safety, 96. DOI: 10.1016/j.strusafe.2022.102202	Ketahanan berbasis reliabilitas sistem; pisahkan probabilitas sistem dari skor deterministik.

Tabel 2.12 (lanjutan)

RIS source	Theme and comparison limits
Abbracciavento dan De Biagi (2025). *Damages on complex structural systems: Effects on load paths*. International Journal of Solids and Structures, 311. DOI: 10.1016/j.ijsolstr.2024.113213	Perubahan jalur beban akibat kerusakan; uji apakah skor graf menangkap transisi perilaku sistem.
Sio dan Izzuddin (2024). *Enhanced rational horizontal tying force method for practical robustness design of building structures*. Engineering Structures, 301. DOI: 10.1016/j.engstruct.2023.117268	Metode gaya pengikat; bandingkan dasar mekanis tanpa menyamakan tujuan desain dan penyaringan.
Gu et al. (2022). *Physics-informed neural networks for analysis of 2D thin-walled structures*. Engineering Analysis with Boundary Elements, 145, 161–172. DOI: 10.1016/j.enganabound.2022.09.024	PINN struktur dinding tipis; metode dan target berbeda dari GNN prediksi struktur.
Gade et al. (2021). *A continuum-mechanical theory of redundancy in elastostatic structures*. International Journal of Solids and Structures, 226–227, 110977. DOI: 10.1016/j.ijsolstr.2021.01.022	Teori redundansi kontinu; bedakan operator mekanis dari skor graf diskret ARI.
Borraccia et al. (2025). *Green metrics for AI: A hybrid strategy for environmental impact assessment*. Array, 28. DOI: 10.1016/j.array.2025.100528	Metrik dampak AI; metrik energi/emisi perlu diukur terpisah dari kinerja mekanis.

Tabel 2.12 (lanjutan)

RIS source	Theme and comparison limits
Jeanquartier et al. (2026). *Assessing the carbon footprint of language models: Towards sustainability in AI*. Resources, Conservation and Recycling, 226. DOI: 10.1016/j.resconrec.2025.108670	Jejak karbon model bahasa; konteks pelaporan tidak dapat ditransfer sebagai biaya GNN struktur.
von Scheven et al. (2021). *Quantification of the redundancy distribution in truss and beam structures*. International Journal of Solids and Structures, 213, 41–49. DOI: 10.1016/j.ijsolstr.2020.11.002	Distribusi ketidaktentuan statis; bandingkan konstruk mekanis dengan skor jalur ARI.

2.14.2 Matriks Kesesuaian dan Perbedaan Metode

The detailed matrix is available in `results/literature/reference_matrix.csv`. Setiap baris Each row contains the citation key, group, role, theme, assumptions or domain, inputs, method, outputs, similarities to ARI, differences and their rationale, required evidence, citation location, review status, and compatibility assessment. The method and domain columns summarize how the source is used in this dissertation; they are not a complete extraction of each article. A citation location identifies one file where the source is used, not every occurrence.

Review status preserves the trace in Section 2.11: `T6` denotes a focused review recorded on September 6, 2026; `T8` denotes a limited review on September 8, 2026, within the scope stated there; `T8-additional` marks eight sources from the new catalog integrated on the same date, with review scope recorded in the catalog CSV; and `W` denotes an inherited source without a focused-review record in that inventory. These codes do not indicate a full-article audit, methodological approval, or reproduction of results. Compiling this matrix does not add claims of source

review that have not been performed.

The initial compatibility assessment for every row is recorded as *undecided*. After comparing the source evidence with its application, the decision is recorded using the following categories.

Consistent within the reviewed domain.

The adopted definitions, assumptions, and steps agree with the source; equation or section numbers and review results are recorded. This decision is not a guarantee across all structural families.

Deliberate adaptation.

Changes to inputs, formulas, algorithms, or targets are explained together with their rationale and consequence tests.

Unvalidated new proposal.

The methodological difference is a research hypothesis requiring evidence or experiments, not automatically an error or an advantage.

Outside the comparison domain.

Assumptions, objectives, or information access differ, so results cannot be compared directly without additional mapping.

Inconsistency requiring correction.

The implementation conflicts with the assumptions or steps claimed to be adopted; the issue and corrective action must be traceable.

One source may support consistency for one component but not another. As review becomes more detailed, rows may be split by component or claim without increasing the count of unique sources. Compatibility is not calculated as the percentage of articles that appear to agree: a conflict over one important assumption must be examined even when many other sources support the general context.

2.15 Statistical Foundations and Uncertainty

2.15.1 Ranks and Target Quantities

Spearman correlation is defined as the Pearson correlation between the midranks of CI and CRC. Midranks are used for tied scores. If either vector is constant, the correlation is undefined; it is not replaced with zero or one. The proportion of models with undefined correlations is also reported.

For rank analysis, Kendall's τ_b is used as a complementary check:

$$\tau_b = \frac{n_c - n_d}{\sqrt{(n_c + n_d + t_x)(n_c + n_d + t_y)}}. \tag{2.21}$$

n_c and n_d are the numbers of concordant and discordant pairs; t_x and t_y count pairs tied only on each respective variable. A zero denominator yields an undefined value. The Jaccard index for two priority sets $\mathcal{T}_1$ and $\mathcal{T}_2$ is written as $|\mathcal{T}_1 \cap \mathcal{T}_2|/|\mathcal{T}_1 \cup \mathcal{T}_2|$; the case in which both sets are empty is recorded separately.

2.15.2 Data Structure and Intervals

Roberts et al. (2017) emphasize the need for validation strategies that account for data-dependence structure. In this study, all derivatives of a parent model are kept together. One model with hundreds of scenarios does not substitute for hundreds of independent models.

Bootstrap is used as a resampling framework to estimate statistical uncertainty (Efron, 1979). The resampling unit and scheme must follow the specified estimand. Per-model, per-family, and weighted pooled estimates have different meanings; reporting weights must not be chosen after seeing which group is most favorable.

2.15.3 Risk Control and Claims

Angelopoulos et al. (2024) discuss controlling the expected value of a monotone loss function through conformal calibration. This reference distinguishes

guarantees in expectation from high-probability risk bounds. The ARI protocol has not adopted conformal methods as a substitute for Gate D: the exchangeability unit must be the parent model, the loss function must be appropriate, and calibration assumptions must first be checked. CRC in this dissertation continues to mean the reference criticality index, not that conformal method.

2.16 Research Gap and Novelty Position

The gap under investigation is whether a combination of multiple alternative-path scores and initial utilization can serve as a more useful pre-inference filter than simple measures, without accessing post-disruption responses at deployment. This is a domain-bounded question, not a claim that none of ARI's components have been studied before.

Novelty will be assessed across three elements: the index definition and properties, a cross-family validation protocol capable of falsifying the candidate, and measurement of net benefit with a fixed GNN. If the same components already exist in the literature, the contribution must be described as a new combination, adaptation, or evaluation—not as discovery of those components. The final comparison matrix must include data requirements and comparator computational costs.

2.17 Conceptual Framework

The research framework separates the operational-information chain from the evaluation chain. The intact model and disruption definition produce ARI and the forwarding decision. Separately, post-disruption analysis provides the reference, which is joined only after the decision has been stored. Failures in the evaluation chain must not silently change the filter inputs. Figure 2.4 shows where the two chains are joined. Reference responses may be used for development according to the data split, but they are not operational filter inputs when evaluating a frozen configuration.

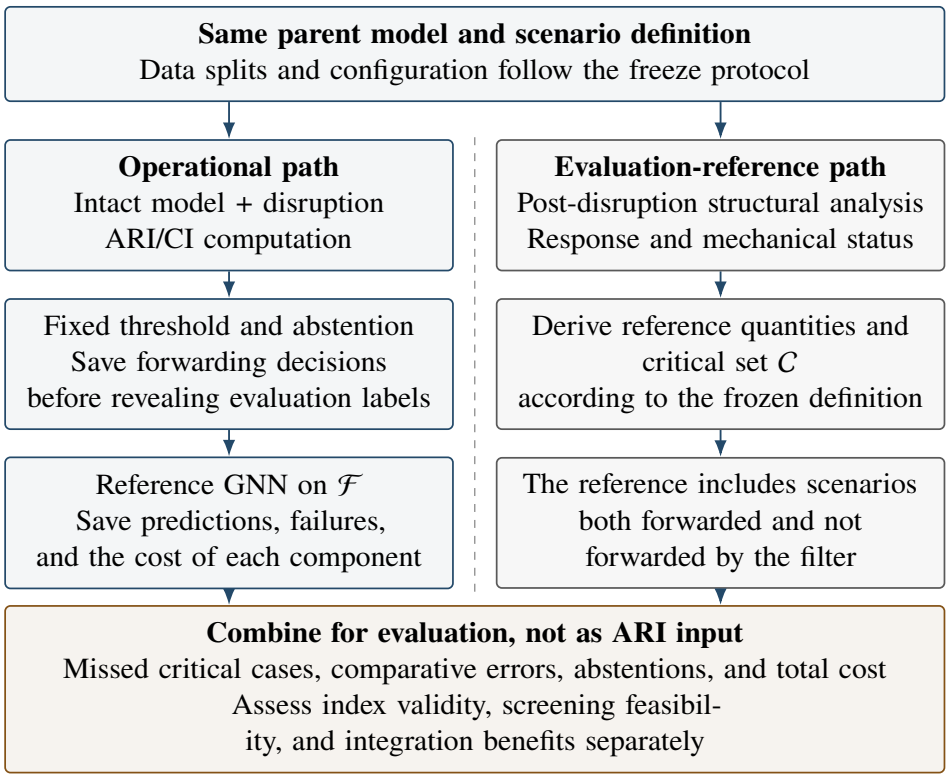


Figure 2.4: Separation of operational and reference information when evaluating a frozen configuration. No post-disruption response flows from the reference path into ARI inputs. The diagram describes the design, not a GNN integration result.

2.18 Research Hypotheses

The following empirical hypotheses are tested after implementation verification is complete.

1. H1: for the specified families, CI agrees with mechanical rankings and adds value over the frozen comparator set.
2. H2: rankings and priority sets remain stable under perturbations within the specified range, without excluding unfavorable cases.
3. H3: the frozen threshold meets Gate D recall and reduction criteria on independent parent models.
4. H4: integration meets final-coverage and recall-degradation limits and achieves net time savings with the same GNN.

H1–H4 are not considered true merely because the formula lies between zero and one. Unsupported hypotheses are reported with their effect sizes and uncertainty. H4 may fail even if H1–H3 are supported, for example, if path search costs more than the inference it saves.

Chapter III

RESEARCH METHODS

3.1 Research Design and Methods Implemented

The research underlying the evidence presented in this dissertation is a method-development study with deterministic computational verification. Its objects are the ARI formula, the path-search algorithm, and the relationship between index outputs and the response of canonical axial-bar models. The structural robustness methodology study available in the journal file combines a modelling strategy with a robustness index and compares its models against selected experimental tests (Setiawan et al., 2024). This dissertation has a different scope: four models are selected as cases that can be traced analytically and numerically, rather than as a random sample representing a population of structures. The results are therefore used to check the implementation and identify limiting examples, not to estimate screening performance across a population.

3.1.1 Materials, Units of Analysis, and Configuration

The computational materials comprise two parallel 1D bars, a planar triangular truss, a square with one diagonal, and a square with two diagonals. Geometry, constraints, loads, materials, cross-sections, and capacities are taken from `examples/verification_models.json`. Full definitions and the configuration that produced the tables are presented with the results in Chapter 4. Each model has one prescribed load case. No field measurement data, nonlinear models, or trained GNN parameters are used to produce these verification results.

The canonical model is the unit of mechanical analysis. A single-element disruption scenario is the unit of observation derived from it. The four models contain 16 elements in total; each element is examined at residual factors $\lambda \in \{0, 0.25, 0.50, 0.75\}$, yielding 64 disruptions. The control $\lambda = 1$ is recorded for each element, yielding 16 controls, not additional independent models. No train–

validation–test split is applied to these four cases because all are used for method development and checking.

3.1.2 Calculation and Checking Procedure

1. The intact model is read, its stiffness matrix assembled, and its elastic response calculated using `examples/truss_reference.py`. The reduced-matrix rank, equilibrium, member forces, utilization, and energy are checked against the corresponding reference cases.
2. Intact-model attributes are normalized using a fixed scale for each case. Baseline and disruption scores are calculated by `examples/ari_core.py` through simple-path enumeration and edge-disjoint path-combination optimization. The score inputs do not include post-disruption mechanical responses.
3. In the physical model, element area and capacity are multiplied by the disruption factor. In the graph, that factor multiplies the normalized weight. These two operations are recorded separately; using the same factor does not imply that the physical response and graph score change in the same way.
4. Post-disruption responses are checked separately. A matrix with insufficient rank is recorded as having no unique solution, without replacing it with a pseudoinverse solution or filling an undefined response with zero.
5. The algorithm is compared with a reference enumeration; the formula and solver are checked against analytical values, invariance, monotonicity, and boundary cases. An uncertified search results in abstention.
6. Outputs, check statuses, configuration, library versions, and source hashes are archived by `examples/run_verification.py` in `results/verification`. The result tables and figures are generated from those outputs, not values inserted during manuscript editing.

3.1.3 Use of Claude Code in the Development Workflow

Claude Code supports narrowly scoped changes to the prototype, debugging, and maintenance of LaTeX files. Its use is part of the software development process, not a structural analysis method, a data source, or a means of producing verification results. Unlike a GNN designed to analyze or predict structural responses, such as the StructGNN framework for static structural analysis (Chou et al., 2024), Claude Code is not used to predict structural responses or generate mechanical results. The researcher remains responsible for defining the formulas, mechanical assumptions, model configuration, and verification criteria, and for interpreting the results.

The workflow consists of the following steps:

1. The researcher defines the task, identifies the relevant files and scope of change, and sets acceptance criteria based on the research formulation and numerical contracts.
2. Claude Code receives the relevant code context and targeted instructions, then proposes or applies software or manuscript changes. The researcher reviews each proposed change against the equations, inputs, outputs, and established domain limits before accepting it.
3. Code changes are tested using the deterministic runner `examples/run_verification.py`. Results are compared with analytical values, reference enumeration, solver checks, and relevant boundary cases. Failures and uncertified outcomes are retained in the verification outputs rather than altered to match expectations.
4. Once the checks pass, tables and figures are generated from the runner outputs. The researcher checks the code hashes, configuration, library versions, and results against the manifest, then confirms that the manuscript values match the output files before using them as evidence.

Claude Code is therefore limited to reviewable implementation and editing support. It does not determine whether a formula is correct, replace the solver or independent checks, or provide result values. The verification manifest records the

configuration, library versions, and source hashes, but not the Claude Code version or model, prompts, or session identity. Details that were not recorded are not inferred or reconstructed in this report.

3.1.4 Results Analysis and Limits of Inference

The analysis matches calculated values to expected values and tolerances, then compares graph connectivity, RR, ARI, and CI with mechanical rank and response. Cases in which the score does not distinguish a rank reduction are examined through geometry, initial forces, and displacement modes. Passing software checks is not treated as passing the index's mechanical validity assessment.

The number of automated checks is reported separately from the number of models and scenarios. No generalization interval is calculated under the assumption that disruptions from the same model are independent samples. This experiment makes no claims about population-level screening recall, superiority over all comparators, or GNN inference savings. The protocols for obtaining such evidence are described in later sections of this chapter and distinguished from the methods already implemented.

3.1.5 Guard Development and Diagnostic Evaluation

After a failure to distinguish mechanical cases was identified, a kinematic guard based on intact-model stiffness was implemented separately from the ARI formula. Its design, algebraic basis, and results are presented in Section 4.2. The four canonical models are used for regression testing, with 16 additional parametric models for development diagnostics. None are blind-test data. The four canonical models and the 16 parametric models do not count toward the parent-model quotas for S1–S5. The main dataset generation, threshold calibration, comprehensive ablation, and blind validation have not been conducted; the procedures for carrying them out are presented separately in Section 3.9.

The ARI-only, guard-only, and combined policies are compared at the diagnostic CI threshold of 0.5. Decisions are stored before the reference disruption

responses are calculated by the runner. Rank deficiency, utilization exceeding one, and a displacement ratio exceeding two are counted separately and in combination. Outputs include critical cases not forwarded, not merely successful detection of mechanisms. Preparation and decision times are recorded, but are not used to claim GNN savings.

3.2 Further Validation Protocol and Data Splitting

This section and the subsequent evaluation protocols document the planned S1–S7 validation, which has not yet been conducted. The sample sizes, data splits, calibration, statistical inference, and GNN integration described here are not accounts of experiments that produced the tables in Chapter 4. The study uses a quantitative computational design. Development configurations are separated from final evaluation to limit selection bias (Cawley dan Talbot, 2010), and all derived scenarios are split by parent model to avoid ignoring data dependence (Roberts et al., 2017). The independent unit is the parent model, not each scenario, related geometric variant, or random-seed repetition.

Figure 3.1 shows the operational forwarding rule. The abstention pathway is separated so that an unavailable score is not mistaken for low priority. The diagram is an integration plan, not evidence that a GNN model is available or that screening benefits have been achieved.

Each S1–S5 family is planned to contain at least 60 parent models. Its adequacy has not been demonstrated; the subtotal of 300 parent models is a plan, not a generated dataset. The S1–S5 validation columns represent blind validation, not a subset for threshold selection. Calibration uses the internal development subset described in Section 3.9. Sample adequacy is assessed in a pilot through precision and interval coverage, after which final counts and sampling rules are frozen before the final test is opened. The number of scenarios depends on the element inventory and disruption levels; the figure of 1,800 from the original manuscript illustration is not used as a result or fixed target. Pilot data used to select the method are

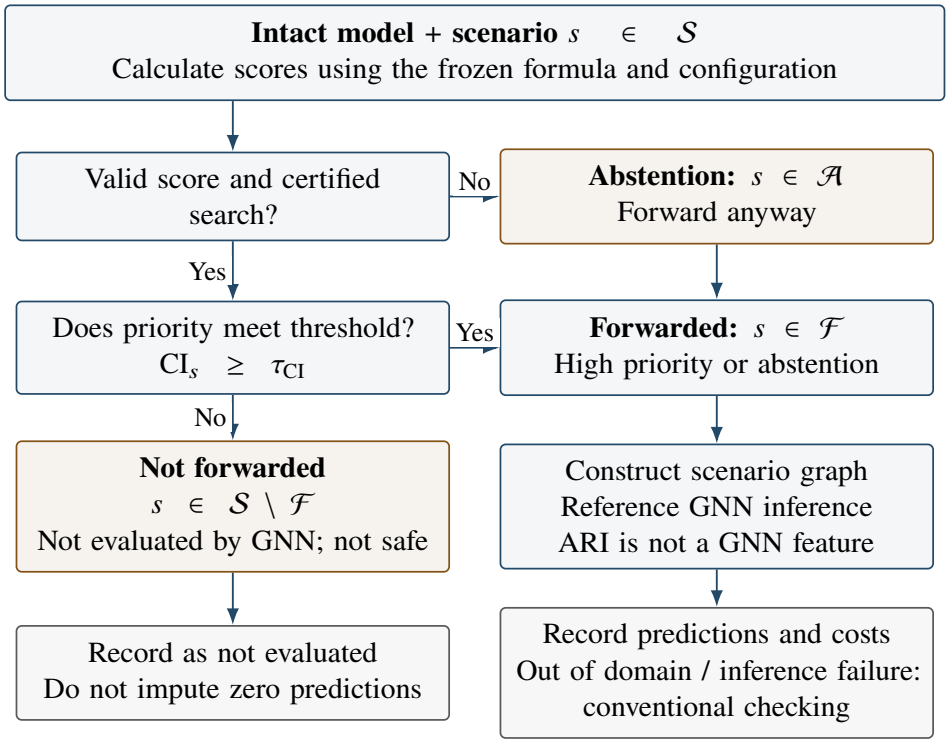


Figure 3.1: Screening workflow before reference GNN prediction. Abstentions count toward the forwarded total. Reference outputs are still required for all scenarios to evaluate missed critical cases, including those not forwarded.

Table 3.1: Initial sample plan; not research-result data.

Code	Family or function	Development	Validation	Final test
S1	Simple and continuous beams	36	12	12
S2	Planar trusses	36	12	12
S3	Planar frames	36	12	12
S4	Braced frames	36	12	12
S5	Simple spatial structural systems	36	12	12
	Planned subtotal S1–S5	180	60	60
S6	GNN integration on new S2 parent models	60	20	20
S7	External evaluation, at least 25 new models	Not for calibration		

development data, not blind-evaluation data.

Intact models with ambiguous boundary conditions or untraceable attributes are rejected before splitting. In contrast, loss of stability caused by a disruption is retained as an important case. Numerical failures that cannot yet be distinguished from mechanisms are marked unresolved and are not silently removed from evaluation.

3.3 Development and Freezing Protocol

1. Define parent-model identities and data splits before calibration. All variants and scenarios derived from a parent follow the same split.
2. Use internal cross-validation within the development subset to select normalization, formula, coefficients, and search configuration. Set the screening threshold after selecting the formula, using either an internal calibration subset or documented out-of-fold predictions.
3. Freeze all decisions, including critical-case definitions, tolerances, ranking tie rules, and thresholds. Blind validation evaluates this configuration; it does not select the most favorable threshold.
4. If validation prompts a revision, designate that set as development data. Prepare a replacement independent validation set before claiming blind evaluation of the revised version.
5. Open the final test after all final decisions are frozen. Changes after the final test require new independent test data.

All post-disruption responses in development may be used according to their data roles. ARI does not select GNN training data. Final-test responses, including scenarios not forwarded, are opened only for evaluation.

The split in Figure 3.2 is made before calibration. Each parent model and all its variants belong to a single group, so variations in disruption to the same structure do not become test data that are incorrectly treated as independent of development.

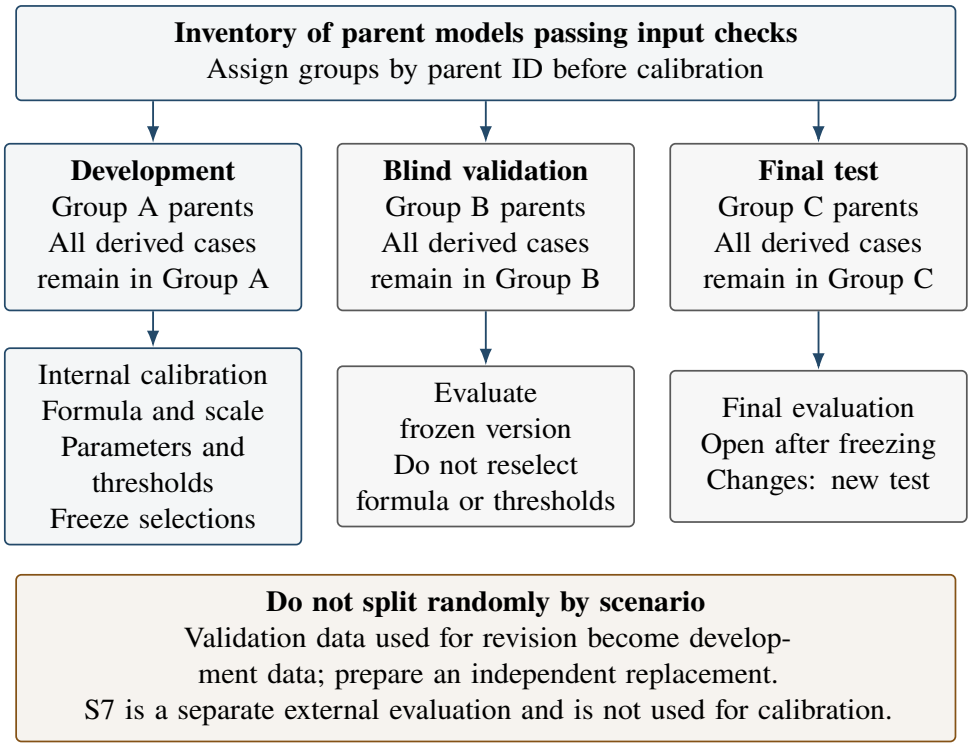


Figure 3.2: Parent-model data-splitting plan. Letters A–C in the diagram label data groups only, not Validation Gates A–C. Model counts follow the sample plan and post-pilot determination, not collected results.

3.4 Stages and Prerequisites for Further Validation

The verification already conducted is described in Section 3.1. This section defines stages for further validation, rather than a record of experiments conducted. Hardware, operating system, solver, numerical libraries, and inference devices must be recorded in the manifest before the corresponding performance measurements. Hardware identities that have not yet been determined are not treated as research facts. Table 3.2 presents a prerequisite-based sequence, not a semester schedule or dates assumed to have occurred.

Table 3.2: Further validation stages and transition criteria.

Stage	Main activity	Deliverable and transition criterion
Audit	Model sources and literature coverage	Traceable data and explicit domain
Revision	Address canonical-model limitations	Versioned candidate and re-verification
Pilot	Model generation and precision	Manifest, data split, and sample adequacy
Freeze	Development and ablation	Final configuration before blind evaluation
Evaluation	Final test, confirmation, and integration	A–D evidence, domain limits, and total cost
Reporting	Analysis and documentation	Reproduction package and evidence-based conclusions

Stage transitions are not determined solely by the passage of time. Verification failures or data-audit issues must be resolved before final-test labels are opened. Delayed integration because the GNN is not yet compatible is reported as implementation status, not addressed by changing the evaluation target after results are known.

3.5 Variables and Operational Definitions

Research variables are distinguished by their roles in model generation, index calculation, experiment control, and evaluation. This separation prevents evaluation responses from being incorporated as operational features.

Table 3.3: Operational variables and information-access rules.

Group	Variable	Use rule
Geometry	Coordinates, connectivity, lengths, and supports	Available in the intact model and frozen for each scenario.
Initial mechanics	E_e, A_e, I_e, R_e, q_e, initial demands and reactions	Valid input if its definition, units, and acquisition cost are recorded.
Disruption	Target element, λ, physical-change rule	Defined before post-disruption responses are opened.
ARI parameters	Scale, weights, K, γ, β, η, regularization	Selected only during development; final version remains fixed.
Reference response	Displacement, force, utilization, stiffness, and mechanism	Development use follows the split, or evaluation use occurs after forwarding decisions are stored.
GNN prediction	Target, output, inference status	Not used to retune ARI on test data.
Outputs	Ranking, recall, reduction, error, time, memory	Summarized per model with explicit denominators and uncertainty.

The initial condition q_e and disruption factor λ must not substitute for one another. If initial degradation is already incorporated into effective stiffness or capacity, applying q_e a second time must be prevented through the data definition. Similarly, safety or capacity reduction factors are not applied twice if the input capacity already includes them.

3.6 Model and Scenario Generation

3.6.1 Model Sources and Families

Each base model comes from a reproducible example or parameterized generation with stored rules. Models from external sources must include adequate geometry, materials, cross-sections, loads, and boundary conditions. A structural image without mechanical data is not sufficient to qualify as an external-validation model.

Figure 3.3 shows example idealizations of the five structural families in the main validation. The schematic supplements the family list; it does not replace the geometry, boundary conditions, or feasibility checks for each generated model. The five planned families are distinct from the four canonical models used for initial verification in Chapter 4.

The initial variation design is shown in Table 3.4. Values in the table are candidate pilot configurations, not an inventory of actual models. Each configuration must pass intact-model feasibility checks before it is used to test disruptions. Continuous parameters may be stratified-sampled within ranges approved in the

Table 3.4: Initial model-family variations for pilot design.

Code	Initial topology variations	Paired mechanical variations
S1	Single-span and continuous beams with 2–4 spans	Span ratio, relative stiffness, supports, and load locations
S2	Trusses with 2, 4, or 6 panels	Relative height, diagonal pattern, area, symmetric and asymmetric loads
S3	Frames with 1, 2, or 4 stories	Number of bays, beam–column stiffness, and load pattern
S4	Frames with X, V, or K bracing	Brace stiffness, placement, and lateral-load direction
S5	Simple spatial frameworks	System depth, spatial connectivity, and load distribution

pilot. Ranges are not selected based on final-test outputs. Variants from closely related templates receive the same group identity; independence is not claimed solely

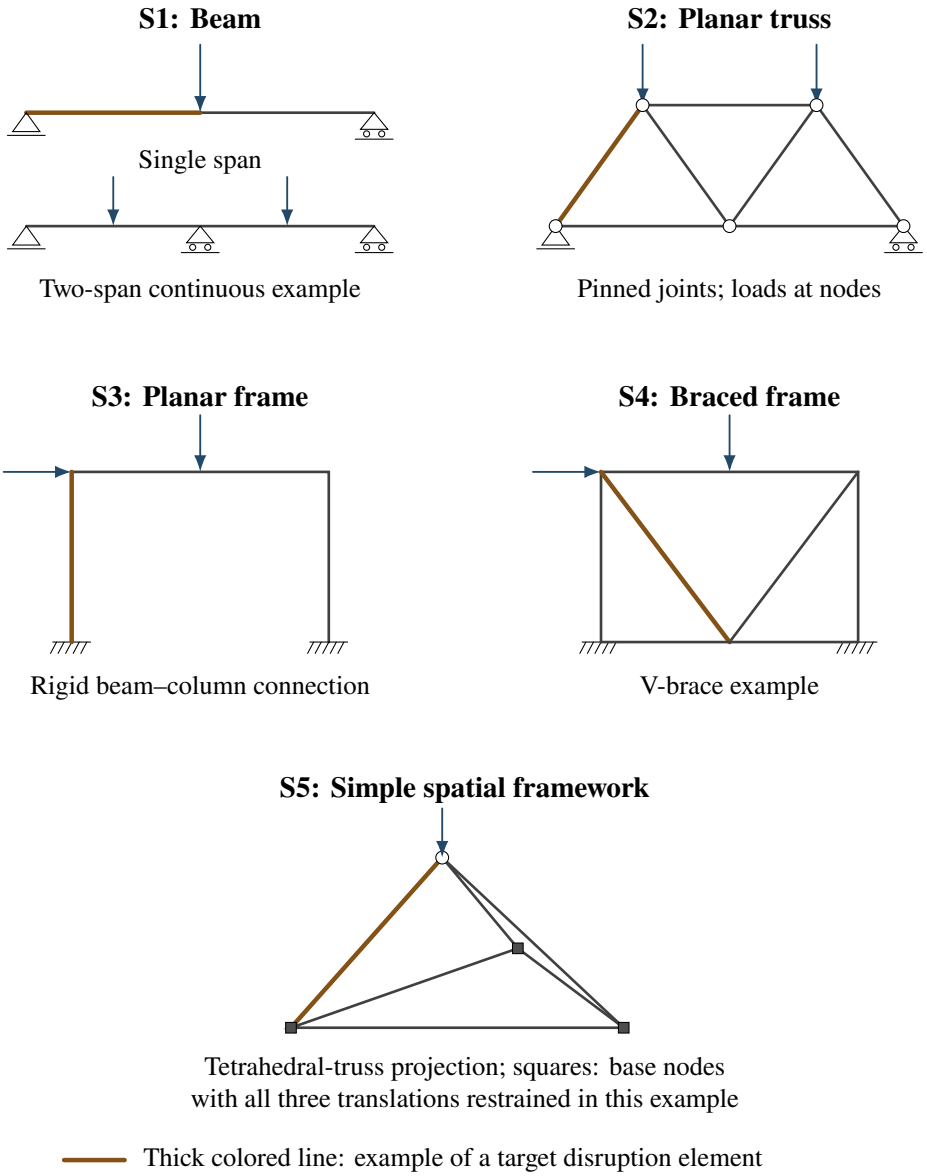


Figure 3.3: Schematic of the five S1–S5 structural families for the main validation plan, not models that have already been analyzed. Loads, supports, and target elements are examples only; final geometry and boundary conditions are set during model generation. S1 includes two topology examples, while S4 shows only the V-brace variant. Not to scale; the figure does not indicate that mechanical checks have been passed.

because file names differ.

3.6.2 Disruption Rules

The initial disruption grid is $\lambda \in \{0, 0.25, 0.50, 0.75\}$, while $\lambda = 1$ is retained as a control. For n_e eligible elements, the maximum number of candidates for one load case is $4n_e$ before justified exclusions. If the number of load cases or grid changes, the scenario denominator is rebuilt from the manifest rather than calculated from a fixed assumption.

The physical definition of weakening is distinguished from its operation on ARI weights. For an axial pilot, the illustrative rules $k_{e,\lambda} = \lambda k_e$ and $R_{e,\lambda} = \lambda R_e$ may be used as a defined disruption family. These rules do not automatically represent corrosion, cracking, or a particular material loss. For frames, changes to A, I, modulus, and capacity are determined according to the specified disruption type; changing area alone does not automatically apply the same factor to the moment of inertia.

Each scenario stores the reason for inclusion or exclusion. Elements without valid capacity do not receive fabricated scores. A disrupted model that becomes a mechanism is not removed merely because the static solver fails; its status must be distinguished from input errors and numerical failures.

3.7 Data Collection and Instruments

3.7.1 Data Sources, Extraction, and Merging

Data are separated into intact-model files, scenarios, ARI outputs, reference responses, GNN outputs, and cost logs. Merging uses IDs, not row order. Before merging, checks cover key uniqueness, unit consistency, load case coverage, and the association of each element with its parent model.

Literature collection uses an extraction sheet recording source identity, research question, formula, assumptions, model examples, code/data availability, and reason for use. Structural-model extraction records unit conversions and changes in

idealization relative to the source. Any modification affecting the response must be traceable.

3.7.2 Instruments and Review

Table 3.5: Instruments, outputs, and acceptance checks.

Instrument	Output	Check
Model sheet	Geometry and attribute inventory	Required data, units, and source
Structural solver	Intact and disrupted responses	Reference solution, equilibrium, and stability
ARI script	Scores and certification status	Unit tests, invariance, and boundary cases
Experiment manifest	Split, parameters, and versions	No parent IDs across splits
GNN interface	Predictions and inference status	Identical domain, target, and parameters
Cost logger	Time, memory, and failures	Complete and separate cost components

The instruments are planned for review by at least two experts representing structural mechanics and graph computing or machine learning. This review has not been reported as completed. The feedback form records issues, importance, revision actions, and acceptance decisions. Expert agreement does not replace numerical verification or empirical validation.

3.8 Candidate Calibration and Complexity Control

The initial coefficient candidates use $\alpha_k \in \{0, 0.25, 0.50, 0.75, 1\}$, $\alpha_R = 1 - \alpha_k$, and $\eta \in \{0, 0.25, 0.50, 0.75, 1\}$. Candidates for K, γ, and β follow Chapter 2. These combinations define the development search space, not optimization results. Exploration proceeds in stages so that selection costs are not hidden.

First, candidates that fail mathematical tests or canonical cases are excluded. Second, weight forms and path counts are compared on internal splits. Third, coefficients and thresholds are examined using out-of-fold predictions or the designated

calibration subset. When performance cannot be distinguished at the prespecified precision, simpler and less costly forms are preferred. The selection rule must be written before comparison, not chosen to make a particular candidate win.

Reference responses from blind validation and final testing are not used to select normalization or thresholds. If the implementation changes because a bug is found, all affected outputs receive a new version and are reevaluated under controlled conditions. A change prompted by unfavorable results differs from a bug fix demonstrable against an independent reference case.

3.9 Main Dataset Implementation, Calibration, and Blind Validation Plan

3.9.1 Priority for Independent Evaluation in a Restricted Domain

Before expanding to S1–S5, the next stage prioritizes the planar axial-bar domain supported by the available solver and guard. This stage does not replace cross-family validation, but allows the added-value question to be answered without simultaneously changing the mechanical idealization, solver, and GNN interface. All canonical and parametric models already examined remain development data.

The implementation sequence is as follows.

1. **Verify the enhancement.** Implement and check the capacity and displacement bounds in Section 4.4, including near-singular cases and abstention. Bound violations must be addressed before benefit testing; post-disruption labels are not decision inputs.
2. **Diagnose sensitivity.** For limiting cases, vary K, weights, and regularization while keeping the mechanical model fixed. Test changes to loads, geometry, and supports separately, using intact models within the domain. Record RR, CI, reference rank, and search certification so score changes are not confused with incomplete searches.
3. **Ablate and freeze.** On development data, compare ARI, the guard, the guard with each consequence channel, both channels without ARI, and the complete policy. Freeze the formula, thresholds, model ranges, critical-

case definitions, tie rules, and analyses before independent evaluation. Linear rank-one recomputation is included as a cost comparator.

4. **Test on new parent models.** Define partitions by parent and all its variants before labels are opened. Determine the number of parents from target precision and the pilot, not from the number of derived scenarios. A model used to revise the method after evaluation no longer counts as independent evidence for the revised version.
5. **Decide added value.** Report coverage of mechanisms, capacity, and displacement, including their intersections, reduction, abstention, and paired cost differences. Test ranking at an equal forwarding budget and cost at a target coverage achieved by both methods. If the target is not achieved or mandatory triggers exceed the budget, report the operating point as infeasible rather than excluding those cases.

The output of this stage is a decision on whether ARI provides additional benefit in the restricted domain, not a claim of generalization or GNN savings. A result showing no added value is still an acceptable research result. Family expansion and integration follow after method readiness and cost have been assessed in this stage.

3.9.2 Evidence Status and Prerequisites by Family

This plan operationalizes Table 3.1; it does not add research results. Available evidence remains deterministic verification of four canonical models and diagnostics on 16 additional parametric models. There is no main S1–S5 dataset manifest, calibrated threshold, comprehensive ablation result, or blind evaluation that can establish successful main validation. The CI threshold of 0.5 in the guard package is a development diagnostic point, not an approved operational threshold.

The available 1D/2D axial-bar solver does not automatically meet the requirements for beams, frames, or space frames. Capacity definitions for each mode, force combinations, displacement measures, and weakening operations must be established before producing data for the relevant family. Failure to meet a

Table 3.6: Prerequisites for implementing the five main dataset families; not a pass status.

Code	Required mechanical model	Minimum evidence before data production
S1	Beam elements with translation and rotation	Reference deflection and reactions, moments, shear, and end conditions
S2	Planar axial bars	Axial forces, compatibility, equilibrium, and reference mechanisms
S3	Planar frame elements with axial and bending response	Local–global transformation, joint rotation, and end releases
S4	Braced planar frames	Brace connections, axial–bending interaction, and stated tension/compression idealization
S5	Space frame with specified element idealization	Spatial components, constraints, and forces; torsion and bending if space-frame elements are used

prerequisite delays production for that family; it is not replaced with an unsuitable axial-bar response. For S5, the choice between a space truss and a space frame is frozen first, and the two are not mixed without subfamily identities.

The guard and enhancement domains in Section 4.4 are also not automatically extended to the five families. If a mandatory policy component does not support a model, its status is abstention and the scenario is forwarded. ARI alone may still be evaluated as a separate comparator. Claims about the complete policy must identify the domain actually supported; forwarding every out-of-domain case does not establish screening ability across families.

3.9.3 Generation and Splitting of Parent Models

The proposed implementation sequence is as follows.

1. **Freeze generation specifications.** Define topology classes, parameter ranges with units, physical bounds, load cases, disruption levels, sampling methods, and seeds for each family. Record model sources and initial rejection rules before examining scores or disruption labels. Intact models must satisfy the specified domain and numerical criteria; mechanisms

arising after disruption are retained.

2. **Define grouping units.** Each parent has an ID and specification hash; variations in geometry, material, load, and disruption inherit the group ID. Duplicate or near-duplicate models from related templates are not treated as independent parents merely because their seeds or file names differ. State the grouping rule; if it yields fewer than the target number of independent groups, the quota is not considered met.
3. **Separate pilot and evaluation.** Pilots used to select ranges, solvers, formulas, and sample sizes belong to development. Split parents within each family without using critical labels. The initial plan of 36/12/12 parents per family remains a starting point, not evidence of adequate precision. Final numbers may be revised through a pilot before blind data are opened, with the rationale and protocol version recorded.
4. **Separate internal calibration.** The proposed split of 36 development parents is 24 for formula selection through grouped cross-validation and 12 for threshold calibration. Across five families, the plan for 300 parents comprises 120 for formula selection, 60 for calibration, 60 for blind validation, and 60 for final testing. These numbers do not establish an adequate number of models containing critical events; the pilot must assess this separately. The four canonical models and earlier diagnostic models remain labelled as development data and are not moved into independent-evaluation quotas.
5. **Build a complete manifest and reference set.** The inventory records parents, families, partitions, attributes, elements, load cases, scenarios, controls, quality status, and exclusion reasons. Reference responses must be available for every evaluation scenario, including those not selected by the filter, so missed critical cases can be counted. This reference calculation is a validation-experiment cost, not an operational filter input.

Models that fail to solve are not selectively replaced to make performance look better.

Candidate counts, pre-analysis rejections, solver success, and unresolved cases are reported by family and partition. Adding samples after blind results are known is not a continuation of the same test unless a sequential rule was established in advance.

3.9.4 Auditable Threshold-Selection Rule

The formula, normalization, ARI parameters, guard components, and critical event definitions are selected during formula selection and frozen before calibration. Mechanical-consequence thresholds are not changed to make recall easier to achieve. The calibration stage below selects only the CI forwarding threshold; mandatory guard and abstention rules remain active.

The initial candidate grid is $\mathcal{T} = \{0, 0.01, \ldots, 1\}$, with forwarding at CI $\geq \tau$. The grid is set before calibration labels are opened and may be changed only through a protocol revision before that stage. For each τ, calculate mean reduction per parent $\overline{\text{Reduksi}}_{\text{cal}}(\tau)$, mean recall per parent with critical cases, and the one-sided lower bound $L_{\text{cal}}(\tau)$ under the frozen inference procedure. Models with no critical cases remain in reduction calculations and model counts, but their recall is not set to one or zero. The number of models contributing to each metric is always shown.

Given the initial Gate D targets, the candidate rule is

$$\mathcal{T}_{\text{eligible}} = \{\tau \in \mathcal{T} : L_{\text{cal}}(\tau) \geq 0.95, \quad \overline{\text{Reduksi}}_{\text{cal}}(\tau) \geq 0.30\}, \tag{3.1}$$

$$\tau^* \in \arg\max_{\tau \in \mathcal{T}_{\text{eligible}}} \overline{\text{Reduksi}}_{\text{cal}}(\tau). \tag{3.2}$$

If several candidates have the same reduction, select the smallest threshold. If the eligible set is empty or inference is inadequate, the stage output is "no eligible threshold yet," not a point that violates the targets. A temporary strategy forwarding every scenario may be recorded, but is not described as having achieved reduction.

An interval bound on data also used to select a threshold is an internal selection criterion, not a post-selection coverage certificate. Passing Gate D still requires independent evaluation of τ^*, including the comparators and other conditions. Degenerate intervals, families with too few critical models, and uncertainty due

to unresolved references result in an insufficient-evidence status. Results must be reported by family and event type; an aggregate mean does not automatically establish that every family passes. Family-specific thresholds are a different method and must be selected before calibration, with adequate calibration data for every family.

3.9.5 Ablation, Freezing, and Opening Blind Data

The comparator ablation list is supplemented with the guard alone, ARI with the guard, and capacity and displacement channels if they have been verified within the supported domain. Use the same models and scenarios for all methods. Two questions are distinguished: ablation at a fixed configuration assesses the effect of removing a component, whereas comparison of calibrated methods selects each method's threshold only on the same calibration partition. Their results are not combined as if they came from identical operating points. The number of comparisons and the primary versus exploratory analyses are specified before blind data are opened.

The frozen package contains code and manifest hashes, partitions, feature lists, normalization, tolerances, thresholds for each method, label definitions, tie rules, failure handling, seeds, interval procedures, and the reporting plan. An evaluator or separate process stores blind-validation and final-test reference responses. The filter accesses only permitted inputs, then stores its decisions and their hashes before labels are opened for assessment.

Blind validation evaluates the frozen version without retuning. If its results are used to change the method, that data becomes development data and must be replaced by new independent validation; the final test remains closed. The final test is opened after the final configuration is set, and its results, including failures, are reported rather than used to reselect a threshold. Changes after opening the final test require a new version and evaluation data.

Minimum outputs from this stage are a versioned dataset manifest, solver-verification logs by family, a threshold-selection file containing all examined candidates, a frozen configuration package, decisions stored before labels are opened, and

a paired-evaluation report. The planned archive location follows Section 4.10. Until these artifacts are available, this remains an implementation plan, not a validated dataset or evidence of passing the gates.

3.10 Mechanical Reference and Critical Cases

The structural-analysis software must be verified against analytical solutions or reference cases, with force equilibrium satisfied and response tolerances set before use. A candidate verification tolerance is a maximum relative difference of 1% for reference responses not close to zero; responses near zero use a recorded absolute tolerance with units.

For stable scenarios, response-change indicators include

$$\delta_u^{(s)} = \frac{|u_{\max}^{(s)} - u_{\max}^{(0)}|}{|u_{\max}^{(0)}| + \epsilon_u}, \tag{3.3}$$

$$\delta_U^{(s)} = \frac{|U_{\max}^{(s)} - U_{\max}^{(0)}|}{|U_{\max}^{(0)}| + \epsilon_U}, \tag{3.4}$$

$$\delta_F^{(s)} = \max_{g \in \mathcal{G}_F} \frac{\max_{a \in g} |F_a^{(s)} - F_a^{(0)}|}{\max_{a \in g} |F_a^{(0)}| + \epsilon_{F,g}}, \tag{3.5}$$

$$\delta_k^{(s)} = \max\left\{ \frac{k_{\text{sys}}^{(0)} - k_{\text{sys}}^{(s)}}{k_{\text{sys}}^{(0)} + \epsilon_k}, 0 \right\}. \tag{3.6}$$

k_{sys} uses consistently defined controlling load and displacement, not K, which is reserved for the number of paths. Internal force components are grouped into $\mathcal{G}_F$ by mode and units; forces and moments are not combined directly into one dimensioned maximum. The component mapping and treatment of missing elements are frozen before evaluation. For modes not applicable to a family, the corresponding indicator is not assigned an artificial value; the active indicator set must be the same for comparators within that family.

With positive scales t_h selected and frozen during development,

$$\widehat{\delta}_h^{(s)} = \frac{\delta_h^{(s)}}{\delta_h^{(s)} + t_h}, \qquad y_s = \mathrm{CRC}_s^{\mathrm{conv}} = \max_{h \in \mathcal{H}} \widehat{\delta}_h^{(s)}, \quad \mathcal{H} \subseteq \{u, U, F, k\}. \tag{3.7}$$

The maximum operator is compared with fixed-weight aggregation in the sensitivity analysis. Scales are not re-estimated from test-model responses.

For a model with $N = |\mathcal{S}| > 0$, define

$$n_{\mathrm{ref}} = \min\{N, \max\{3, \lceil 0.10N \rceil\}\}, \tag{3.8}$$

$$C = C_{\mathrm{rank}} \cup C_{\mathrm{consequence}} \cup C_{\mathrm{mechanism}}. \tag{3.9}$$

C_{rank} contains cases with the largest CRC values up to rank n_{ref}, including all ties at the boundary, among cases with defined CRC. If fewer cases are available, all of them are included. $C_{\mathrm{consequence}}$ contains cases with at least one $\delta_h^{(s)} \geq \tau_h$, using fixed development thresholds. $C_{\mathrm{mechanism}}$ contains mechanically confirmed mechanisms. CRC is not forced to a finite value for a mechanism.

Critical status for screening evaluation is distinguished from availability of a GNN regression target. Mechanism cases remain in the coverage denominator, while regression error is calculated only when both target and prediction are valid. Unresolved reference cases are reported separately and analyzed using best- and worst-case bounds; no pass claim is made if this uncertainty determines the decision. This critical-case definition is a research rule, not a safety limit.

3.11 Comparators and Ablations

Pre-analysis comparators include initial utilization, stiffness participation, capacity limitation, topological edge betweenness, weighted betweenness, and simple hybrids. All use information available before post-disruption analysis. For $g_s = 1 - \lambda$, the utilization comparator is $g_s U_e^*$; the same severity factor is explicitly applied to the other attribute-based comparators.

Active-learning methods such as AK-MCS (Echard et al., 2011) are not included as pre-analysis comparators with identical input access: model updates require performance-function evaluations. If tested later, they must be treated separately by accounting for additional evaluation, retraining, and prediction costs. Including this source in the literature review does not mean that an AK-MCS experiment has been conducted.

The comparator list is frozen before the final test. Correlation is calculated per model, with both high CI and high CRC indicating high priority. Ablations include $\eta = 0$, a single path, alternative weight forms, uniform support weights, and regularization variations. Performance is not inferred from statistical significance alone; effect size, uncertainty, and cost are also reported.

3.12 Validation Gates

Gate A: mathematical and algorithmic.

Check value range, monotonicity, unit and node-label invariance, the no-disruption control, agreement of the algorithm with an exact reference, and independence of the score from post-disruption response columns. The last check changes or removes those columns and verifies that ARI remains unchanged.

Gate B: mechanical agreement.

Initial targets follow the original design: median Spearman correlation of at least 0.70, a 95% interval lower bound of at least 0.50, median correlation gain of at least 0.10 over the best comparator in the fixed list, and a lower bound on the gain greater than zero. Coverage of the top-ranked group targets at least 0.80; group size equals n_{ref}, with ranking ties recorded. The top-ranked CI case must be among the three highest-priority reference cases; all boundary ties are reported. Normalized rank error is calculated as the mean absolute difference between midranks of CI and CRC divided by $N_v - 1$, where $N_v > 1$ is the number of scenarios with both valid scores; the median across models targets no more than 0.15. Unavailable scores and mechanisms are reported separately, not discarded without record.

Gate C: stability.

Initial targets for baseline–perturbation rank correlation and Jaccard index of critical sets are each at least 0.80. Perturbation magnitudes, tested parameters, and aggregation rules are set during development, not after results are observed.

Gate D: screening.

Initial targets are a one-sided 95% lower bound of at least 0.95 for mean recall across models, mean reduction of at least 0.30, mean recall advantage of at least 0.15 over random screening at the same budget, and a lower bound greater than zero for recall difference against the prespecified comparator.

All targets are initial operational criteria, not safety thresholds or proof of passing. Precision, feasibility, and control of multiple comparisons are examined in a pilot; the final configuration is frozen before blind evaluation. Failure of A–C limits support for the index. Failure of D does not automatically invalidate mathematical properties, but limits application claims.

3.13 Statistical Inference

The primary estimand for Gate D is mean recall per parent model, not the pooled proportion across all scenarios. If related variants exist, variant recall is first summarized according to a fixed rule. For M independent models,

$$\overline{R} = \frac{1}{M}\sum_{m=1}^{M} R_m, \qquad L_{0.95} = Q_{0.05}\left(\overline{R}^{*(1)}, \ldots, \overline{R}^{*(B)}\right), \quad B \geq 10\,000. \tag{3.10}$$

Bootstrap resampling draws parent models together with all their variants. The quantile interpolation rule and seed are recorded. This interval is an approximation, not a guarantee for every structure. Method comparisons use paired resampling of the same models.

A degenerate bootstrap distribution, for example when all recall values are one, is not interpreted as certainty about population recall. Sample adequacy and interval coverage are checked in the pilot. If the inference method is inadequate, the

status is insufficient evidence; multiplying the number of scenarios on the same model is not a substitute for independent models. Simultaneous claims across families require prespecified control of the error rate.

3.14 Local Sensitivity, Ablation, and Boundary-Case Analysis

Local sensitivity examines changes in the index when one parameter varies around the baseline configuration. For a positive parameter p_j and a small perturbation that remains within the domain, the descriptive measure is

$$D_j(s) = \frac{\mathrm{ARI}_s(p_j + \Delta p_j) - \mathrm{ARI}_s(p_j - \Delta p_j)}{2\Delta p_j}. \tag{3.11}$$

A normalized version is calculated only if the scale denominator is not close to zero. Because path optimization and ranking can change non-smoothly, finite-difference derivatives are not considered a complete description of sensitivity. Changes in rank, top-ranked set, and threshold decision are also examined.

The pilot grid for attribute perturbations may use relative changes of ±1%, ±5%, and ±10% from positive baseline values. Changes that violate physical conditions are rejected with a reason, not silently clipped. Final perturbation magnitudes are selected based on the model uncertainty objective and frozen before evaluation. Attribute uncertainty, model-sampling uncertainty, and algorithmic error are presented separately.

Ablation uses the same models and scenarios so that differences can be calculated in paired form. Required results include removal of utilization adjustment, a single path, alternative weight forms, uniform support weights, and regularization variations. A component is considered useful only if its benefit can be distinguished from uncertainty and is not offset by failure on important cases.

3.15 Mechanical Confirmation and External Validation

3.15.1 Additional Testing against a Nonlinear Reference

Comparison with nonlinear analysis is positioned as an additional test to examine the limits of ARI applicability, not as a change to the index concept or formula. Linear comparison remains the baseline; the additional test asks to what extent ARI ranking and screening performance persist when disruption consequences are assessed against a nonlinear reference. In this context, robustness of index performance is not equivalent to a claim that the structure withstands collapse. This section is an unimplemented protocol, not a nonlinear-validation result.

The confirmation set is selected from high-, medium-, and low-criticality strata after the formula is frozen. Selection considers model family, element location, load direction, and proximity to a mechanism change. Analysis is extended in stages according to the question: material nonlinearity, geometric effects, or connection models are not activated simultaneously without justification, because the source of response changes would be difficult to distinguish.

For this set, parent models, disruption definitions, and load cases are paired between linear and nonlinear analyses. The formula, parameters, normalization, intact-model inputs, and ARI threshold remain frozen; the same scores and forwarding decisions are evaluated against both references. Post-disruption nonlinear responses are not inputs to or a basis for retuning ARI. Consequence measures and critical-case criteria for each reference are established before the confirmation results are opened.

Additional reporting includes agreement between CI ranking and consequence severity, recall under the same forwarding policy, and low-CI scenarios that are critical according to the nonlinear reference. The number of forwarded cases remains the same because the policy does not change; the assessment concerns changes in coverage when the reference differs. Results are presented by consequence type and limited to the confirmation set, rather than directly generalized to all scenarios. Convergence failures are recorded as numerical statuses requiring examination, not automatically treated as collapse or removed. Results supporting or limiting ARI are

reported without changing the linear verification results already available.

3.15.2 External Validation

S7 tests transportability on new models with different sources or attribute ranges. S7 is not used to reselect the formula. Each external model is identified as being within the development range, exceeding the range of one attribute, or having a different topology. Separate reporting prevents averages from concealing failures on out-of-distribution models.

If ARI fails confirmation, conclusions are limited to the supported idealization. If external data are incomplete, a model is not considered validated merely because its geometry can be visually reproduced. The number of unusable models and reasons for exclusion are still reported.

3.16 Quality Control and Validity

Internal validity is maintained through paired comparisons, configuration freezing, label-access audits, and consistent treatment of failures. Construct validity is examined using examples that distinguish a connected graph from a stable structure, and high capacity from effective stiffness. External validity is limited by the model families, data sources, and sampling assumptions actually used.

No solver output is accepted as correct merely because a run terminates without an error message. Checks include equilibrium residuals, reactions, expected matrix symmetry or other properties, and energy consistency for cases that satisfy the assumptions. Similarly, a path search terminated because of a time limit is not labelled optimal without a certificate.

To test for leakage, an operational-data copy is created without all post-disruption response columns. ARI and forwarding decisions must match the outputs obtained before those columns were removed. With a fixed GNN, inference on the same graph must agree across strategies within numerical tolerance. These checks assess different issues and both must be reported. For the enhancement

in Section 4.4, the audit also starts from an empty cache with access to reference files denied. Randomizing or replacing labels must leave features and decisions unchanged. This check is supplemented by audits of parent-model splits and the origins of normalization and thresholds; independence from label columns during inference alone does not rule out leakage through selection of a configuration on test data.

3.17 Ethics, Transparency, and Reproducibility

No human-subject experiments are planned. Ethical responsibilities primarily concern source traceability, data/code licenses, failure reporting, and preventing unsupported safety claims. Closed project data are not published without permission; use of open data must comply with the source license.

The reproducibility package will distinguish raw data, transformations, configurations, outputs, and reporting scripts. Failure logs are not removed to make experiments appear clean. Table values come from stored outputs, not manual editing. A manuscript change that alters a metric definition must be accompanied by a new analysis version so that tables and text do not refer to different procedures.

3.18 ARI–GNN Integration Protocol

3.18.1 Reference GNN Prerequisites

The GNN identity has not yet been selected; the integration experiment is not ready to run. Selection is based on domain and target compatibility, not ARI performance on test data. Before integration, the following records must be complete.

1. Implementation name, primary source, code version, license, and computing environment.
2. Model domain, node and edge features, units, normalization, and support for element loss and partial weakening in S2 trusses.
3. Output definition: direct CRC prediction or a documented transforma-

tion of all required responses. Displacement predictions alone do not automatically provide CRC, which also depends on utilization, internal forces, and stiffness.

4. Training-data provenance, model-overlap audit, identity of the trained-parameter file, prediction-equivalence tolerance, and handling of out-of-domain inputs.
5. Rules for cases without valid regression targets, inference failures, and separation of GNN results from conventional structural checks.

If compatible parameters are unavailable, the reference architecture may be trained once without ARI-based selection, using the S6 preparation subset. Training termination uses an internal split of that subset. The 20 S6 validation models are used for compatibility validation and output decision calibration; this subset is not a final blind test. The 20 S6 test models remain closed until all decisions are complete. Retraining by strategy or threshold is not permitted. If no compatible implementation exists, integration remains deferred rather than fabricating results.

3.18.2 Candidate Audit and Selection Decision

A targeted review of the StructGNN publication and repository (Chou et al., 2024) indicates a representation with pseudo-nodes for floor diaphragms in multistory structures and outputs for displacement, moment, and shear. This information does not establish compatibility with S2 trusses and CRC. Therefore, this candidate is not immediately designated the final reference model.

Table 3.7: Initial audit of GNN candidates; not a compatibility test result.

Candidate	Rationale	Unresolved decision
StructGNN (Chou et al., 2024)	Multistory-structure implementation	Adaptation of S2 representation, target, license, and parameters
StructGNN-E (Song et al., 2023)	Physics-based elastic analysis	Access to implementation, solution type, and CRC interface
Other implementations	Only if consistent with the prespecified domain	Primary source and the same complete audit

The final decision follows this order of criteria: legality of use, domain compatibility, adequacy of outputs, availability of an inspectable implementation, and ability to produce fixed predictions. Runtime or accuracy on ARI test data is not used as a selection criterion. If changes to the output head or preliminary training are required, they are documented as reference-model preparation, not as an ARI architecture contribution.

Before freezing, the interface is tested on small models with traceable targets. Tests cover edge deletion, weakening attributes, different node counts, components without valid targets, and predictions after node IDs are reordered. A mismatch that requires changing the research domain must be resolved through a protocol revision, not concealed by mapping outputs that are unavailable.

3.18.3 Model Card and Target Reconstruction

The GNN model card records source identity, version, parameter hash, training data, normalization, features, outputs, hardware, and use domain. The manifest also records whether the target is predicted directly or calculated from predicted responses. A reconstruction pathway must not call post-disruption reference responses when used on a new model.

For example, if CRC is obtained from predicted displacements and force calculations through element relations, all matrices, capacities, and boundary conditions required for reconstruction must be information available in the scenario. The cost of these operations is included in prediction cost. If outputs are insufficient to calculate any active indicator, the missing value is not filled using reference-solver responses; the scenario is marked unresolved or the target domain is revised before freezing.

3.18.4 Forwarding Rules and Treatments

For a frozen threshold τ_{CI}, define

$$\mathcal{F} = \{s \in \mathcal{S} : \mathrm{CI}_s \geq \tau_{\mathrm{CI}}\} \cup \mathcal{A}, \tag{3.12}$$

where $\mathcal{A}$ contains invalid scores or uncertified searches. Abstentions count as forwarded so they do not artificially increase reduction. Scenarios in $\mathcal{S} \setminus \mathcal{F}$ are marked not evaluated by the GNN, not safe and not assigned zero predictions.

The four treatments are no screening, random screening, initial-utilization screening, and ARI. Random and utilization comparisons use the same number of forwarded cases as ARI for each model, including abstentions. As a prespecified control, ARI abstentions are forwarded in every screened treatment; the remaining budget is selected randomly or by utilization score. Thus, the added value of ranking is tested beyond the cases that must be forwarded. The cost of creating the shared abstention list in the experiment is recorded and is not hidden as zero cost for the comparators.

The GNN, parameters, features, normalization, and inference configuration are identical across treatments. ARI is not a GNN feature. The graph for a scenario contains the same physical disruption condition in all treatments. Predictions for the same scenario must agree within the specified tolerance. Out-of-domain inputs or inference failures are referred for conventional checking, with results and costs reported separately.

3.18.5 Performance Measures

For $C \neq \emptyset$ and $\mathcal{S} \neq \emptyset$,

$$\text{Recall}_{\text{screen}} = \frac{|\mathcal{F} \cap C|}{|C|}, \qquad \text{Reduksi} = 1 - \frac{|\mathcal{F}|}{|\mathcal{S}|}. \tag{3.13}$$

Models without critical cases are not automatically assigned recall of one; they are reported separately. For a fixed output threshold c_{GNN}, let $\mathcal{H}_{\mathcal{F}}$ contain forwarded scenarios with successful inference and $\widehat{y}_s \geq c_{\text{GNN}}$. Final coverage is

$$\text{Recall}_{\text{final}}(\mathcal{F}) = \frac{|\mathcal{H}_{\mathcal{F}} \cap C|}{|C|}. \tag{3.14}$$

Missed critical cases are separated into those not forwarded, inference failures or out-of-domain cases, and predictions below threshold. Conventional-check results are

not added to the GNN-recall numerator. All critical cases remain in the denominator, including mechanisms unsupported by the GNN.

Errors are calculated only on set Q with valid targets and predictions:

$$\mathrm{MAE}(Q) = \frac{1}{|Q|}\sum_{s\in Q}|\widehat{y}_s - y_s|, \qquad \mathrm{RMSE}(Q) = \sqrt{\frac{1}{|Q|}\sum_{s\in Q}(\widehat{y}_s - y_s)^2}. \tag{3.15}$$

If Q is empty, the metric is unavailable. Paired comparisons use the intersection of the same scenarios. Error on the selected set is not directly compared with error on all candidates to claim model improvement. The number of cases without predictions or valid targets is always reported.

3.18.6 Total Cost and Benefit Decision

Net time savings are defined as

$$T_{\mathrm{saved}} = T_{\mathrm{without}} - T_{\mathrm{ARI-GNN}}, \tag{3.16}$$

$$T_{\mathrm{ARI-GNN}} = T_{\mathrm{intact}} + T_{\mathrm{ARI}} + T_{\mathrm{selection}} + T_{\mathrm{graph},\mathcal{F}} + T_{\mathrm{inference},\mathcal{F}} + T_{\mathrm{handling}}. \tag{3.17}$$

ARI time includes path search, certification, and searches that reach the time limit. The intact-model cost is not assumed to be free. The no-screening treatment includes preprocessing, all graphs, inference, and applicable failure handling. Evaluation-label generation and GNN preparation costs are reported separately from operational cost, not claimed as inference savings.

Wall-clock time is measured over ten repetitions with warm-up, device synchronization, and balanced treatment order. No-cache conditions are the primary comparison; cached results are reported separately. Peak memory, candidate counts, graph size, abstentions, and failures are also recorded. Timing repetitions are not treated as additional independent models.

A benefit claim requires passing A–D in the relevant domain, a one-sided 95% lower bound of at least 0.95 for final recall, a one-sided 95% upper bound no greater than 0.02 for the decrease in final recall relative to no screening, and a

one-sided 95% lower bound greater than zero for paired time savings. The 0.02 margin is an initial operational choice to be examined in the pilot. If the GNN without screening does not meet coverage, scenario reduction alone is insufficient to claim benefit. Inadequate samples or degenerate intervals result in an insufficient-evidence status.

3.18.7 Energy Measurement and Emissions Estimation Protocol

This additional protocol has not yet been conducted. Measurements are planned after a compatible reference GNN is available and integration treatments are frozen. Energy and emissions reporting principles follow the traceability requirements discussed by Henderson et al. (2020); the following comparison design is specified for the ARI study and is not presented as a result from that source.

System boundary and measured quantities. Use $\mathcal{E}$ to denote electrical energy so that it is not confused with the elastic modulus E_e. For treatment $v \in \{\text{without}, \text{with}\}$, measured power $P_{\text{el},v}(t)$ in watts and time in seconds give

$$\mathcal{E}_v = \frac{1}{3.6 \times 10^6} \int_{t_{0,v}}^{t_{1,v}} P_{\text{el},v}(t)\,\mathrm{d}t \quad [\text{kWh}]. \tag{3.18}$$

Both treatments process the same initial candidate set with identical GNN model and parameters. The primary boundary includes energy for the entire machine during one operational run, including input preparation, required intact-model cost, data transfer, screening, additional mechanical checks, inference, and failure handling. Idle power during the run remains included; results subtracting idle power, if needed, are reported as a separate analysis with its own procedure.

For separate, non-overlapping stages, energy attribution for the treatment with screening can be written as

$$\begin{aligned}\mathcal{E}_{\text{with}} = \mathcal{E}_{\text{preparation}} + \mathcal{E}_{\text{ARI}} + \mathcal{E}_{\text{guard}} \\ + \mathcal{E}_{\text{inference},\mathcal{F}} + \mathcal{E}_{\text{handling}}.\end{aligned} \tag{3.19}$$

Preparation energy includes graph construction and all preprocessing used. If stages run concurrently, the sum of stage energies must not count the same interval twice; total measurement is the primary reference. One-time per-model costs are distinguished from per-scenario costs. Amortization uses only the number of scenarios actually processed. Training, hyperparameter search, and solver-label generation are recorded separately as development costs, not claimed to be eliminated by screening. Hardware manufacturing and structural life-cycle impacts are outside the boundary.

Paired measurement design. Use comparable hardware, power limits, software versions, and batching policies. Follow the time protocol's ten repetitions, warm-up, synchronization, and balanced order; warm-up energy is recorded separately from the operational measurement window. Monitoring starts before preparation and ends after device work is complete. The primary cache condition matches the time protocol. Record background processes, meter resolution, sampling interval, missing samples, and monitoring overhead. Results smaller than the meter's resolution are not used to claim measured savings.

If only CPU or GPU telemetry is available, label the result as component energy, not whole-machine energy. Do not add machine energy to its component energy. Direct measurement and power-model-based estimation are distinguished; nominal power specifications are not treated as records of actual consumption. Without adequate meters or telemetry, report energy as unavailable rather than imputing it from runtime.

Emissions estimation and benefit decision. For measurement interval j, with energy $\Delta\mathcal{E}_{v,j}$ in kWh and electricity emissions intensity ι_j in kg CO_2e/kWh, estimated operational emissions are

$$\widehat{M}_{\mathrm{CO_2e},v} = \sum_j \Delta\mathcal{E}_{v,j}\,\iota_j. \tag{3.20}$$

The source, location, period, accounting method, and uncertainty of the emissions factor are recorded. A period-average factor is an estimation assumption, not a direct emissions measurement or evidence of avoided marginal emissions. To isolate computational differences, use the same reference emissions factor in the primary comparison; location or time variations are analyzed separately. If a suitable factor is unavailable, report energy only. Cooling or facility energy is added only if data are available and it is not already included, with explicit boundaries and assumptions.

For $\mathcal{E}_{\text{without}} > 0$, report

$$\Delta\mathcal{E} = \mathcal{E}_{\text{without}} - \mathcal{E}_{\text{with}}, \qquad r_{\mathcal{E}} = \frac{\Delta\mathcal{E}}{\mathcal{E}_{\text{without}}}. \tag{3.21}$$

Negative values are still reported as additional consumption. Present total energy, energy per initial candidate, and the paired-difference interval together with final recall, reduction, abstention, and missed critical cases. Resampling for statistical inference follows parent models; instrument repetitions are not independent model samples. An energy-benefit claim requires the frozen integration-quality criteria and a positive one-sided 95% lower bound for the energy difference, accounting for meter uncertainty. Energy savings do not replace passing the gates or meeting the time criteria. Store instrument configuration, timestamped power records, energy integration, emissions factors, quality decisions, and artifact hashes. All remain planned outputs until measurements are actually available.

Chapter IV

RESEARCH RESULTS AND DISCUSSION

This chapter presents the available formulation and numerical verification results, then discusses their implications for mechanical validity and the limits of ARI applicability. The material proceeds from actual results to interpretation, proposed developments, and supporting documentation. It is organized into four groups:

1. **Results and discussion.** The canonical verification in Section 4.1, the kinematic-guard diagnostics in Section 4.2, and the discussion in Section 4.3 present the available evidence and its interpretive limits.
2. **Analytical proposal.** Section 4.4 formulates an enhancement for non-mechanism consequences; its empirical benefit has not been tested.
3. **Supporting documentation.** Sections 4.5 through 4.8 contain the implementation, data structure, checklists, and analytical examples. This material supports traceability; it does not add validation samples.
4. **Follow-up protocols.** Sections 4.9 and 4.10 specify remaining finalization needs and reporting formats for experiments that have not yet been conducted.

These four groups form Chapter IV. The literature-search documentation and source catalogue are presented in Section 2.11 of Chapter 2. Results from development models are not used to claim successful independent validation or benefits from GNN integration.

4.1 Numerical Verification Results for Canonical Models

4.1.1 Verification Objectives and Scope

This section presents the numerical results obtained for four synthetic canonical models. The input files, implementation, checks, and outputs are available in the project. The objective is to check algorithmic consistency and identify

limitations of the formulation before the main experiment, not to estimate performance across a population of structures.

The prototype generated 64 disruption scenarios and 16 controls across the four models. The verification script recorded 342 automated checks that passed their respective expected-value and tolerance criteria. These are software and arithmetic checks, not independent structures. Passing checks also include detection of rank deficiency in the reduced stiffness matrix; this does not mean that ARI assigned high priority to every such case.

4.1.2 Canonical Models and Configuration

The four models are two parallel 1D bars, a planar triangular truss, a square with one diagonal, and a square with two diagonals. The parallel-bar model has one active translational degree of freedom. It is not modeled as a planar frame with an added lateral support at the loaded node, since that change would alter the support-node set used in the ARI score.

The triangular model has nodes $(0, 0)$, $(1\,000, 0)$, and $(500, 800)$ in mm. Both square models have nodes $(0, 0)$, $(1\,000, 0)$, $(1\,000, 1\,000)$, and $(0, 1\,000)$. In the planar models, the lower-left node is restrained in both translational directions, and the lower-right node is restrained vertically. A 30 kN load acts downward at the apex of the triangular model; a 30 kN load acts rightward at the upper-right corner of each square. The two diagonals do not have a joint at their intersection.

The modulus of all bars is $200\,\mathrm{kN/mm^2}$. The two parallel bars have areas of $100\,\mathrm{mm^2}$ and $50\,\mathrm{mm^2}$, respectively, while each bar in the planar models has an area of $100\,\mathrm{mm^2}$. The capacities of the two parallel bars are 40 kN and 20 kN, respectively. Each bar in the triangular model has a capacity of 40 kN, while each bar in both square models has a capacity of 80 kN. These capacities are verification-case parameters chosen to keep initial utilization below one; they are not the result of a design-code check or measurements of a real structure.

The primary configuration uses $\alpha_k = \alpha_R = 0.5$, $q_e = 1$, $K = 2$, $\gamma = 1$, $\eta = 0.5$, and $\beta = 0.1$. Weight and path-cost regularization are both 10^{-9}; the



Figure 4.1: The four canonical models analyzed. The two 1D bars are drawn apart for visibility; squares mark constrained nodes, with restraint directions as defined in the text and input files.

baseline-score tolerance is 10^{-12}, and capacity regularization is 10^{-6} kN. The search limit is 200,000 states and 10 s per call, including both intact and disrupted scores. This limit is a prototype safeguard, not evidence of efficiency.

Stiffness and capacity scales are explicitly specified for each case: $(15, 30)$ for the parallel-bar model, $(20, 40)$ for the triangle, and $(20, 80)$ for both squares, with units of kN/mm for stiffness and kN for capacity. These scales remain fixed across all related disruptions. Case-specific scales for canonical models are not treated as valid calibration for generalization claims; the main study must still follow the data split and scale-freezing procedure in Chapter 3.

4.1.3 Algorithm and Solver Checks

The core implementation enumerates simple paths without repeated nodes and checks edge-disjoint combinations. The comparator uses paths from `all_simple_edge_paths` in NetworkX, which preserves edge keys in multigraphs (NetworkX Developers, t.t.); the comparator's optimum is obtained by checking all small subsets. Five graph cases are compared at $K \in \{1, 2, 3\}$ and $\gamma \in \{0.5, 1\}$, producing 30 check pairs. The cases include parallel edges, a disconnected graph, an empty path, a counterexample to a greedy approach, and a weighted complete graph with four nodes.

The solver assembles the axial-bar stiffness matrix and solves the reduced system using `numpy.linalg.solve` (NumPy Developers, t.t.). Rank is checked using the eigenvalues of the symmetric matrix, with a relative tolerance of 10^{-10} with respect to the largest eigenvalue in absolute value. If the rank is lower than the number of free degrees of freedom, the solver does not use a pseudoinverse to produce an artificial finite displacement.

The parallel-bar model is checked against a displacement of 1 mm, bar forces of 20 kN and 10 kN, a reaction magnitude of 30 kN, and a strain energy of 15 kN mm. For the triangular model, bar forces are obtained from joint equilibrium and displacements from virtual work. Equilibrium residuals and energy consistency are checked for every scenario with a unique solution. The unit-conversion test from

kN–mm to N–m also converts the modulus, area, capacity, normalization scales, and dimensional regularization; ARI values remain within tolerance.

4.1.4 Intact-Model Response as the Reference

The intact-model response serves as the reference for interpreting changes caused by disruption. Table 4.1 summarizes the solver outputs and graph scores for the same initial conditions.

Table 4.1: Solver results for four intact canonical models; not the main validation dataset.

Model	Bars	Rank/DOF	$u_{\max}$ (mm)	$U_{\max}$
Parallel bars	2	1/1	1.000000	0.500000
Triangle	3	3/3	1.154460	0.442218
One diagonal	5	5/5	5.935311	0.530330
Two diagonals	6	5/5	3.582277	0.320083

The displacement in Table 4.1 is the maximum nodal displacement norm; it is not a particular vertical or horizontal component. The Rank/DOF column compares the rank of the reduced stiffness matrix with the number of free degrees of freedom, while $U_{\max}$ denotes the maximum utilization among the bars in the intact model. A unique-solution status does not imply that all capacity or serviceability limits are satisfied after disruption.

4.1.5 Disrupted Responses and a Mechanical Counterexample

Each bar is tested at $\lambda \in \{0, 0.25, 0.50, 0.75\}$; the control $\lambda = 1$ is stored separately as the control. In the solver, weakening multiplies both area and capacity by λ, so axial stiffness changes by the same factor. In the graph, the previously computed weight is multiplied by λ without renormalization. These are distinct operations, specified explicitly rather than assumed to be identical implementations.

All 64 disrupted graphs remain connected. However, three full removals in the triangular model and five full removals in the square with one diagonal produce rank deficiency. Thus, eight full- removal scenarios do not have a unique

Table 4.2: Disruption status in the canonical models; controls are excluded from disruption counts.

Model	Disruptions	Unique solution	Rank deficient	Connected
Parallel bars	8	8	0	8
Triangle	12	9	3	12
One diagonal	20	15	5	20
Two diagonals	24	24	0	24

solution, while the other 56 have a unique solution within the solver tolerance. The "Connected" column in Table 4.2 reports the number of connected graphs, not the number of models satisfying mechanical criteria. These examples computationally demonstrate that connectivity alone is insufficient to establish stability under the axial-bar idealization used here.

Table 4.3: Single-bar removals in the square with one diagonal: graph scores and mechanical rank.

Removed bar	RR	CI	Rank/DOF
e1	0.983152	0.016848	4/5
e2	0.964724	0.216162	4/5
e3	1.000000	0.000000	4/5
e4	1.000000	0.000000	4/5
e5	0.717285	0.472914	4/5

The key finding in Table 4.3 concerns the loss of e_3 and e_4 in the square with one diagonal. RR remains close to one and CI close to zero, while the rank drops from five to four. These bars carry no initial force under the applied load case, and removing their paths does not reduce the optimal score with a two-path limit. Nevertheless, their removal releases a displacement mode at a node that remains part of the mechanical system.

Figure 4.2 illustrates the loss of e_3. Node D remains connected through bar AD, but an infinitesimal horizontal displacement at D produces no first-order axial strain. The mode arrow is not a load-induced displacement solution; the solver does not return a unique solution when the rank is deficient.

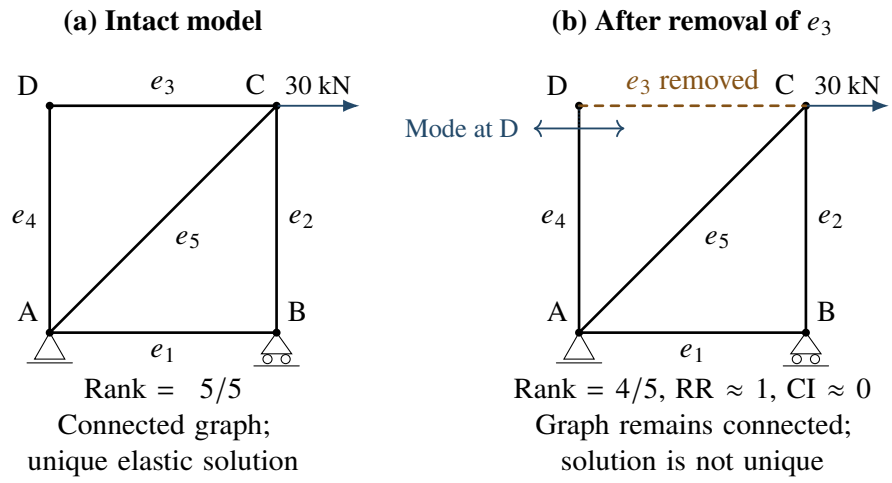


Figure 4.2: Geometric explanation of the case missed by CI in the square with one diagonal. The dashed line is the removed bar, not an active member. The double-headed arrow indicates the direction of an infinitesimal mode, not the magnitude of load-induced deformation or a prediction of collapse. Score and rank values are from Table 4.3; the diagram is not to scale.

This is not a violation of the formulation's range or monotonicity. The score was computed as defined, but the aggregated information is insufficient to distinguish this mechanical consequence. Replacing CI with one after accessing the solver status would leak an evaluation label into the operational score and is not used as a correction in these results.

The two square models also have nearly identical baseline scores in this configuration, although their maximum displacements differ. The path-count limit and pair weights mean that adding a diagonal does not automatically increase the score. This observation reinforces the need not to interpret a graph score as a measurement of physical stiffness or capacity.

4.1.6 Implications for Guard Development

The initial results support implementation agreement on the cases checked, but do not support accepting candidates across the entire mechanical domain. The failure to distinguish the canonical counterexample must be resolved or made an explicit domain restriction before claiming that Gate B has been passed or applying Gate D. Increasing the number of models without addressing this construct problem



Figure 4.3: CI versus the remaining factor for the four canonical models. Crosses indicate cases with solver rank deficiency, not ARI outputs. The point at $\lambda = 1$ is the control.

is not a substitute for methodological improvement.

Development options include a richer representation of degrees of freedom, a separate kinematic check based on geometry and the disruption definition, or a restriction of the application domain. If added, the kinematic check must be reported as a separate forwarding-policy component, computed without response labels, and included in total cost. Agreement with the same rank criterion cannot also be described as independent validation of the index's predictive capability.

The initial verification tables retain the original formulation so that the findings can be reproduced. The kinematic guard subsequently implemented and tested is reported separately in Section 4.2; this change concerns the forwarding policy, not a replacement of CI. These four canonical models have already been used in development and must not be presented as blind-test data.

4.1.7 Output Traceability and Verification Limits

Outputs are stored in `results/verification`. The manifest records the execution date, Python and library versions, configuration, and source hashes. Tables and figures in this section are generated directly by the verification script rather than populated manually. Instructions for rerunning it are provided in Section 4.5.6.

The results do not include a complete CRC for the main study, uncertainty estimates across independent models, comparison with all mechanical indices, nonlinear confirmation, or GNN predictions. No ARI–GNN time savings have been measured. The search limits only control the prototype process; they are not a substitute for measuring application costs.

4.2 ARI Development Results with a Kinematic Guard

The counterexample in Section 4.1 motivated the addition of a kinematic check. This section presents the mechanical basis, forwarding rules, diagnostic design, and guard test results in sequence.

4.2.1 Purpose and Scope of the Change

The implemented correction retains the ARI and CI formulations but adds a kinematic guard to the forwarding policy. Thus, score failures for zero-initial-force elements are not concealed by replacing CI based on a solver label. The guard uses geometry, supports, intact-model stiffness, and the definition of single axial-bar weakening. This check adapts mechanical-redundancy information; it does not claim a new discovery in redundancy theory or mechanical-redundancy matrices (Gade et al., 2021; von Scheven et al., 2021).

The implementation domain is limited to linear-elastic 1D/2D axial bars, zero-displacement supports, fixed geometry, and single-element disruptions represented by a stiffness factor $\lambda \in [0, 1]$. The area reduction in the experiment satisfies these assumptions. Changes to supports, geometry, connections, buckling, plasticity, and dynamic effects are not covered. Multiple-element disruptions are not analyzed by summing single-guard results; they produce an abstention instead.

4.2.2 Mechanical Basis from the Intact Model

For the free degrees of freedom, let $\mathbf{a}_e$ be the bar-elongation compatibility row and $k_e = E_e A_e / L_e$ its axial stiffness. For an intact model with positive-definite $\mathbf{K}_0$,

$$\mathbf{K}_0 = \sum_e k_e \mathbf{a}_e^\mathsf{T} \mathbf{a}_e, \qquad h_e = k_e \mathbf{a}_e \mathbf{K}_0^{-1} \mathbf{a}_e^\mathsf{T}. \tag{4.1}$$

The quantity h_e requires neither initial force nor post-disruption response. The corresponding redundancy-matrix diagonal is $1 - h_e$; the quantification of redundancy distribution in frame structures is discussed by von Scheven et al. (2021). For the remaining stiffness factor λ, define

$$\mathbf{K}_{e,\lambda} = \mathbf{K}_0 - (1 - \lambda) k_e \mathbf{a}_e^\mathsf{T} \mathbf{a}_e, \qquad m_e(\lambda) = 1 - (1 - \lambda) h_e. \tag{4.2}$$

Dengan $\mathbf{v}_e = \sqrt{k_e}\mathbf{K}_0^{-1/2}\mathbf{a}_e^\mathsf{T}$, congruence transformation gives

$$\mathbf{K}_0^{-1/2}\mathbf{K}_{e,\lambda}\mathbf{K}_0^{-1/2} = \mathbf{I} - (1-\lambda)\mathbf{v}_e\mathbf{v}_e^\mathsf{T}. \tag{4.3}$$

The matrix on the right has one eigenvalue equal to m_e and all other eigenvalues equal to one. Therefore, in exact arithmetic within this domain, $m_e = 0$ indicates loss of positive definiteness, whereas $m_e > 0$ preserves it. Under complete removal, $m_e = 1 - h_e$. An element with zero initial force can still have $h_e = 1$ and cause a mechanism when removed. This argument explains why the guard handles the counterexample that initial utilization does not distinguish.

The implementation does not form an explicit inverse. The weighted compatibility matrix has rows $\sqrt{k_e}\mathbf{a}_e$ and is factorized using SVD. For a full-rank intact model, h_e is computed as the sum of the squared entries in the corresponding row of the left-singular-vector matrix. The initial stiffness ratio $\rho_0 = \lambda_{\min}(\mathbf{K}_0)/\lambda_{\max}(\mathbf{K}_0)$ is also recorded. In exact arithmetic,

$$\frac{\lambda_{\min}(\mathbf{K}_{e,\lambda})}{\lambda_{\max}(\mathbf{K}_{e,\lambda})} \geq m_e\rho_0. \tag{4.4}$$

This bound follows from $\mathbf{K}_{e,\lambda} \succeq m_e\mathbf{K}_0$ and $\mathbf{K}_{e,\lambda} \preceq \mathbf{K}_0$. Numerical calculations use a margin tolerance of 10^{-8}, a relative rank tolerance of 10^{-10}, and a rounding allowance recorded in the code. Values near the boundary or with inadequate initial conditioning are forwarded; they are not a universal numerical stability certificate. These tolerances are a verification configuration, not empirically calibrated safety thresholds.

4.2.3 Forwarding Rules and Information Separation

Define $g(s) = 1$ when the guard requires forwarding because the margin is small, the conditioning bound is inadequate, or the calculation cannot be used. The candidate policy is

$$F(s) = \mathbf{1}\{\mathrm{CI}(s) \geq \tau \ \vee \ g(s) = 1 \ \vee \ \text{ARI abstains or is invalid}\}. \tag{4.5}$$

The guard status `cleared` means only that this check did not trigger forwarding. It does not mean that capacity, displacement, or structural safety has been checked. A scenario that is not selected is assigned the status `not_selected`, not deemed safe. Forwarded disruptions require evaluation using a solver or GNN within its supported domain; forwarding alone does not ensure that the GNN can handle a mechanism.

The guard is computed once per intact model, and its margin is then updated for each single disruption. SVD, input preparation, and decision costs are recorded separately from ARI and reference-solver times. These local timing records do not support a claim of GNN time savings. In this evaluation, the runner saves `results/revision/decisions.csv` and its hash before computing post-disruption reference responses. Reference responses are merged only after the decisions have been saved.

4.2.4 Regression-Test Design and Development Diagnostics

The four canonical models are reused as regression tests, not blind tests. A total of 16 additional parametric models were generated; they include variations of the triangle, square with one diagonal, square with two diagonals, and a three-bar arrangement connecting one free node to three supported nodes. Four deterministic configurations per shape vary geometry, area, and load magnitude. The configurations and actual models are stored in the code and in `results/revision/models.json`. Related variants do not constitute additional independent population samples.

The combined package contains 336 disruptions and 84 controls. The CI threshold is set to $\tau = 0.5$ as a diagnostic configuration, not as a calibrated result or deployment threshold. Three policies are compared: ARI only, guard only, and the combined policy. In addition to rank deficiency, diagnostic critical events include maximum utilization exceeding one or maximum displacement exceeding twice its intact-model value. The union of the three events is counted without double-counting scenarios that meet multiple criteria. This relative-displacement limit is not a normative serviceability limit. These diagnostic targets are not a complete CRC and do not replace Gates B or D. Rank-deficient responses are not assigned artificial

utilization or displacement values; any other unresolved reference status stops the runner.

4.2.5 Actual Results and Remaining Failures

Table 4.4: Regression tests for two zero-initial-force element removals.

Element	CI	h_e	m_e	Rank	ARI	Combined
e3	1.11e-16	1.000000	1.11e-16	4/5	No	Forward
e4	0.00e+00	1.000000	2.22e-16	4/5	No	Forward

Table 4.5: Development diagnostics at a CI threshold of 0.5; controls are excluded.

Group	Policy	N	Forwarded	Rank missed	Critical missed
Canonical	ARI	64	5	6/8	14/17
Canonical	Guard	64	8	0/8	9/17
Canonical	Combined	64	11	0/8	8/17
Parametric	ARI	272	17	24/32	54/71
Parametric	Guard	272	32	0/32	39/71
Parametric	Combined	272	41	0/32	30/71

All 1475 automated checks passed, including analytical values for parallel bars, an isostatic case, unit and ordering invariance, the guard's independence from load and capacity, a near-singular case, and forwarding when SVD fails or ARI abstains. The combined policy forwards all 40 rank-deficient disruptions in the development package. The two zero-initial-force element removals in Table 4.4 are no longer missed by the policy at the tested threshold, although their original CI remains low.

However, the final column of Table 4.5 shows that critical events without rank deficiency are still missed. The kinematic guard does not measure capacity exceedance or displacement amplification; this limitation is also tested explicitly using the two-parallel-bar example. The list of remaining misses is stored in `results/revision/remaining_misses.csv`. The combined results are better at capturing the mechanisms examined, but do not yet support use as a general

mechanical screen. Passing checks that ensure failures are recorded does not mean that the policy passes the main-study criteria.

The table values are development-diagnostic counts, not estimates of population recall, reliability intervals, or evidence that Gate D has been passed. Agreement between the guard and solver on rank is also not independent validation of ARI's ability to predict capacity. ARI's additional contribution must continue to be compared with the guard alone at comparable costs and forwarding budgets. The next step is to address remaining critical misses, then calibrate the policy on development data and evaluate it on independent models before GNN integration.

4.3 Discussion of the Main Findings

This discussion connects the canonical verification with the guard diagnostics to address three questions: the adequacy of the score representation, its added value in the forwarding policy, and the limits of applicability claims. The reported values come from the two preceding sections, not from a new experiment.

4.3.1 Algorithmic Consistency and Construct Validity

The passing checks show that the implementation produces reference values for the tested configurations. This answers whether the algorithm computes the specified formulation, but does not establish whether that formulation adequately represents mechanical consequences. The distinction matters because the square with one diagonal shows algorithmically consistent scores that do not assign high priority to two element removals that cause rank deficiency.

This finding limits the ARI construct to a compact measure of alternative paths adjusted for initial utilization. Mechanical edge attributes do not turn a path score into a solution of equilibrium and compatibility equations. The contribution of the formulation therefore lies in an explicit, verifiable definition, not in equating the score with mechanical redundancy in its entirety.

4.3.2 Stability Guarantees and Ranking Utility

Three claims require different evidence. First, the reported canonical counterexample rejects the claim that a low CI guarantees stability. Second, that example alone cannot determine whether CI is useful for ranking mechanical consequences; a ranking method may be imperfect yet still provide value under a particular budget. Third, a claim that ARI adds value once a guard is available requires a policy comparison with and without ARI. The guard's success in forwarding a mechanism cannot be attributed to an unchanged ARI score.

Accordingly, the negative result does not eliminate the ranking question; it rejects interpretations stronger than the evidence. The decision to retain ARI in a final screen must depend on its incremental benefit, not solely on its status as the index developed in this study.

4.3.3 Explaining Insensitivity to Specific Elements

For the removal of e_3 or e_4 in the square with one diagonal, two components of the formulation do not change enough to signal rank loss. First, the affected element carries no force under the initial load case, so the initial-utilization adjustment does not indicate its role in restraining a displacement mode. Second, removing the element does not reduce the optimal path score for $K = 2$. Consequently, RR remains close to one and CI close to zero.

This interpretation follows the reported case; it does not prove that all elements with zero initial force are critical. The example shows that a contribution to kinematic restraint is not always reflected in the initial force magnitude or a change in the optimal path score. Broader causal relationships require purpose-designed variations in geometry, loading, and boundary conditions.

This distinction is consistent with the vector-space analysis of trusses by Pellegrino dan Calladine (1986). The contribution of the ARI case is not the discovery of the mechanism concept, but tracing how the formulation and examined path configuration failed to assign high priority to a loss of restraint detected by the

mechanical reference.

Because the example uses $K = 2$, it does not establish insensitivity across all choices of path count, weights, or regularization. Further tests should separate changes in score parameters for a fixed mechanical model from changes in loading, geometry, or supports that alter the mechanical case. If analytically verified, pairs of cases with the same score but different stability status would strengthen evidence of lost discriminative power in this representation. Similarity in floating-point values alone is not treated as proof of exact equality, and no such pair is claimed as a result in this manuscript.

4.3.4 Diagnostic Added Value and Forwarding Cost

Table 4.5 shows the change from the guard-only to the combined policy. In the canonical group, forwarding increases from 8 to 11 scenarios, while missed critical cases decrease from 9 to 8. Because the combined policy uses an OR operation, the three additional forwarded scenarios capture one additional critical case. In the parametric group, forwarding increases from 32 to 41 and missed critical cases decrease from 39 to 30, so the nine additional forwarded scenarios capture nine additional critical cases. These counts are derived from the same diagnostic table, not from a new experiment.

This additional coverage shows a policy difference at the tested threshold, but does not prove ARI's superiority: forwarding counts are not matched, and all models were used in development. The combined policy still misses 8 of 17 canonical critical cases and 30 of 71 parametric critical cases. Forwarding all rank-deficient cases is therefore insufficient to establish screening adequacy for all mechanical consequences.

The next primary comparison should retain mandatory guard triggers and abstentions in both policies, then assess whether ARI selects additional cases better than a comparator under the same remaining budget. Fixed- budget ranking analysis must be separated from the conservative-bound policy: pruning mandatory triggers to meet a budget would change the policy and invalidate the conditional coverage

property being tested. The costs of intact-model preparation, the guard, path search, and subsequent evaluation must all be counted.

4.3.5 Implications for Screening before GNN Evaluation

A CI close to zero in a rank-deficient case indicates a conceptual risk if low priority is used directly as a reason not to evaluate a scenario. However, this result is not yet a false-negative count for a screening policy in the initial verification. Policy evaluation at the diagnostic threshold of 0.5 was conducted only in the separate development package in Section 4.2; GNN evaluation has not been conducted. Missed-case counts in that package are not estimates of population recall.

The additional kinematic guard has now been implemented as a separate component with recorded data requirements and costs. It handles rank-loss cases in the test package but does not eliminate all missed critical capacity or displacement events. Setting CI based on the post-disruption solver status would change the index inputs and violate the information separation underlying pre-analysis screening.

Section 4.4 formulates a follow-up based on bounds for force redistribution and displacement derived from the intact model. These channels distinguish kinematics, capacity reserve, and flexibility without changing CI. This remains an analytical proposal, not a demonstrated recall improvement; bound conservatism and preparation cost are part of the feasibility questions that must be tested.

4.3.6 Potential Contribution to Sustainable Computing

ARI screening could support sustainable computing if the reduction in inference energy exceeds the additional energy used for screening and mechanical checks without violating the specified critical-case coverage criteria. This potential has not been demonstrated in this study. Considering efficiency alongside prediction quality is consistent with the *Green AI* perspective in Section 2.10.

The available verification and diagnostic packages do not include measurements of GNN-integration energy or emissions estimates supported by power records. ARI and guard timing logs are insufficient to infer such benefits. Path search,

intact-model preparation, and mandatory forwarding may add computational load; net benefit must be tested through Section 3.18.7, including null or negative results.

Green-technology implications are limited to computational energy use within the stated measurement boundary. Reduced construction material, extended structural service life, and lower building life-cycle carbon are not inferred from scenario reduction. Scenarios not forwarded remain unevaluated by the GNN; they are not declared safe for the sake of efficiency.

4.3.7 Synthesis of Answers to the Research Questions

Table 4.6: Research questions, available evidence, and limits of the answers.

Research question	Evidence in the dissertation	Limits of the answer
ARI formulation	Definition, disruption operation, mathematical properties	Priority score, not physical capacity
Algorithm verification	342 automated checks in the canonical package	Applies to the cases and configurations examined
Mechanical validity	Eight rank-deficient disruptions; two cases with near-zero CI	Does not establish mechanical adequacy or superiority
Sensitivity and domain	Weakening variations and guard diagnostics on parametric models	Comprehensive ablation and cross-family generalization remain unavailable
Screening feasibility	ARI, guard, and combined-policy diagnostics	Critical cases remain missed; main validation is unavailable
ARI–GNN integration	Comparison protocol and cost separation	Trained model and inference results are unavailable

Table 4.6 shows that the strongest outputs are the formulation, limited verification, and identification of a counterexample. A single model-consistent example that the score fails to distinguish is sufficient to reject the claim that connectivity and path scores always guarantee stability within that domain. Conversely, four canonical models are insufficient to estimate failure frequency in a structural population.

4.3.8 Status of the Findings and Next Steps

The negative finding on representational adequacy is part of the research results, not a reason to remove an unfavorable model. Its direct implication is to

withhold claims of general applicability until a domain restriction or methodological revision receives independent support. This conclusion does not mean that all graph-based screening is unhelpful; it concerns the candidate and configuration reported here.

S1–S7 validation, nonlinear confirmation, equal-budget comparisons, and integration experiments remain unavailable. Required outputs and error- analysis procedures are compiled in Section 4.10. The analytical follow-up that directly addresses non-mechanism failures is first described in Section 4.4, followed by documentation and implementation protocols.

4.4 Proposed Pre-Analysis Enhancement for Non-Mechanism Consequences

4.4.1 Objective and Evidence Status

The kinematic guard determines whether weakening may remove a restraint, not whether forces in other bars exceed capacity or displacements increase. The following proposal therefore retains CI as a path score and adds a separate mechanical channel. The equations below are analytical derivations for a development design; they have not been implemented in the `results/revision` package and have no recall–reduction results. The relationship between redundancy and flexibility changes caused by element removal follows the mechanical-redundancy framework (Gade et al., 2021; von Scheven et al., 2021); the following consequence bounds are formulated for screening in this study's domain and are not claimed as a new redundancy index.

The counterexample must be identified precisely: the near-zero-CI cases in Table 4.3 involve removal of edge bars $e_3 = CD$ or $e_4 = DA$ in the square with one diagonal, not removal of diagonal $e_5 = AC$. Zero initial force alone does not explain the low CI; the optimal path score also changes very little. The guard must still be applied first, since zero initial force does not rule out a mechanism.

4.4.2 Intact-Model Information and Graph Representation

The proposed domain is the same as that of the guard: linear-elastic axial bars, positive-definite $\mathbf{K}_0$, fixed geometry and supports, and weakening of a single bar. The nodal load $\mathbf{f}$ does not change due to disruption; element-load release, initial strain, and changes in load direction are not covered. For each load case specified before screening, the intact-model analysis provides

$$\mathbf{K}_0\mathbf{u}_0 = \mathbf{f}, \qquad N_{j,0} = k_j\mathbf{a}_j\mathbf{u}_0, \qquad D_j = |N_{j,0}|. \tag{4.6}$$

Unlike the absolute demand D_j, $N_{j,0}$ retains the sign of the axial force. Capacity $R_{j,s}$ is determined from initial attributes and the frozen disruption rule, not estimated from post-disruption solver output. For the area weakening in the current package, $R_{e,s} = \lambda R_e$ and the capacities of the other bars remain unchanged; this rule does not automatically apply to all types of damage.

The proposed representation stores a vector of attributes instead of collapsing all information into a single path weight. Edge attributes include bar direction, k_j, R_j, $N_{j,0}$, initial utilization, and h_j; node attributes include a restraint mask, load, initial displacement, and intact-model local flexibility. The disruption definition adds the element ID, λ, and the capacity-change rule. Global information includes the intact model's numerical conditioning and load-case identity. Feature scales are selected from physical units or development data only. These features can be used directly by deterministic rules; their use in a GNN requires a compatible architecture and training, and is not assumed to be available in the fixed reference GNN.

4.4.3 Force-Redistribution Bound from the Intact Model

Write $\alpha_s = 1 - \lambda$ and $m_s = m_e(\lambda)$ as in Equation (4.2). Only if the guard clears the case and the margin is sufficiently far from zero does the rank-one update identity give

$$\Delta\mathbf{u}_s = \frac{\alpha_s N_{e,0}}{m_s}\mathbf{K}_0^{-1}\mathbf{a}_e^{\top}. \tag{4.7}$$

Here $\Delta\mathbf{u}_s$ is the exact response change under the linear idealization, used to derive the bound; it is not an input field from the reference solver. The identity can be checked by multiplying $\mathbf{u}_0 + \Delta\mathbf{u}_s$ by $\mathbf{K}_{e,\lambda}$. For an unaffected bar $j \neq e$, the Cauchy–Schwarz inequality in the $\mathbf{K}_0^{-1}$ metric gives

$$\left|k_j \mathbf{a}_j \mathbf{K}_0^{-1} \mathbf{a}_e^{\mathsf{T}}\right| \leq \sqrt{\frac{k_j}{k_e} h_j h_e}. \tag{4.8}$$

Therefore, an upper bound on axial demand can be computed from only the intact-model cache and the disruption definition:

$$\overline{D}_{j,s} = \begin{cases} |N_{j,0}| + \dfrac{\alpha_s |N_{e,0}|}{m_s} \sqrt{\dfrac{k_j}{k_e} h_j h_e}, & j \neq e, \\ \dfrac{\lambda |N_{e,0}|}{m_s}, & j = e,\ \lambda > 0. \end{cases} \tag{4.9}$$

The second row accounts for the stiffness change in the weakened bar; using the unaffected-bar formula for that bar would be incorrect. Removed bars are not included in the active-bar set $\mathcal{E}_s$. For positive capacity in all active bars,

$$\overline{U}_s = \max_{j \in \mathcal{E}_s} \frac{\overline{D}_{j,s}}{R_{j,s}}. \tag{4.10}$$

This bound captures potential redistribution to bars with little capacity reserve even when m_s remains positive. A value of $\overline{U}_s \geq 1$ triggers forwarding but does not prove that capacity is actually exceeded; the inequality may be loose. Zero, invalid, or unknown capacity in an active bar results in abstention, not division by a regularized value that conceals the issue.

4.4.4 Displacement-Amplification Bound

Let $\mathbf{P}_i$ extract the translational components of node i from the free-degree-of-freedom vector, with restrained components set to zero. The intact-model cache can store

$$c_i = \lambda_{\max}\left(\mathbf{P}_i \mathbf{K}_0^{-1} \mathbf{P}_i^{\mathsf{T}}\right), \qquad d_0 = \max_i \|\mathbf{P}_i \mathbf{u}_0\|_2. \tag{4.11}$$

The operator norm and Equation (4.7) give

$$\|\mathbf{P}_i \Delta \mathbf{u}_s\|_2 \le \frac{\alpha_s |N_{e,0}|}{m_s} \sqrt{c_i \frac{h_e}{k_e}}, \qquad \overline{d}_s = \max_i \left\{ \|\mathbf{P}_i \mathbf{u}_0\|_2 + \frac{\alpha_s |N_{e,0}|}{m_s} \sqrt{c_i \frac{h_e}{k_e}} \right\}. \tag{4.12}$$

Thus, maximum nodal displacement is bounded by $\overline{d}_s$. If d_0 is sufficiently far from zero, the amplification-factor bound is $\overline{A}_s = \overline{d}_s / d_0$. The amplification threshold A_{lim} must match the event definition used in the evaluation and be frozen before testing. If d_0 is close to zero, the ratio is declared unusable; use the absolute threshold d_{lim} specified in advance by the protocol or forward the case as an abstention. No new denominator is introduced that would silently change the critical-event definition.

The quantity c_i distinguishes flexible nodes even when the graph remains connected. A single factorization of the intact model can be used to solve the required right-hand sides; an explicit inverse is unnecessary. However, obtaining all c_i still incurs time and memory costs. An approximation that may underestimate c_i cannot be treated as a guaranteed upper bound without error control.

4.4.5 Layered Policy and Pre-Analysis Limits

Define $g_C(s) = 1$ if $\overline{U}_s \ge 1$, and $g_D(s) = 1$ if $\overline{A}_s \ge A_{\text{lim}}$ or, when the absolute criterion is enabled, $\overline{d}_s \ge d_{\text{lim}}$. The candidate policy is

$$F_+(s) = \mathbf{1}\{F(s) = 1 \ \vee \ g_C(s) = 1 \ \vee \ g_D(s) = 1 \ \vee \ \text{bound is unusable}\}. \tag{4.13}$$

The kinematic guard is checked first; cases with a small margin or invalid status are forwarded immediately without evaluating formulas that divide by m_s. This ordering prevents cases with $N_{e,0} = 0$ and $m_s = 0$ from being incorrectly treated as zero change. When $m_s > 0$, zero initial force does imply zero response change for a fixed load in this linear domain, but that conclusion does not apply to mechanisms or other load cases.

In exact arithmetic and within the stated domain, a capacity-exceedance or displacement-amplification event, as defined, cannot exceed its upper bound. This

is a conditional analytical property, not an empirical recall result or a guarantee of real structural safety. Implementation requires numerical error bounds or forwarding when the bound cannot be established. Loose bounds may forward nearly every scenario, so coverage alone does not establish screening feasibility. For multiple predefined load cases, decisions are combined using OR; the cost per load case is still counted.

Not reading labels is different from performing no additional analysis. Computing the full vector in Equation (4.7) to obtain all responses is a rank-one linear reanalysis, even if the disrupted matrix is neither formed nor factorized. If used, this method must be reported as a reanalysis comparator, not presented as a low-cost pre-analysis feature. The candidate screen here uses only bounds derived from intact-model quantities. Pre-analysis means before full response evaluation of each disruption, not without mechanical computation.

4.4.6 Leakage Audit, Ablation, and Acceptance Criteria

The implementation design must separate the following three stages.

1. **Intact-model preparation.** Build a cache from the geometry, supports, material, initial capacity, and predefined loads. The cache identity includes hashes of all these inputs and the algorithm and tolerance versions. Changes to the model or loading invalidate the corresponding cache.
2. **Operational decision.** The process receives only the cache, disruption definition, and frozen configuration. Save features, bounds, abstention reasons, decisions, and hashes before running the reference process. Rank, forces, utilization, displacement, and post-disruption critical labels are unavailable to this process.
3. **Separate evaluation.** The reference solver then generates responses and labels for comparison with the saved decisions. Development labels may be used to select a configuration, but not as operational features; each parent model and all its variants remain in the same partition.

Normalization, thresholds, and learned weights must not be adjusted using final-test labels.

Run the audit from an empty cache: deleting, randomizing, or replacing the label column must produce identical features and decisions, and the screening process must continue to run when access to the reference directory is denied. An empty-cache test is needed to ensure reference responses are not hidden in intermediate artifacts.

Further verification compares the bounds with the solver for stable disruptions, the $\lambda = 1$ control, complete removal, margins close to zero, zero-initial-force bars, capacity changes without stiffness changes, and changes in load case. Capacity-only changes are tested with $\lambda = 1$ and a separate $R_{j,s}$ rule; this is a protocol extension, not a scenario already present in the current package. Every bound violation or numerical abstention is reported. Multiple-element disruptions and nonlinear models remain outside the domain and are not handled by summing single bounds.

Paired ablations compare ARI, the kinematic guard, the guard with only the capacity channel, the guard with only the displacement channel, both channels without ARI, and the full policy F_+. Mechanism, capacity, and displacement recall are reported separately, along with their overlap, reduction, and abstention. Equal-forwarding-budget comparisons are reported as a separate ranking experiment: pruning bound triggers to meet the budget invalidates the conditional coverage property of the OR policy. Cache, per-disruption feature, ARI, and subsequent evaluation costs are counted in full, with rank-one reanalysis as a comparator. The proposal is worth developing only if gains in non-mechanism coverage are not offset by excess forwarding or added cost; no result in this manuscript establishes that it has met this criterion.

The following documentation complements these results and proposals with the implementation contract, data records, and verification tools. Analytical examples accompany the verification documentation so that derivations remain traceable without being conflated with independent experimental results.

4.5 ARI Implementation Documentation

This section documents the algorithm contract and prototype that support the verification results. Operational requirements are distinguished from the list of implementations already available in Section 4.5.6.

4.5.1 Inputs, Outputs, and Algorithm Contract

Inputs consist of the intact model, disruption definition, frozen configuration, and numerical tolerances. The intact model contains nodes, elements, boundary conditions, loads, reactions, and initial mechanical attributes. Post-disruption response data are not inputs. Outputs include the baseline score, disruption score, RR, ARI, CI, certification status, runtime, and forwarding decision.

The algorithm contract requires unique edge IDs, including for parallel elements. A removed edge is deleted rather than assigned a small weight and treated as still present. All scales, node weights, load components, and path parameters use the same configuration for the intact and disrupted models. Outputs must record why a score is undefined.

4.5.2 Operational Computation Sequence

1. Check the data schema, ID uniqueness, units, boundary conditions, and positive-valued attributes. Reject invalid inputs with a traceable reason code.
2. Analyze the intact model with a verified solver. Record cost, reactions, demands, and stability status.
3. Form normalization attributes and mechanical weights; compute node weights and load components once from the intact model.
4. Compute the certified baseline score. If it is invalid or too small, mark all disruptions of that model as abstentions.
5. For each scenario, modify the graph according to the disruption definition without re-estimating normalization or node weights.

6. Compute the disruption score and check that it does not exceed the baseline beyond tolerance. Treat a violation as an implementation failure.
7. Compute RR, ARI, and CI when the score is valid. Otherwise, store a null value and status; do not replace it with an artificial priority value.
8. Form the forwarding set using the frozen threshold and all abstentions. Save decisions before merging evaluation-reference data.
9. Run the GNN only on forwarded scenarios within its supported domain. Use conventional analysis separately for failures or out-of-domain inputs.
10. Join reference labels by ID only during evaluation; compute metrics with complete denominators and total cost.

4.5.3 Exact Search on the Reference Graph

For distinct node pairs, paths are enumerated using depth-first search without repeated nodes. Each path stores its ordered edge IDs and total cost. In an undirected graph, traversing an edge in the opposite direction still counts as using the same physical edge. Parallel single-edge paths are distinguished by their element IDs.

Once the path list is available, all combinations of size zero through K are checked. A combination is accepted only if its edge-ID sets are disjoint. Costs within each combination are sorted, rank weights are applied, and the maximum value is retained. ID ordering only selects a representative when the objective is tied; it must not change the objective value.

Kode 4.1: Pair-score pseudocode for a small reference graph.

```
pair_score(graph, source, target, config):
    if source == target:
        return omega[1] / epsilon_d, CERTIFIED
    paths = enumerate_all_simple_paths(graph, source, target)
    best_score = 0
    for count in 1 .. min(K, number_of(paths)):
        for candidate in combinations(paths, count):
```

```
            if shared_physical_edge(candidate):
                continue
            costs = sort_ascending(path_costs(candidate))
            score = sum(omega[rank] / (costs[rank] + epsilon_d))
            best_score = max(best_score, score)
    return best_score, CERTIFIED
```

The pseudocode expresses the logic accompanied by the prototype `examples/ari_core.py`; it is not a production-ready implementation. Runtime monitoring includes path and combination enumeration. If either stops before completion, the output label changes to uncertified. The best value found so far may be stored for diagnosis, but must not be treated as an exact score for excluding a scenario.

4.5.4 Certification, Pruning, and Caching

A faster algorithm may replace enumeration if it has an equivalence argument or optimality certificate appropriate to the objective function. Agreement on several small graphs demonstrates agreement only for those graphs. If pruning uses an upper bound, the bound must be proven valid for all pruned choices.

Baseline-score caching is permitted for all disruptions of the same model within one operational run. Cache keys include model identity, normalization, weights, and parameters. A configuration change must invalidate the corresponding cache. Comparisons with and without caching are separated; reusing prior computations is not presented as cost-free.

4.5.5 Minimum Status Codes

Table 4.7: Recommended operational status codes.

Code	Meaning and action
`scored`	Valid and certified score; apply the threshold.

Code	Meaning and action
`invalid_input`	Invalid input; record the reason and do not impute a score.
`invalid_baseline`	Intact model/score is unusable; abstain on all related scenarios.
`search_timeout`	Time limit reached; forward as an abstention.
`uncertified`	The result lacks the required optimality basis.
`not_forwarded`	No GNN prediction; this does not mean safe.
`gnn_failed`	Inference failed; report and handle separately.
`out_of_domain`	Input is outside the frozen GNN domain.
`reference_unresolved`	Reference label is undetermined; do not omit the case.

Score, forwarding-decision, and solver statuses are stored in separate columns. A scenario can have a valid ARI score but an invalid regression target because it becomes a mechanism. Combining all statuses into a single pass/fail column would discard this information.

4.5.6 Available Prototype Implementation

The included implementation extends the two-bar example to small-graph enumeration and a 1D/2D axial-bar solver. It is a development reference, not a production tool for all structural families. The functions of each file are as follows.

`examples/ari_core.py`

Provides graph validation, path enumeration, optimization of edge-disjoint combinations, weighted-pair aggregation, ARI calculation, and forwarding decisions with abstention when the search is incomplete.

`examples/truss_reference.py`

Provides axial-bar matrix assembly, zero-constraint application, rank checks, displacement solutions, reactions, bar forces, utilization, energy, and graph-input generation from the intact model.

`examples/verification_models.json`

Contains the geometry, loads, constraints, modulus, areas, capacities, and scales for the four canonical models. These values are synthetic verification inputs.

`examples/run_verification.py`

Runs algorithm comparisons, analytical solutions, invariance and error- handling checks, then generates CSV files, a manifest, LaTeX tables, and figures for the manuscript.

4.5.7 Solver Numerical Contract

For a bar from its start node to its end node, the unit direction vector $\mathbf{n}_e$ membentuk baris transformasi $\mathbf{b}_e = [-\mathbf{n}_e^\top, \mathbf{n}_e^\top]$. Kontribusi matriks elemen ditulis

$$\mathbf{K}_e = \frac{E_e A_e}{L_e} \mathbf{b}_e^\top \mathbf{b}_e. \tag{4.14}$$

Positive force denotes tension according to the element's local orientation. Utilization uses the absolute force and positive capacity. A removed element has zero matrix contribution and force, and is excluded from demand divided by zero capacity.

The solver supports only zero-displacement constraints, nodal loads, and linear-elastic materials. It does not include bending, connections, initial strain, nonzero support displacement, buckling, or post-yield response. Numerical rank deficiency is not assigned a pseudoinverse solution. Energy values and residuals are additional checks, not proof that every mechanical idealization is correct.

4.5.8 Score Interface and Search Budget

Core inputs consist of nodes, physical edges with unique IDs, and node pairs with positive weights summing to one. Pair weights represent the aggregation

$\pi_c a_{ic} b_{jc}$ formed from the intact model. The graph core does not receive displacement, force, rank, or disrupted-model labels as arguments.

In the prototype, intact and disrupted scores are computed in a single call so that the denominator cannot be confused with that of another configuration. This repeats baseline computation across scenarios and is not yet cost-optimized. State-count and time limits include both calculations. The `search_limit` status combines state-limit and time-limit events and maps them to abstention, not a zero score.

Exact search in this manuscript means that enumeration of all required choices is complete for the graph in question. Calculations still use floating-point numbers. Configurations that cause rank weights to cease being representable as positive values, infinite costs, or invalid numeric results are rejected rather than reported as certified.

4.5.9 Reproduction

Run the following commands from the project root using the available libraries, without creating a new environment.

Kode 4.2: Commands to reproduce the examples and manuscript.

```
python examples/ari_tutorial.py
python examples/run_verification.py
python examples/run_guard_verification.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The runner is designed for the four canonical cases in the stated order. Replacing a case requires deriving the corresponding reference values, not merely changing expected results so that checks pass. The core module and solver can be called separately to develop new cases, with additional checks appropriate to their domain.

4.5.10 Outputs and Version Checks

Table 4.8: Actual outputs in `results/verification`.

File	Contents
`baseline.csv`	Responses and scores for four intact models.
`scenarios.csv`	64 disruptions and 16 controls with separate statuses.
`checks.csv`	342 automated checks, expected values, and statuses.
`manifest.json`	Configuration, library versions, timing, and code/input hashes.
`.tex` tables	Summaries inserted directly into Chapter 4.
`models.pdf`	Diagrams of the four canonical models.
`ci_by_damage.pdf`	CI curves and reference-rank status markers.

Rerunning replaces generated outputs in that directory. Code or input changes must be followed by regeneration before compilation. Manifest hashes are compared with source files to identify stale outputs. Reproducibility checks do not require byte-for-byte identical PDFs because time metadata may differ; they check inputs, configuration, values, and statuses.

4.5.11 Kinematic-Guard Development Package

The module `examples/mechanical_guard.py` constructs the guard from the intact model, while `examples/run_guard_verification.py` runs the analytical tests, regression tests, and parametric diagnostics reported in Section 4.2. The guard cache is used only for identical geometry, material, cross-sections, and supports; changes to the model require recomputation. A single disruption is represented by the remaining stiffness factor. Multiple-element disruptions result in abstention.

The `results/revision/` directory contains the actual models, decision

logs, reference responses, summaries, list of remaining critical misses, timings, check results, and manifest. The runner saves the decision log before evaluating post-disruption responses. Hashes of code, inputs, decisions, and artifacts support version tracing but do not replace preregistration or blind validation. Tables and check counts in the manuscript are generated directly by the runner, separately from the initial verification package.

The `cleared` status applies only to the guard check; it is not a statement of mechanical safety. Input errors must not be mapped to permission to discard a scenario. This package does not provide GNN parameters, integration-savings measurements, or evidence that Gate D has been passed.

4.6 Data Structure and Experiment Manifest

4.6.1 Identity and Relationships among Files

Minimum keys are the IDs for parent model, variant, load case, element, and scenario. Each scenario ID refers to exactly one disruption definition. Filenames are not used as the sole source of identity because they may change when data are moved. Missing, zero, and not-applicable values must be distinguished.

Table 4.9: Minimum schema for research files.

Logical file	Core columns	Checks
Model	Parent/variant ID, family, source, split, units	Parent does not cross splits.
Node	Node ID, coordinates, restrained degrees of freedom	Finite coordinates and explicit restraints.
Element	Edge ID, two nodes, type, E, A, I, capacity, condition	Unique IDs; comparable capacity and demand.

Table 4.9 (continued)

Logical file	Core columns	Checks
Load	Case ID, node/element, component, value, units	Forces and moments are not mixed.
Scenario	ID, target element, λ, physical rule	Controls and disruptions are distinguished.
ARI	Baseline/disruption scores, RR, ARI, CI, certification	Parameter and algorithm versions recorded.
Reference	Solver status, response, CRC, critical status	Access separated from score computation.
GNN	Parameter ID, prediction, valid target, status	No zero imputation for scenarios not forwarded.
Cost	Stage, time, memory, device, repetitions	Total components are not double-counted.

4.6.2 Configuration Manifest

The manifest distinguishes planned, pilot, and final configurations. The following example shows only the data structure; its values do not mean that the main experiment has been frozen.

Kode 4.3: Example manifest structure, not a final configuration.

```
{
  "protocol_version": "dissertation-validation-draft",
  "status": "not_frozen",
  "parent_model_split": [0.60, 0.20, 0.20],
  "disturbance_factors": [0.0, 0.25, 0.5, 0.75],
  "control_factor": 1.0,
  "path_limit_candidates": [2, 3],
```

```
  "gamma_candidates": [0.5, 1.0],
  "beta_candidates": [0.1, 1.0],
  "bootstrap_repetitions": 10000,
  "timing_repetitions": 10,
  "reference_solver": null,
  "reference_gnn": null,
  "trained_parameter_hash": null,
  "screening_threshold": null,
  "search_time_limit_seconds": null
}
```

Fields with a `null` value must be filled through a development decision before the corresponding run, not assigned an undocumented default. Capacity scales, normalization, tolerances, and force-mode definitions are stored by family and unit system. A unit change must also convert all related dimensional quantities, including capacity regularization.

4.6.3 Decision and Version Log

Each change records the date, responsible person, reason, data accessed, and affected outputs. A formula change after validation data are opened changes the status of those data to development data. A change after the final test requires independent evaluation for new claims. The log records unsuccessful versions as well as successful ones.

The identities of GNN parameter files and normalization configurations should be determined using content hashes. Two files with the same name may not contain the same parameters. When parameters or features change, all strategies must use the same new version, and its preparation cost must be recorded separately from inference savings.

4.6.4 Pre-Analysis Checks

1. Ensure that duplicate keys do not cause a scenario to be counted twice after a join.
2. Reconcile the number of candidates in the manifest with forwarded, not-forwarded, and unresolved statuses.
3. Check that every output uses the solver, ARI, and GNN-parameter versions appropriate to its treatment.
4. Separate reference responses from operational features and test label-column deletion before saving final scores.
5. Check infinite values, units, and ranges without silently changing them to allow computation to continue.
6. Save the check summary as an auditable output.

4.7 Verification and Validation Checklist

4.7.1 Canonical Cases

The following cases were designed before the main experiment. Some checks have been run through the prototype and recorded in `results/verification/checks.csv`. This log does not mean that the entire cross-family checklist has passed. Identifying a mechanism case with low CI is a limitation, not a mechanical pass.

Table 4.10: Minimum test cases and expected outcomes.

ID	Case	Check
A01	$\lambda = 1$ control	RR equals one; CI equals the initial priority.
A02	Two identical parallel paths	Removing one path halves the score for equal weights.

Table 4.10 (continued)

ID	Case	Check
A03	Two nonidentical paths	Removal ordering matches the analytical calculation.
A04	Progressive weakening	Score does not increase as λ decreases.
A05	Edge removal	A removed edge does not remain as a high-cost path.
A06	Node relabeling	Score remains unchanged within tolerance.
A07	Unit conversion	Score is invariant when all dimensional scales are converted.
A08	Disconnected graph	A pair without a path contributes zero.
A09	Invalid baseline score	All related scenarios are assigned abstentions.
A10	Misleading greedy solution	Optimization selects the maximum-objective combination, not the first path.
A11	Utilization saturation	$\eta = 1, U^* = 1$ gives zero ARI without implying equal consequences.
A12	Label access removed	ARI and forwarding remain identical.

Table 4.10 (continued)

ID	Case	Check
B01	Connected graph becomes a mechanism	The reference detects instability; connectivity is not treated as stability.
B02	Low stiffness, high capacity	Additive-weight compensation is checked against mechanical response.
B03	Different loads and supports	Directional consistency and weight dominance are checked.
G01	Identical graph, different strategies	GNN predictions agree within the frozen tolerance.
G02	Failed or out-of-domain inference	Status and cost remain in the appropriate evaluation.

4.7.2 A Connected Graph That Becomes a Mechanism

The distinguishing example uses four nodes at the corners of a square, four edge bars, and one diagonal. The lower-left support restrains both translations, and the lower-right support restrains vertical translation. With axial elements and nondegenerate geometry, there are five free degrees of freedom. The model with a diagonal can be checked using a five-row compatibility matrix; after the diagonal is removed, only four rows remain, so rank cannot reach five.

The square graph remains connected after the diagonal is removed, but a free displacement mode is not restrained by the four edge bars under this linear idealization. This example tests a representational limit; it does not prove that every square frame fails under all conditions. Supports, element types, and removal must follow the example definition.

4.7.3 Numerical Verification Form

Each test case stores inputs, expected outputs, computed outputs, tolerances, differences, code version, and failure reason. Relative tolerances are used only for reference values not close to zero. Near-zero cases require an absolute tolerance with appropriate units.

If the solver reports singularity, diagnosis includes boundary conditions, zero elements, connectivity, rank, and sensitivity to numerical tolerance. A failed result is not automatically labeled a mechanism until modeling and numerical errors can be distinguished according to the protocol.

4.7.4 Pre-Freeze Checklist

1. Constructs, inputs, targets, and domains are stated consistently in the abstract, theory, methods, and data documentation.
2. All applicable mathematical and algorithmic cases have been run with traceable logs.
3. Every parent model has a single split, and test labels are not accessed during configuration selection.
4. Comparators, parameter grids, tie rules, and thresholds are predefined using development data.
5. Bootstrap units, final model count, interval method, and simultaneous claims have been reviewed in the pilot.
6. The GNN, target, features, parameters, and domain are specified.
7. All cost components and failure-handling rules are agreed upon.
8. Post-freeze changes follow versioning and independent-evaluation rules rather than being made silently.

4.8 Analytical Examples for Checking the Formulation

4.8.1 Two Parallel Bars

This example is a mathematical exercise, not an independent validation model. Two parallel bars connect a loaded block to a support; each has a length of 1 000 mm and a modulus of 200 kN/mm^2. Their cross-sectional areas are 100 and 50 mm^2, so their stiffnesses are 20 and 10 kN/mm. Their capacities are set to 40 and 20 kN, respectively; the total load is 30 kN.

From compatibility and equilibrium,

$$k_{\mathrm{sys}}^{(0)} = 30\,\mathrm{kN/mm}, \quad u^{(0)} = 1\,\mathrm{mm}, \quad D_1 = 20\,\mathrm{kN}, \quad D_2 = 10\,\mathrm{kN}. \tag{4.15}$$

The unregularized utilization of both bars is $1/2$. Take exercise scales $s_k = 15\,\mathrm{kN/mm}$ and $s_R = 30\,\mathrm{kN}$, a condition factor of one, and $\alpha_k = \alpha_R = 1/2$. Normalization gives $w_1^M = 4/7$ and $w_2^M = 2/5$. These scales are for this example only; they are not calibrated from the research data.

For $K = 2$, $\gamma = 1$, and $\eta = 1/2$, the limit as regularization tends to zero gives

$$S(G^M) = \frac{1}{2}\left(\frac{4}{7} + \frac{2}{5}\right) = \frac{17}{35}. \tag{4.16}$$

When the first bar is removed, the score becomes $1/5$; when the second is removed, it becomes $2/7$. The analytical results are: These values are algebraic results in the limit,

Table 4.11: Analytical example of two parallel bars in the zero-regularization limit.

Condition	RR	ARI	CI
Intact control	1	3/4	1/4
First bar removed	7/17	21/68	47/68
Second bar removed	10/17	15/34	19/34

not experimental results or exact values for positive regularization. Removing the first bar gives an elastic displacement of 3 mm and a demand of 30 kN in the second bar, exceeding its specified capacity of 20 kN. This elastic value only indicates capacity

exceedance under the example's idealization; it is not a prediction of post-yield response.

4.8.2 Runnable Checks

The file `examples/ari_tutorial.py` provides the two-bar example calculations, an undisturbed control, weakening monotonicity, ARI bounds, utilization saturation, and the threshold table for the eight scenarios in this section. The script uses rational arithmetic for the zero- regularization limit and floating-point numbers for positive regularization. It is not a general path-optimization implementation, a structural solver, or evidence that Gates A–D have been passed.

4.8.3 Partial Weakening and Physical Changes

In the zero-regularization limit, weakening the first bar in the two-bar example gives the following weight and score ratio:

$$w_{1,\lambda} = \lambda\frac{4}{7}, \qquad \mathrm{RR}_1(\lambda) = \frac{10\lambda + 7}{17}, \qquad \mathrm{CI}_1(\lambda) = 1 - \frac{3}{4}\frac{10\lambda + 7}{17}. \tag{4.17}$$

For weakening the second bar,

$$\mathrm{RR}_2(\lambda) = \frac{10 + 7\lambda}{17}, \qquad \mathrm{CI}_2(\lambda) = 1 - \frac{3}{4}\frac{10 + 7\lambda}{17}. \tag{4.18}$$

At $\lambda = 0$, the edge is removed; the limiting formula still gives the score from the remaining bar. This formula checks the implementation of the two-path example; it does not replace optimization on a general graph.

If the exercise's physical rule is $k_{1,\lambda} = 20\lambda$ and $R_{1,\lambda} = 40\lambda$, then

$$u_1(\lambda) = \frac{30}{20\lambda + 10}\ \mathrm{mm}. \tag{4.19}$$

The force in each bar is its stiffness multiplied by the shared displacement. At $\lambda = 0.25$, displacement becomes 2 mm, the forces in the two bars are 10 and 20 kN, respectively, and both utilizations equal one. At $\lambda = 0$, only the second bar is active

and its elastic utilization becomes 1.5. The removed element is not evaluated by dividing zero force by zero capacity.

4.8.4 Renormalization Is a Different Operation

Suppose attribute x is normalized to $x/(x + s_x)$. Weakening the already normalized weight gives $\lambda x/(x + s_x)$, whereas renormalizing the weakened physical attribute gives $\lambda x/(\lambda x + s_x)$. For $0 < \lambda < 1$, these values differ. This difference follows from the definitions; it is not a rounding error.

For the first-bar example, $k_1 = 20$ and $s_k = 15$. Halving the weight gives $2/7$, whereas renormalizing the physical stiffness of 10 gives $2/5$. The primary candidate implementation must use the operation specified by the formulation. Renormalization may only be treated as a separate candidate calibrated and evaluated independently.

4.8.5 Screening-Threshold Example

Use the eight disruptions of the two bars at $\lambda \in \{0, 0.25, 0.50, 0.75\}$; exclude the intact control. For illustration only, define a case as critical when the maximum elastic utilization of an active element is at least one. Under this rule, critical cases are weakening of the first bar at $\lambda = 0.25$ and $\lambda = 0$. This exercise rule is not a safety definition or a substitute for the main study's critical-case definition.

Table 4.12: Threshold counts in the analytical example with eight disruptions.

CI threshold	Forwarded	Critical captured	Recall	Reduction
0	8	2	1	0
0.55	3	2	1	5/8
0.60	1	1	1/2	7/8

The 0.55 threshold forwards removal of either bar and weakening of the first bar down to 0.25. The 0.60 threshold forwards only removal of the first bar. The table illustrates the coverage–reduction trade-off, but comes from a single small model with two critical cases under the exercise rule. A recall of one in this example

does not establish that Gate D has been passed.

4.8.6 Why Greedy Path Selection Can Fail

Consider a graph with nodes L, a, b, T and edge costs $c_{La} = 2$, $c_{aT} = 5$, $c_{Lb} = 5$, $c_{bT} = 2$, and $c_{ab} = 2$. The shortest path $L - a - b - T$ has cost six. After its edges are removed, no second path from L to T remains. For $K = 2$ and equal weights, the strategy's score in the regularization limit is $1/12$.

By contrast, paths $L - a - T$ and $L - b - T$ are edge-disjoint and each has cost seven. Their score is $1/7$, greater than $1/12$. This example is enough to disprove equivalence between the greedy algorithm and the ARI objective. It does not claim that all shortest-path algorithms are useless; the unjustified step is replacing the objective without proof.

4.8.7 Cost-Arithmetic Example

Suppose there are $N = 1\,000$ candidates and $M = 300$ are forwarded. If graph and inference costs are assumed fixed at $t = 0.004$ s per candidate, the avoided work is $(N - M)t = 2.8$ s. With an additional screening cost of 0.5 s, the difference is 2.3 s; with an additional cost of 5 s, the difference is -2.2 s.

Both cases have the same reduction, 70%. These values are an arithmetic example, not measurements on a device or GNN. In real experiments, batching, warm-up, synchronization, and failures may make runtime nonlinear in the number of candidates.

After the supporting documentation, the final two sections summarize unfinished work and its reporting rules. Their criteria and formats guide future implementation; they are not test results.

4.9 Manuscript and Data Finalization Protocol

This section summarizes manuscript preparation status, outstanding evidence needs, and completion criteria. A detailed format for reporting future experimental results is provided in Section 4.10.

4.9.1 Manuscript Origin and Limits of Completeness

The manuscript was developed from `prism-uploads/main.pdf`, the original 161-page PDF. Its complete LaTeX source was unavailable in the project when revision began. This version reconstructs and expands the necessary scientific sections rather than copying every sentence or mechanically preserving the page count. The original PDF is retained.

Structural completeness of the dissertation is distinguished from empirical completeness of the research. The five chapters, including supporting documentation in Chapter IV, provide the structure, definitions, algorithms, data formats, and decision procedures. The prototype, canonical verification results, and kinematic-guard diagnostics are included. The package does not yet provide solver results on the main dataset, compatible trained parameters, or evidence that the gates have been passed. Missing identifiers are not fabricated to make the manuscript appear complete.

The abstract and conclusions reflect actual results, while completed methods are distinguished from follow-up protocols. A complete presentation does not mean that academic or administrative requirements have been met.

4.9.2 Replacing Placeholders with Actual Results

1. Freeze the manifest and save forwarding decisions before opening evaluation labels.
2. Generate tables and figures from versioned outputs; do not manually edit output values in the document.
3. Reconcile all model and scenario counts in the abstract, methods, results, and supporting documentation against the same manifest.
4. Replace an unavailable status only when the actual data and their provenance can be inspected.
5. Distinguish analytical results, pilot outputs, blind validation, final testing, and external confirmation in table titles and text.

6. Write conclusions after gate decisions, intervals, and failures are established; do not adjust criteria to fit an initial conclusion.

4.9.3 Decisions Still Requiring Implementation

Table 4.13: Implementation prerequisites not satisfied by manuscript completeness.

Decision	Required evidence
Solver and baseline model	Device identity, data, and reference verification
Final GNN	Implementation, features, target, license, and trained parameters
Model count	Pilot precision and interval-coverage results
Thresholds and tolerances	Development calibration and freeze record
Passing Gates A–D	Test logs and outputs on independent models
Integration benefit	Final coverage and paired total runtime
Energy and emissions estimate	Measurement boundary, power records, paired energy, emission factor, and uncertainty

4.9.4 Revision Priorities and Completion Criteria

Editorial revision does not replace research implementation. The first scientific priority is to address cases with near-zero CI when stiffness- matrix rank decreases. The guard in Section 4.2 handles rank-loss cases in development, but critical capacity and displacement events remain missed. Remaining failures and independent evaluation are the next priorities. Candidate representation revisions or domain restrictions must be stated explicitly and retested before use in the main validation. Cases that revealed weaknesses remain regression tests; success on those cases alone is not evidence of generalization. A domain restriction must not be defined by discarding failures after final-test results are known.

Table 4.14 specifies outputs to inspect. `results/revision/` contains completed guard-development results. Other locations in the table are proposed output locations, not claims that the files already exist. Initial verification evidence remains in `results/verification/`. Task completion is distinguished from

method success: a completed evaluation may yield a failure decision or insufficient evidence. Numerical thresholds follow the protocol frozen before evaluation and are not adjusted to preserve a desired result.

Table 4.14: Finalization tasks and their completion criteria.

Task and status	Output and proposed location	Completion criterion
1. ARI policy revision. The guard has been tested in development; independent validation is unavailable.	Guard, rank regression tests, parametric diagnostics, and remaining critical misses. `results/revision/`. Evidence available: Section 4.2.	Compare with the initial version; retain failures in the record; domain rules do not require evaluation labels. Report inputs and costs for any additional mechanical checks.
2. Main dataset and solver. Primary evidence is unavailable.	Parent-model manifest, data provenance, units, boundary conditions, solver verification, and data split. `results/dataset/`.	Models are reconstructible; model-count adequacy is justified by a pilot; parent models and variants do not leak across data partitions; failed solver responses are not fabricated.
3. Comparison and stability. Main validation is unavailable.	Comparator results, ablations, sensitivity, uncertainty, and Gates A–C decisions. `results/validation/`.	Comparable evaluation references; frozen configuration; family-level results and failures reported against predefined gate criteria.

Table 4.14 (continued)

Task and status	Output and proposed location	Completion criterion
4. Operational screening. Development diagnostics are available; main evaluation has not been conducted.	Frozen threshold, forwarding log, recall, reduction, abstentions, missed critical scenarios, intervals, and Gate D decision. `results/screening/`.	Forwarding is saved before evaluation labels are opened; metric denominators are explicit; uncertainty accounts for parent models; unresolved references are handled according to protocol.
5. GNN integration. A compatible trained model and results are unavailable.	Model card, parameter provenance and hash, features, target, license, predictions, and paired timing. `results/integration/`.	Processes with and without screening use comparable models and devices; coverage and final error are reported with total ARI, additional-check, and inference costs.
6. Final reporting. Awaiting research evidence and decisions.	Tables and figures from versioned outputs, a claim–evidence matrix, and check records. `results/finalization/`.	Claims are linked to actual results; title, abstract, conclusions, and domain limits are consistent; cross-references and compilation are checked without manually editing result values.

Each completed task is recorded with its responsible person, actual implementation date, code and data versions, artifact location, and review decision. Evidence status in Table 4.16 is updated only after the artifact can be inspected. The existence of a script or a completed checklist alone is not sufficient to claim success.

4.9.5 Claim Limits in the Final Version

If main validation or GNN integration has not been conducted, the title, stated achievements, abstract, and conclusions must continue to reflect the formulation, initial verification, and development diagnostics. Guard success in rank tests does not replace main validation. GNN integration is presented as an untested plan, not a demonstrated benefit. Evaluation failures remain reported to limit claims rather than being removed to make the method appear successful. A complete report does not by itself establish the academic or administrative acceptability of the dissertation.

4.9.6 Evidence for Sustainable-Computing Claims

The energy protocol in Section 3.18.7 is an additional evaluation that has not been conducted. Before making a benefit claim, the archive must include the device and measurement boundary, timestamped power records, total energy for both treatments, the emission- factor source, uncertainty, final recall, and missed critical cases. Timing records alone do not meet this prerequisite. Component-level measurement is not described as whole-machine energy, and an emissions estimate is not described as a direct measurement.

Evidence remains unavailable until measurement artifacts can be inspected. An energy-saving claim must pass the frozen quality and energy criteria; negative results are not omitted. The term green technology is not used to claim material or construction-carbon savings that this study did not measure.

4.9.7 Final Editorial Check

The check covers consistency of symbols, equation numbers, table and figure labels, cross-references, citations, and reference ordering. The phrase “the results show” is used only for results that are actually available. “Will be tested” is retained for designs not yet executed. Unavailable, undefined, and not applicable are not used interchangeably because they have different analytical meanings.

4.10 Follow-Up Validation Reporting Protocol

4.10.1 Main Validation and Integration Outputs

This section provides a format for the S1–S7 experiments that have not yet been conducted; it does not present additional dissertation results. Every entry must come from traceable actual outputs. Minimum requirements include:

1. Inventory of parent models and scenarios by data split, physical parameters, model source, exclusions, and analysis-completion status.
2. Numerical verification, invariance, monotonicity, mechanism cases, utilization saturation, and path-search certification.
3. Correlation by model and family, comparison with simple methods, ablation, effect sizes, and uncertainty intervals.
4. Recall–reduction curves, missed critical cases, abstentions, and the Gate D decision using the frozen threshold.
5. GNN and parameter identity, prediction equivalence, error on comparable sets, final recall, component costs, and net runtime.
6. Nonlinear confirmation and external evaluation with extrapolation limits.

For implementation of Section 3.9, archive the dataset manifest in the proposed location `results/dataset/`; calibration results and threshold package in `results/screening/`; and blind evaluation and ablation in `results/validation/`. These locations are an archive plan, not claims that the files already exist. Reports must distinguish the initial target of 300 S1–S5 models from the actual count and present formula selection, internal calibration, blind validation, and final testing as four separate data roles.

4.10.2 Model Inventory and Verification

The format in this section is specific to the main study. The four canonical models and reported initial checks are not used to fill in the number of independent S1–S7 models or to claim that a gate has been passed.

Reporting begins with the model flow, not a table of ARI advantages. The numbers of models sought, generated, passing checks, excluded, and used in each split are presented with reasons. Scenario counts are taken from the manifest and distinguished by complete removal, weakening, controls, and load cases. Unresolved models are not omitted from the flow diagram.

Table 4.15: Actual-results inventory format to be completed after the experiment.

Set	Models and scenarios	Quality status	Source
Development	Unavailable	Not evaluated	Manifest and solver log
Blind validation	Unavailable	Not evaluated	Manifest and frozen configuration
Final test	Unavailable	Not evaluated	Sealed evaluation log
S6	Unavailable	GNN not finalized	Integration manifest
S7	Unavailable	Source to be determined	External-model audit

The phrase “unavailable” in Table 4.15 does not mean zero models, zero failures, or zero performance. When actual data become available, every entry must point to an output file and an analysis version. Do not fill the format with numbers merely to make the table appear complete.

Report verification in groups covering units and attributes, intact-model solutions, disruptions, graph algorithms, and ID integration. For each reference case, report the expected value, computed value, tolerance, difference, and status. Claim that all tests passed only when the test list is complete and actual logs support the claim.

4.10.3 Mechanical Validity and Ablation

The main summary includes correlation distributions by model, intervals, and differences from comparators. Models with constant rankings, undefined targets, or mechanisms are shown in separate categories. CI–CRC scatter plots should include family identity; combining all scenarios into one correlation can conceal between-structure variation.

Present ablation results in pairs on the same models. Each row explains the component removed, parameters held fixed, and changes in recall, reduction, runtime, and sensitivity. If ARI does not outperform initial utilization or a single path, discuss the reasons and domain limits rather than merely selecting another metric that appears better.

4.10.4 Coverage and Screening Efficiency

Recall–reduction curves show all thresholds predefined for descriptive analysis. The primary operating point remains the threshold selected during development. Results tables distinguish forwarding due to score, forwarding due to abstention, and out-of-domain handling. Recall is not reported without the number of models and critical cases that compose it.

Table 4.16: Components of the gate-decision report by family.

Gate	Required evidence	Evidence status
A	Mathematical tests, algorithm, invariance, and certification	Initial verification available; incomplete
B	Correlation, comparator differences, and ranking error	Initial limitation identified; main validation unavailable
C	Perturbations, ablations, and set stability	Unavailable
D	Recall, reduction, abstention, and intervals	Unavailable
Integration	Final recall, comparable error, and net runtime	Unavailable

Table 4.16 summarizes required evidence and its availability, not final pass/fail decisions. Final decisions use the categories pass, fail to meet criteria, not tested, and insufficient evidence. The last category includes inadequate precision or an unresolved reference that could change the decision. It is not combined with pass merely because the point estimate appears favorable.

4.10.5 GNN Integration and Cost Reporting

Show the four treatments side by side with the reference model and the same test set. Report errors for all successful predictions, the forwarded set, critical cases, and clearly defined paired intersections. Report the number of scenarios without predictions alongside the metrics rather than burying it in a footnote.

Table 4.17: Format for separating operational integration costs.

Component	Measurement description
Intact-model response	Includes initial analysis required by the screen
ARI and selection	Includes searches that fail or reach the time limit
Graph and inference	Candidate count, batching, and device held fixed
Handling	Checks for failed and out-of-domain cases
Total and difference	Paired measurement without caching as primary
Separate preparation	GNN training and evaluation-label generation

Populate the components in Table 4.17 from actual measurements, not estimates based only on scenario reduction. The cost report explains whether realistic operations require an intact- model response that was not previously available. If that response is available from other work, cached analysis may be shown as an additional scenario; benefits are not generalized to uncached use. Negative savings are reported as results, not removed as outliers without a predefined reason.

4.10.6 Confirmation and External-Validation Discussion

The confirmation discussion links ranking changes to mechanism changes rather than merely reporting a new correlation. For S7, describe performance according to its proximity to the development domain. If an external model differs in topology, loading, and material simultaneously, do not attribute the result to one factor without a comparator design.

Passing all families is not assumed. If supported by evidence, conclusions may be restricted to a specific domain; utility in small axial systems and limitations in portal frames must still be tested. The discussion must explain whether a limitation arises from the representation, formulation, algorithm, data uncertainty, or inadequacy of the reference GNN.

4.10.7 Error-Analysis and Follow-Up Protocol

A false negative is a critical scenario according to the evaluation reference that is not forwarded; a false positive is a noncritical scenario that is forwarded. Forwarding due to abstention is recorded separately from forwarding due to score so that the reason for each decision is traceable. These definitions do not yield actual error counts without applying a threshold and evaluating the policy.

Patterns to examine include elements that are not dominant in the intact model but become important after disruption, mechanism changes, subsystem interfaces, and paths with nearly equal stiffness. The canonical cases in Chapter 4 motivate examination of kinematic effects but do not establish the dominant error patterns in the main validation.

Map each false negative to geometry, deformation, internal forces, reactions, and post-disruption utilization when those quantities are defined. Analyze rank-deficient cases through their status and modes, not through artificial finite responses. Examine false positives to understand the cost of unnecessary forwarding. A correction motivated by final-test results is recorded as a new research version requiring independent evaluation.

With this separation, the conclusions in Chapter 5 remain grounded in the reported formulation, canonical verification, and guard diagnostics. The proposed enhancement and follow-up protocols inform future research recommendations; they do not demonstrate successful screening or GNN integration.

Chapter V

CONCLUSIONS AND RECOMMENDATIONS

5.1 Conclusions

The main finding of this dissertation is a limit on the adequacy of the ARI representation: graph connectivity and a mechanically weighted path score are insufficient to guarantee structural stability. Canonical counterexamples show that the index can be computed as defined even when a disruption removes a necessary kinematic constraint. This limit applies to interpreting the score as a guarantee of stability for the configurations examined; it is neither an impossibility theorem for all graph methods nor a rejection of every use of ARI as a ranking measure. The following six conclusions follow the sequence of problem identification, problem formulation, and research objectives, while distinguishing evidence-supported findings from further validation work.

1. **ARI formulation.** The ARI candidate has been formulated as a ratio of alternative-path scores adjusted for initial utilization. Its definition specifies the domain, normalization, regularization, and disruption operation, with inputs restricted to the intact model and the scenario definition. Mathematical properties are discussed as a basis for checking the formula; the score remains a relative priority measure, not a measured residual capacity or a failure probability.
2. **Algorithm verification.** A path-search prototype and an axial bar solver are available and can be rerun. A total of 342 automated checks in the verification package support the algebraic consistency and implementation that were tested. Search results that are not certified are treated as abstentions. This verification provides a basis for examining the limits of score interpretation separately from implementation correctness. The evidence is limited to the cases and checks performed; it does

not guarantee algorithmic success across all graph sizes and structural families.

3. **Mechanical adequacy and added value.** Initial verification identified eight full-loss scenarios with reduced matrix rank despite the graph remaining connected. For the loss of e_3 or e_4 in the single-diagonal square model, CI approaches zero even though the rank falls from five to four. These cases refute the sufficiency of connectivity and path scores as guarantees of stability for the representation tested. A guard based on the intact-model stiffness matrix has been added to the policy. Without changing the ARI score, it forwards all 40 rank-deficient cases in the regression package and additional development tests. This result demonstrates the distinct role of a kinematic check. Critical capacity or displacement events that remain undetected confirm that mechanism detection does not replace the assessment of other mechanical consequences. Whether ARI adds value over simpler pre-analysis comparators remains an empirical question for the main validation. The S1–S5 plan, with an initial subtotal of 300 parent models, has not been implemented. The diagnostic threshold of 0.5 is not a calibrated result, and no blind evaluation of the final configuration, as specified in Section 3.9, is yet available.
4. **Sensitivity and applicability limits.** Canonical cases show that alternative paths in a graph and kinematic constraints are distinct aspects that cannot be interchanged when interpreting ARI. This finding sets limits on the claims but does not map the index's entire domain of usefulness. Comprehensive sensitivity evaluation, ablation, and generalization across five structural families with parent-model-level splitting are not yet available. Representation enrichment, differentiated forwarding rules, or domain restrictions are therefore methodological implications of the findings; independent evaluation is required before extending the claims.
5. **Screening feasibility.** ARI diagnostics, the guard, and their combination

at a threshold of 0.5 clarify that successfully forwarding mechanism cases is a different outcome from covering all critical cases. Therefore, the ability to preserve coverage while reducing the number of forwarded cases requires separate evidence. Recall–reduction evidence, comparisons at equal budgets, and the effects of abstentions and failures in the main validation are not yet available to establish that Gate D has been passed. Scenarios that are not forwarded are not evaluated by the GNN; they are not thereby declared safe.

6. **Benefits of ARI–GNN integration.** The research output on this point is an evaluation protocol that separates representation validity, scenario reduction, total cost, and final coverage. Overall time savings and final coverage have not been demonstrated because no integration experiment using a compatible reference GNN with the same trained parameters is yet available. A decision requires measuring screening and inference costs and testing final coverage against the specified tolerance.

These six conclusions distinguish three levels of evidence: index validity (A–C), screening feasibility (D), and the benefits of GNN integration. Success at one level does not replace evidence at another. Counterexamples are substantive findings that establish limits on the claims; results that do not meet the criteria are still reported to help map the domain of usefulness. A high ARI does not indicate that a structure is safe, and a low ARI does not by itself indicate structural failure. Safety assessment still requires appropriate structural analysis and engineering judgment.

5.2 Traceability and Open Decisions

Before experiments begin, the archive must include parent-model and scenario IDs, units, geometry, boundary conditions, attributes, data splits, disruption definitions, ARI configuration, thresholds, random seeds, software versions, outputs, and runtimes. Frozen configurations must be assigned a version identifier and date. The small-graph prototype, axial bar solver, and canonical models are available in the

`examples` directory; outputs and manifests are available in `results/verification`. The main validation dataset, implementations for all structural families, and trained GNN models are not included.

Open decisions include the GNN and solver to use for the main validation across structural families, physical parameter ranges, weakening rules for each family, consequence thresholds, numerical tolerances, the final inference procedure, and pilot-based sample sizes. These decisions form an operational agenda for testing domain expansion after the representational limits have been identified. Values that have not been set or measured remain distinct from actual evidence.

5.3 Implications of the Research Findings

The theoretical implication of the canonical results is the identification of a limit on the adequacy of the ARI representation. In the configurations examined, graph connectivity and a low CI coexist with a deficient stiffness matrix rank. These counterexamples show that adding mechanical attributes to scalar weights does not, by itself, make a path score a measure of stability or residual capacity. The findings set limits on the interpretation of the representation tested; they do not establish the impossibility of all graph representations.

The methodological implication is that implementation verification must be separated from construct validation. When implementation checks meet their expected values, mechanical counterexamples provide a basis for examining representational adequacy rather than merely correcting calculations. Reporting these cases and separating reference responses from operational inputs make the development of representations and forwarding rules traceable.

The practical implication is the need to distinguish ARI ranking, kinematic checks, and capacity and displacement assessment. A guard that forwards rank-deficient cases does not make other scenarios safe. The selection of inference volume and computational savings still require testing; comparisons with simpler methods are needed to assess whether the additional information from ARI justifies

its computational cost.

Potential contributions to green technology are limited to computational energy efficiency, not measured environmental benefits of structures. Section 4.3.6 distinguishes this potential from actual results; integration energy measurements and emissions estimates are not yet available.

5.4 Research Limitations

The identified representational limit must be distinguished from the limits of the evidence base. Counterexamples are sufficient to reject a general guarantee that an ARI path score implies stability, but they do not determine how frequently such cases occur or establish performance across all structural families. The evidence is limited by the following factors.

1. Verification is deterministic and computational, using four canonical models and a parametric development model. Related variants do not represent measurement uncertainty or a sample from the population of real structures.
2. The elastic line-element model does not include all failure mechanisms, connection interactions, contact effects, or damage histories.
3. Findings about the adequacy of a scalar graph apply to the ARI representation and interpretation examined; representations containing other mechanical information require separate testing.
4. Path optimization may limit problem size; a high abstention rate may eliminate the benefits of screening.
5. The number of parent models and the quality of their sources determine the precision and scope of conclusions; a large number of scenarios does not remove these limitations.
6. GNN integration has not been performed. Even if implemented, a single-GNN protocol would not support claims across architectures or devices.

7. Available results are limited to verification of synthetic canonical models and policy diagnostics with a kinematic guard on a parametric model. Successfully forwarding rank-deficient cases does not guarantee detection of capacity exceedance or excessive displacement. Main-validation evidence, compatible trained parameters, and a complete independent test dataset are not yet available in this dissertation.

5.5 Recommendations for Future Research

As an additional check, compare the unchanged ARI score with consequences from nonlinear analyses on a set of models and scenarios paired with linear references. The aim is to map the extent to which ARI ranking performance and coverage hold beyond the linear idealization, not to replace the formula or make nonlinear analysis the primary focus of the research. The protocol in Section 3.15.1 separates this additional check from index development and from results already available.

The most urgent evidence need is to test ARI's added value in the planar axial-bar domain using new parent models, rather than extending claims from the same diagnostics. The sequence in Section 3.9.1 prioritizes verification of consequence limits, sensitivity diagnosis, ablation, freezing, and independent evaluation before expanding to S1–S5. The initial target of 300 models across families remains a plan that must be reviewed through a pilot and assessment of solver readiness; generating many scenarios from a small number of parent models does not replace independent evaluation.

The representational-limit findings direct future research toward the additional mechanical information required for each type of consequence. The kinematic guard has forwarded all rank-loss cases in the development package; critical capacity and displacement cases that remain undetected are targets for subsequent testing. Priorities are to assess representation enrichment, auditable supplementary checks, or explicit domain restrictions, rather than assuming that changing the score alone guarantees mechanical adequacy. The concrete design in Section 4.4 uses stiffness

leverage, initial forces and capacity reserves, and nodal flexibility to establish consequence bounds. Implementation priorities are to verify the bounds, ensure that caches are free of post-disruption labels, and then measure recall by event type and computational cost on separate parent models. A rank-one reanalysis should serve as a comparator so that enrichment does not merely shift solver costs into the screening stage. Candidates then need to be calibrated and evaluated on independent models, with data requirements and total costs fully accounted for. Model-source auditing remains necessary. A pilot should then determine precision, search configuration, and the feasibility of candidate GNNs. The formula and all evaluation decisions must be frozen before the final test.

Once integration is ready for testing, add paired energy measurements in accordance with Section 3.18.7. Report the full costs of screening, the guard, and inference, and retain the final-recall criterion. Emissions estimates require a traceable electricity emissions factor; time savings must not be used as a substitute for evidence of energy or emissions savings. Any lack of measurement tools or demonstrated benefits must be reported explicitly.

Further development may examine richer mechanical representations, multiple-element disruptions, attribute uncertainty, or search approaches with auditable guarantees. Such extensions should not be combined with the main results without a separate design. Every change in domain or method must have its own protocol version and independent evaluation.

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APPENDIX: PHILOSOPHICAL REFLECTION ON THE RESEARCH

This appendix reflects on the findings and limitations of the ARI research. Its purpose is not to add empirical evidence, but to consider what the study teaches about the formation of knowledge and the responsible use of technology.

1. Ontology: a model is not reality itself. Physical structures, mechanical models, graphs, and indices are representations at different levels. A graph that remains connected does not necessarily mean that the structural model retains its stability. Every simplification selects some properties of an object and leaves others out. ARI should be understood as a limited representation, not a complete account of structural redundancy or safety.

2. Epistemology: computability is not the same as trustworthiness. Agreement between an implementation and the formula it encodes shows that the calculation follows the definition; it does not show that the definition adequately represents the mechanical consequences. Trust in a method requires evidence appropriate to its claims. Canonical tests support knowledge about the cases examined; generalization requires further testing. Acknowledging the limits of knowledge prevents certainty from exceeding the evidence.

3. Criticism and correction: failures provide lessons. Cases with low CI and rank loss reveal limitations of the index. This finding motivates adding a safeguard, not removing unfavorable cases. Although the safeguard addresses the tested mechanism, it still fails to capture some critical capacity- or displacement-related events. Improvements must be assessed against actual failures and remain open to further correction, rather than being treated as final success.

4. A systems perspective: relationships among components determine meaning. An element with zero initial force can still help restrain structural movement. Its role arises from the relationships among geometry, supports, stiffness, and loading, not from force magnitude alone. This research highlights the importance of examining consequences that a single indicator does not capture, without assuming that adding indicators automatically solves the problem.

5. Axiology: efficiency must be accompanied by responsibility. Reducing the number of evaluations is not enough to establish a method's usefulness if critical cases are missed. Honest reporting, data traceability, and distinguishing results from plans are part of the researcher's responsibility. A scenario excluded from further analysis must not be deemed safe merely because it was not examined. Efficiency must be assessed alongside the consequences of errors, not treated as an end in itself.

Main lesson. The value of research lies not only in a formula that can be computed, but also in clarity about what remains unknown. The scientific attitude learned here is to propose ideas courageously, test them willingly, acknowledge failures honestly, and limit claims until adequate evidence is available.

Five Complete Journal Article Drafts

Development of the Adjusted Redundancy Index for Screening Structural Disruption Scenarios

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September 10, 2026

Editorial Note

The five manuscripts below follow the article structure in the example file `prism-uploads/main-2_2.pdf`: opening page, article-specific contents, title and author details, abstract, keywords, introduction, methods or formulation, results or planned outputs, discussion, conclusion, and references. This section contains narrative manuscripts developed with equations, tables, and mechanical explanations. All text is written in LaTeX for editing rather than inserted as PDF pages. The layout follows the example: A4 paper, 1-inch (2.54 cm) margins, 11-pt Computer Modern font, and single spacing. Titles, abstracts, headings, tables, and equation numbering follow the example's hierarchy; page numbering continues from the main document.

These five drafts are neither five published papers nor five independent empirical studies. Article 1 develops the presentation of the formulation; Article 2 examines a limiting example; Article 3 presents guard diagnostics; Article 4 is an analytical proposal; and Article 5 is a protocol. Reported values are taken from archives already available in the project and were not obtained from new experiments while preparing this section. No GNN results, energy measurements, or successful main-validation outcomes have been added without evidence.

Article	Main focus	Evidence status
1	ARI formulation and computational contract	Methodological and analytical
2	Connected graphs and constraint loss	Results from four canonical models
3	Kinematic guard and forwarding policy	Development diagnostics
4	Demand and displacement bounds	Analytical proposal, not implemented
5	ARI–GNN coverage, time, and energy	Protocol, not yet executed

Articles 1–3 share a foundation and some cases; this reuse is disclosed in each manuscript. Their separation is an editorial arrangement. Manuscripts whose questions and evidence overlap too heavily should be combined before submission. No claim is made that every topic is sufficiently novel to warrant a separate publication. The name and affiliation follow the dissertation cover; additional authors, correspondence details, funding, and conflicts of interest have not been determined.

The provisional reference lists use the dissertation source and several entries already cited in it. Article-specific literature searches, bibliographic checks, development of comparisons, and

adaptation to the target journal's format are still required. Source page numbers from the example PDF are not used because adding sections to the project may change pagination; traceability instead relies on sections and source files.

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An Alternative-Path-Based Adjusted Redundancy Index for Prioritizing Structural Disruptions

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Abstract

Screening disruption scenarios requires a score that can be computed before the full response to each disruption is available, with a clear operational definition and limits on interpretation. This article presents the *Adjusted Redundancy Index* (ARI) as a ratio of alternative-path score retention adjusted for element utilization in the intact model. Stiffness and capacity attributes are normalized before being combined into graph weights. The score is determined by optimizing a combination of edge-disjoint paths, while weakening is applied using fixed scales and paired weights. The analysis describes the score range, conditional monotonicity, the no-disruption control, and the distinction between weakening a weight and renormalizing an attribute. An arithmetic example shows that greedy shortest-path selection is not equivalent to the candidate objective. The 342 automated checks reported in the dissertation are presented as limited verification, not as evidence of mechanical validity or population-level validity. The draft's contribution is an auditable index specification and computational contract. ARI remains a prioritization candidate, not residual capacity, failure probability, or a certificate of stability.

Keywords: ARI; mechanically weighted graph; edge-disjoint paths; normalization; scenario screening.

1.1 Introduction

Evaluating many disruptions creates a need to prioritize scenarios for further examination. In the source study [1], ARI is placed before evaluation of post-disruption responses, including possible prediction by a *graph neural network* (GNN). This positioning requires a separation between information from the intact model and disruption outputs that may only be used as evaluation references.

The issue is not merely producing a number. An index definition must specify its attributes, disruption operation, path-search objective, handling of incomplete computations, and the meaning of extreme values. Changing the normalization or algorithm can produce different candidates even when the index name is retained. Graph representation [2] provides a language for relationships among components, but does not itself solve structural equilibrium and compatibility.

This article addresses two questions: how can ARI be defined operationally, and what properties follow from that definition? Mechanical adequacy is analyzed separately from the formula's internal properties. Accordingly, mathematical regularity is not treated as proof that the index captures every important disruption.

1.2 Materials and Approach

This manuscript reorganizes the formulation, analytical examples, and implementation contract from the dissertation [1]. No experiments, calibration, or GNN tests were rerun. A structure is represented as a graph $G = (V, E)$ whose edge identities preserve element identities. Two bars connecting the same nodes remain distinct edges in a multigraph. The sets of load and support nodes are denoted by V_L and V_T.

Attributes are obtained from the intact model and the scenario definition. Operational inputs do not include rank, displacement, forces, or critical labels produced by the post-disruption solver. Development data may be used to select parameters, but those parameters are frozen before evaluation on independent models.

1.3 Index Formulation

1.3.1 Attributes and edge weights

For an axial bar, the stiffness indicator is $k_e = E_e A_e / L_e$. For a positive attribute x_e and scale $s_x > 0$ with the same units, normalization and edge weight are defined as

$$\widehat{x}_e = \frac{x_e}{x_e + s_x}, \qquad w_e^M = q_e \left(\alpha_k \widehat{k}_e + \alpha_R \widehat{R}_e \right), \tag{1.1}$$

where $\alpha_k + \alpha_R = 1$, both coefficients are nonnegative, and $0 < q_e \leq 1$. Capacity R_e must correspond to the force mode under consideration. Factor q_e represents the modeled initial condition, not a hypothetical disruption factor.

Normalization scales are specified for groups of physically comparable attributes. Unit conversions must consistently convert dimensional scales and regularization terms. Axial and moment capacities are not combined in one group without an additional definition. The additive form in Equation (1.1) permits compensation among attributes; whether that compensation is appropriate is an empirical question, not an automatic consequence of normalization.

1.3.2 Path optimization and aggregation

The cost of an edge and a simple path p are defined as

$$c_e^M = \frac{1}{w_e^M + \epsilon_w}, \qquad d^M(p) = \sum_{e \in p} c_e^M, \qquad \epsilon_w > 0. \tag{1.2}$$

Let $\mathfrak{P}_{K,ij}(G)$ denote the set of all choices of at most K edge-disjoint simple paths between i and j, including the empty choice. Paths in each choice are ordered by nondecreasing cost. The pairwise score is

$$\Phi_{ij}^{(K)}(G) = \max_{\mathcal{P} \in \mathfrak{P}_{K,ij}(G)} \sum_{r=1}^{|\mathcal{P}|} \frac{\omega_r}{d^M(p_{(r)}) + \epsilon_d}, \tag{1.3}$$

$$\omega_r = \frac{\gamma^{r-1}}{\sum_{t=1}^{K} \gamma^{t-1}}, \qquad 0 < \gamma \leq 1, \quad \epsilon_d > 0. \tag{1.4}$$

Unfilled path positions contribute zero. Rank weights are not renormalized when the number of paths decreases. The initial candidate uses $K = 2$ and $\gamma = 1$; these values are not universally optimized for all structures.

For an active load component $c \in \mathcal{D}$, aggregation weights are derived from the loads and reactions in the intact model:

$$F_c = \sum_{i \in V_L} |f_{ic}|, \qquad a_{ic} = \frac{|f_{ic}|}{F_c}, \qquad \pi_c = \frac{F_c / f_{c,\mathrm{ref}}}{\sum_{h \in \mathcal{D}} F_h / f_{h,\mathrm{ref}}}, \tag{1.5}$$

$$B_c = \sum_{j \in V_T} |r_{jc}^{(0)}|, \qquad b_{jc} = \begin{cases} \dfrac{\beta}{|V_T|} + (1-\beta)\dfrac{|r_{jc}^{(0)}|}{B_c}, & B_c > 0, \\ \dfrac{1}{|V_T|}, & B_c = 0. \end{cases} \tag{1.6}$$

Here, $f_{c,\mathrm{ref}} > 0$ is a load scale with compatible units and $0 < \beta \leq 1$. All these weights remain fixed across disruptions of the same model. The graph score is then

$$S(G) = \sum_{c \in \mathcal{D}} \pi_c \sum_{i \in V_L} a_{ic} \sum_{j \in V_T} b_{jc} \Phi_{ij}^{(K)}(G). \tag{1.7}$$

Models without active loads or supports are not scored. A node that is both loaded and supported has a zero-cost empty-path contribution. The potential dominance of this constant contribution must be tested, not ignored when interpreting score retention.

1.3.3 Disruption and utilization adjustment

A scenario $s = (e, \lambda)$ removes an edge when $\lambda = 0$, or multiplies its weight by λ when $0 < \lambda \leq 1$. Other edges, normalization, pair weights, and the intact-model denominator remain fixed. For a valid baseline score, $S(G^M) > \epsilon_S$,

$$\mathrm{RR}_e(\lambda) = \frac{S(G_{e,\lambda}^M)}{S(G^M)}, \qquad U_e^* = \min\left\{\frac{D_e}{R_e + \epsilon_{R,e}}, 1\right\}, \tag{1.8}$$

$$\mathrm{ARI}_e(\lambda) = \mathrm{RR}_e(\lambda)(1 - \eta U_e^*), \qquad \mathrm{CI}_e(\lambda) = 1 - \mathrm{ARI}_e(\lambda), \tag{1.9}$$

where $D_e \geq 0$ and $0 \leq \eta \leq 1$. Capacity regularization has the same units as R_e. An infinite, excessively small, or uncertified baseline score results in abstention, not a zero imputation.

1.4 Reported Analytical Results and Verification

1.4.1 Range, monotonicity, and control

Weakening does not reduce the cost of a path that remains available, whereas deletion reduces the set of feasible choices. After sorting, the cost at each position does not decrease for the same choice. With fixed, nonnegative weights, the maximum objective cannot increase. Therefore, under exact optimization and the stated assumptions,

$$0 \leq S(G_{e,\lambda}^M) \leq S(G^M), \qquad 0 \leq \mathrm{RR}_e(\lambda), \mathrm{ARI}_e(\lambda), \mathrm{CI}_e(\lambda) \leq 1. \tag{1.10}$$

As weakening increases, ARI does not increase and CI does not decrease. This is a regularity property of the score, not evidence of a proportional relationship with physical stiffness loss.

For the control $\lambda = 1$, RR $= 1$ but CI $= \eta U_e^*$. Thus, CI does not represent only the additional effect of disruption; part of it comes from initial utilization. The quantity $1 - \mathrm{RR}$ should be reported separately. If $\eta = 0$, ARI equals RR. If $\eta = 1$ and $U_e^* = 1$, ARI is zero at every disruption level for that element. This saturation removes distinctions among disruption levels in the score; it does not indicate equivalent mechanical consequences.

1.4.2 Two consequential algorithmic differences

Weakening a normalized weight and renormalizing a weakened attribute yield, respectively,

$$\lambda \frac{x}{x + s_x} \quad \text{and} \quad \frac{\lambda x}{\lambda x + s_x}. \tag{1.11}$$

For $x = 20$, $s_x = 15$, and $\lambda = 0.5$, the first value is $2/7$ and the second is $2/5$. Replacing the first operation with the second changes the candidate's definition, not merely its implementation.

The source path example uses nodes L, a, b, T with costs $c_{La} = 2$, $c_{aT} = 5$, $c_{Lb} = 5$, $c_{bT} = 2$, and $c_{ab} = 2$. The shortest path $L - a - b - T$ has cost six and blocks a second path if its edges are removed directly. For $K = 2$ with equal weights, the greedy score in the zero-regularization limit is $1/12$. By contrast, the edge-disjoint choices $L - a - T$ and $L - b - T$, each with cost seven, yield $1/7$. Repeatedly searching for a single shortest path therefore cannot be assumed to solve the ARI objective.

1.4.3 Software evidence and computational contract

The initial archive reports four canonical models, 64 disruptions, 16 controls, and 342 passing automated checks. The number of checks is not the number of independent structures. Checks that successfully detect rank deficiency also do not prove that ARI assigns high priority to those cases.

The computational contract requires edge identities to be preserved, path combinations to be checked, and partial results not to be used as certified scores. Caches must depend on the model, loads, parameters, tolerances, and algorithm version. The availability of source files matching the manifest must be checked before claiming full package reproduction; this article did not rerun the package.

1.5 Discussion

The formulation's strength lies in its clear operations and testable internal properties. However, mechanical weighting does not by itself turn a path optimum into a solution of the structural equations. Limiting the number of paths, compensating among attributes, and adjusting for initial forces may leave out important kinematic information.

A high CI indicates priority only under the candidate's rules. Added value must be tested against simpler comparators with comparable information access and forwarding budgets. Search and abstention costs must also be counted. Showing that fewer scenarios are selected does not establish computational savings.

1.6 Conclusion

ARI can be formulated using fixed normalization, edge-disjoint path optimization, a score-retention ratio, and an initial-utilization adjustment. The range and monotonicity hold under fixed assumptions and exact optimization. The control shows that CI contains initial- utilization information, while the arithmetic example distinguishes the formulation from two non-equivalent algorithmic substitutions. This contribution provides a testable candidate specification, not evidence that mechanical screening has passed validation.

Source Material and Relationship to Other Manuscripts

Methodological and analytical article draft; not screening validation.

The formulation follows the theory section in `main.tex` and the dissertation's implementation documentation and analytical examples. Verification data come from `results/verification`. Article 2 uses the same models to address mechanical adequacy; those values are not counted as new experiments in this article.

Provisional References

[1] Irawan, H. (2026). *Development and Validation of the Adjusted Redundancy Index for Mechanics-Based Screening of Disruption Scenarios Before Graph Neural Network Prediction.* Dissertation draft, Doctoral Program in Engineering, Graduate School, Yogyakarta State University. Project source: `main.tex`, with the sections and result archives cited therein.

[2] Newman, M. E. J. (2010). *Networks: An introduction.* Oxford University Press.

The Graph Remains Connected but the Structure Loses Constraints: Canonical Verification of the Limits of ARI Representation

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Abstract

Alternative paths in a graph are not always sufficient to explain a structure's mechanical constraints. This article revisits verification of four axial-bar models from the ARI study by comparing connectivity, path scores, and the rank of the reduced stiffness matrix. The four models produced 64 disruptions and 16 controls. Every disruption graph remained connected, but eight full removals produced rank deficiency. In the single-diagonal square, removing two edge bars with zero initial force preserved RR near one and CI near zero, while rank fell from five to four. The kinematic explanation shows that the node remains connected in the graph, but one displacement direction is no longer linearly constrained. These findings limit interpretation of the score as a guarantee of stability without rejecting every possible use of ARI as a ranking measure. Arithmetic verification, representational adequacy, and independent validation should be reported as distinct levels of evidence.

Keywords: graph connectivity; kinematic constraints; rank; counterexample; structural redundancy.

2.1 Introduction

A graph states which components are connected, whereas a structural model also specifies deformation directions, stiffness, and boundary conditions. A path that remains available does not automatically restrain every degree of freedom. Therefore, reducing a structure to a graph must be tested against mechanical consequences, not only for consistency of score calculations.

The source study [1] proposes ARI as a candidate for prioritizing disruptions. This article examines its representational limits using canonical examples. The main question is whether a disruption with an almost unchanged score can cause constraint loss in a mechanical model. Stiffness assembly and compatibility checks provide a reference distinct from graph connectivity [2].

The article's contribution is an explainable, traceable limiting example. It is not an estimate of failure probability, the frequency of mechanisms in real structures, or evidence that all graph methods are unusable.

2.2 Methods

2.2.1 Source and model domain

Results are taken from the existing verification archive. The models comprise two parallel 1D bars, a planar triangle, a square with one diagonal, and a square with two diagonals. All

use a linear-elastic axial-bar idealization with zero-displacement supports. Buckling, plasticity, geometric changes, and collapse dynamics are not included.

The triangle has nodes at $(0, 0)$, $(1\,000, 0)$, and $(500, 800)$ mm. The square has corners at $(0, 0)$, $(1\,000, 0)$, $(1\,000, 1\,000)$, and $(0, 1\,000)$ mm. In the planar models, the lower-left node is restrained in both translations and the lower-right node in the vertical translation. The triangle carries a 30 kN downward load at its apex; the square carries a 30 kN rightward load at its upper-right corner. The crossing of the two diagonals is not a central joint.

The bar modulus is 200 kN/mm^2. The parallel bars have cross-sectional areas of 100 and 50 mm^2, respectively, whereas bars in the planar models have an area of 100 mm^2. The capacities of the parallel bars are 40 and 20 kN; the triangle bars have a capacity of 40 kN; and the square bars have a capacity of 80 kN. These are example parameters, not results of design-code checks or field measurements.

2.2.2 Disruptions and comparison measures

Each bar is tested at $\lambda \in \{0, 0.25, 0.50, 0.75\}$; the $\lambda = 1$ control is recorded separately. The solver multiplies area and capacity by the same factor. In the graph, the constructed weight is multiplied by λ without renormalization. These are explicit definitions, not operations assumed to be identical.

The ARI configuration uses $\alpha_k = \alpha_R = 0.5$, $q_e = 1$, $K = 2$, $\gamma = 1$, $\eta = 0.5$, and $\beta = 0.1$. Weight and path-cost regularization are 10^{-9}, the baseline-score tolerance is 10^{-12}, and capacity regularization is 10^{-6} kN. Stiffness and capacity use the fixed scales recorded in the source.

Connectivity is checked on the graph; rank is checked on the reduced stiffness matrix. Displacement is measured as the largest nodal translation norm, not just one component. Responses without a unique solution are not assigned fabricated displacement or utilization values.

2.3 Results

2.3.1 Intact-model condition

All four intact models have full rank at the solver tolerance used. Table 2.1 reproduces the reference outputs. Full rank indicates a unique elastic response in the model, not compliance with all capacity or serviceability limits under disruption.

Table 2.1: Intact-model responses from the verification archive; not the main validation dataset.

Model	Bars	Rank/DOF	$u_{\max}$ (mm)	$U_{\max}$
Parallel	2	1/1	1.000000	0.500000
Triangle	3	3/3	1.154460	0.442218
One diagonal	5	5/5	5.935311	0.530330
Two diagonals	6	5/5	3.582277	0.320083

2.3.2 Connectivity does not detect rank loss

All 64 disruption graphs remain connected. However, three full removals in the triangle and five full removals in the single-diagonal square produce rank deficiency. The other 56 scenarios have

a unique solution within the solver tolerance.

Table 2.2: Canonical disruption status. Controls are excluded from the disruption count.

Model	Disruptions	Unique solutions	Rank deficient	Connected
Parallel	8	8	0	8
Triangle	12	9	3	12
One diagonal	20	15	5	20
Two diagonals	24	24	0	24

The package reports 342 passing automated checks. These include successfully recording rank deficiency as expected. Passing these checks does not mean that ARI captures every mechanism case, and the number of checks does not replace the number of independent models.

2.3.3 Zero-initial-force bars as a limiting example

In the single-diagonal square, the nodes in order are A at the lower left, B at the lower right, C at the upper right, and D at the upper left. The element identities are $e_1 = AB$, $e_2 = BC$, $e_3 = CD$, $e_4 = DA$, and $e_5 = AC$. Table 2.3 shows the removal of one bar at a time.

Table 2.3: Path scores and rank after bar removal in the single-diagonal square.

Removed bar	RR	CI	Rank/DOF
$e_1 = AB$	0.983152	0.016848	4/5
$e_2 = BC$	0.964724	0.216162	4/5
$e_3 = CD$	1.000000	0.000000	4/5
$e_4 = DA$	1.000000	0.000000	4/5
$e_5 = AC$	0.717285	0.472914	4/5

The key cases are the removal of e_3 and e_4, not the diagonal e_5. Neither bar carries initial force under the applied load. Removing either bar also barely changes the path-score optimum. This combination preserves RR ≈ 1 and CI ≈ 0 even though rank falls from five to four. The zeros in the table are rounded; the full numerical outputs remain in the archive.

After CD is removed, node D remains connected through vertical bar AD. An infinitesimal horizontal motion at D produces no first-order axial strain in that bar. After DA is removed, horizontal bar CD does not restrain an infinitesimal vertical motion at D. The graph retains a path, but the model loses one kinematic constraint.

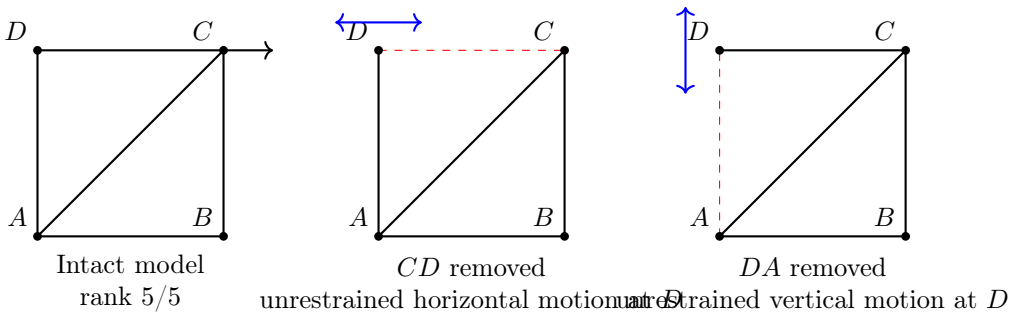


Figure 2.1: Explanatory schematic of the limiting example, not a deformation simulation. Dashed lines indicate removed bars; blue arrows indicate the direction of unrestrained infinitesimal motion. In all panels, A is restrained in both translations and B in the vertical translation.

2.4 Discussion

2.4.1 Claims rejected and questions not yet tested

The example rejects the interpretation that low CI always guarantees stability across the entire candidate domain. The result does not violate ARI's range or monotonicity: a score can be correct by its definition while failing to distinguish certain mechanical consequences. Adding arithmetic checks alone does not resolve the representational problem.

The findings do not state that all zero-force elements are critical, that all graph methods fail, or that ARI cannot be useful in a limited domain. Those claims require other studies. Four canonical models do not provide a basis for inferring failure frequency in a population. The explanation of free motion also refers to linear kinematics, not a dynamic-collapse simulation of a real structure.

2.4.2 Separating the score from reference labels

Changing CI to one after observing solver rank would conceal the representational limit and introduce post-disruption outputs into a pre-analysis decision. Improvement should take the form of an information channel available before reference evaluation, or a restricted domain. The original score should be retained so that the reason for a policy change can be examined.

The findings motivate a kinematic guard based on the intact model. However, handling rank deficiency does not also address excessive utilization or displacement in models that retain full rank. Subsequent evaluation should separate these types of consequences.

2.5 Conclusion

Verification shows that all 64 disruption graphs remain connected, while eight full removals produce rank deficiency. Two edge-bar removals in the single-diagonal square even yield CI near zero as rank falls from five to four. Connectivity and path-score retention are not sufficient guarantees of kinematic stability. The contribution is a limiting example that can be explained mechanically; independent validation is still needed to assess ARI's usefulness as a ranking measure.

Data Source and Relationship to Other Manuscripts

Verification-results article draft; four canonical models, not a population sample.

Tables are reproduced from `results/verification`; the model description follows Section 4.1. The mechanism schematic explains the same geometry and adds no new observations. These canonical cases are reused as regression cases in the guard article.

Provisional References

[1] Irawan, H. (2026). *Development and Validation of the Adjusted Redundancy Index for Mechanics-Based Screening of Disruption Scenarios Before Graph Neural Network Predic-*

tion. Dissertation draft, Doctoral Program in Engineering, Graduate School, Yogyakarta State University. Project source: `main.tex`, with the sections and result archives cited therein.

[2] Zienkiewicz, O. C., Taylor, R. L., & Zhu, J. Z. (2013). *The finite element method: Its basis and fundamentals* (7th ed.). Butterworth-Heinemann.

An Intact-Model-Based Kinematic Guard for Screening Axial-Bar Disruptions: Formulation and Development Diagnostics

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Abstract

Path scores can fail to prioritize the loss of a bar that is kinematically important, even when its initial force is zero. This article presents a guard based on intact-model stiffness that operates on the forwarding policy without changing ARI. Element stiffness contribution is summarized by h_e, and the margin $m_e(\lambda) = 1 - (1 - \lambda)h_e$ is used to check weakening of a single bar. The domain is limited to linear-elastic axial bars, fixed geometry and supports, and an intact model that is positive definite. Diagnostics include 64 canonical regression disruptions and 272 parametric development disruptions. The combined policy forwards all 40 rank-deficient cases, but still misses 8 of 17 and 30 of 71 diagnostic critical events in the respective groups. The results show both the role of a separate kinematic channel and its limits for non-mechanism consequences. Passing automated checks is not equated with population recall or general screening success.

Keywords: kinematic guard; stiffness matrix; redundancy; single-element disruption; forwarding policy.

3.1 Introduction

The limiting example in the ARI study shows that an element with zero initial force can still provide a necessary constraint [1]. When the path score barely changes after that element is removed, CI does not indicate high priority even though rank decreases. The correction must not use post-disruption labels to replace the score.

This article uses information from the intact model as a separate checking channel. The relationship between stiffness contribution and the redundancy matrix follows foundations discussed in the dissertation [2]; the redundancy-matrix concept is not claimed as a new discovery. The focus is its adaptation to the ARI forwarding policy, numerical treatment, and diagnostics against known failure examples.

3.2 Formulation and Domain

3.2.1 Mechanical assumptions

The model consists of linear-elastic 1D/2D axial bars with fixed geometry, zero-displacement supports, and a positive-definite reduced intact stiffness matrix $\mathbf{K}_0$. A single-element disruption changes its axial stiffness by a factor $\lambda \in [0, 1]$. Changes to supports, plasticity, buckling, geometry, joints, and dynamic effects are excluded. Multiple-element disruptions are treated as outside the domain rather than evaluated by summing single-element margins.

Given the elongation-compatibility row $\mathbf{a}_e$ and $k_e = E_e A_e / L_e$, define

$$\mathbf{K}_0 = \sum_e k_e \mathbf{a}_e^\mathsf{T} \mathbf{a}_e, \qquad h_e = k_e \mathbf{a}_e \mathbf{K}_0^{-1} \mathbf{a}_e^\mathsf{T}. \tag{3.1}$$

This quantity uses geometry, boundary conditions, and stiffness, not initial forces or disruption responses. In this domain, $0 \le h_e \le 1$ in exact arithmetic, and the corresponding redundancy diagonal is $1 - h_e$.

3.2.2 Constraint-loss margin

The disrupted matrix and element margin are

$$\mathbf{K}_{e,\lambda} = \mathbf{K}_0 - (1-\lambda) k_e \mathbf{a}_e^\mathsf{T} \mathbf{a}_e, \qquad m_e(\lambda) = 1 - (1-\lambda) h_e. \tag{3.2}$$

Let $\mathbf{v}_e = \sqrt{k_e} \mathbf{K}_0^{-1/2} \mathbf{a}_e^\mathsf{T}$. A congruence transformation gives

$$\mathbf{K}_0^{-1/2} \mathbf{K}_{e,\lambda} \mathbf{K}_0^{-1/2} = \mathbf{I} - (1-\lambda) \mathbf{v}_e \mathbf{v}_e^\mathsf{T}. \tag{3.3}$$

One eigenvalue of the matrix on the right is m_e, while the others are one. Under these assumptions, $m_e = 0$ indicates loss of positive definiteness, whereas $m_e > 0$ preserves it in exact arithmetic. Under full removal, $m_e = 1 - h_e$. A zero-initial-force bar may still have $h_e = 1$, so its role cannot be determined from initial utilization alone.

3.2.3 Implementation, numerical conditions, and decision

The reported implementation does not form an explicit inverse. The weighted compatibility matrix, with rows $\sqrt{k_e}\mathbf{a}_e$, is factorized using SVD. For a full-rank intact model, h_e is obtained from the sum of squares of the entries in the corresponding row of the left singular-vector matrix. This separates the mathematical formulation from the numerical procedure used to compute it.

For $\rho_0 = \lambda_{\min}(\mathbf{K}_0)/\lambda_{\max}(\mathbf{K}_0)$, the following conditional bound holds

$$\frac{\lambda_{\min}(\mathbf{K}_{e,\lambda})}{\lambda_{\max}(\mathbf{K}_{e,\lambda})} \ge m_e \rho_0. \tag{3.4}$$

The bound follows from $m_e \mathbf{K}_0 \preceq \mathbf{K}_{e,\lambda} \preceq \mathbf{K}_0$. The diagnostic configuration uses a margin tolerance of 10^{-8} and a relative rank tolerance of 10^{-10}. Values near the bound, inadequate initial conditions, or computation failure trigger forwarding. These tolerances are not calibrated safety thresholds.

Define $g(s) = 1$ when the guard requires forwarding. The policy is

$$F(s) = \mathbf{1}\{\mathrm{CI}(s) \ge \tau \ \vee \ g(s) = 1 \ \vee \ \text{ARI abstains or is invalid}\}. \tag{3.5}$$

The `cleared` status means only that the guard did not trigger forwarding; it does not mean that capacity or displacement was checked. An unselected case has status `not_selected`, not safe.

3.3 Diagnostic Design

The package comprises four canonical regression models and 16 additional development models, with 336 disruptions and 84 controls. The canonical group contains 64 disruptions and the parametric development group contains 272. Because the canonical models are reused, this package is not added to the initial package as entirely new data.

Three policies are compared: ARI only, guard only, and combined. The CI threshold of 0.5 is diagnostic, not calibrated. A diagnostic critical event is any rank deficiency, maximum utilization above one, or maximum displacement exceeding twice the intact-model value. This displacement bound is not a normative serviceability criterion; the combined event count does not double-count scenarios meeting multiple event types.

3.4 Results

3.4.1 Handling zero-initial-force cases

For the loss of e_3 and e_4 in the single-diagonal square, CI remains near zero. The guard yields h_e near one and a margin near zero, so the combined policy forwards both cases. The decision changes not by replacing CI based on solver labels, but by adding an intact-model check.

Table 3.1: Regression test for loss of two edge bars. Small values are numerical outputs.

Element	CI	h_e	m_e	Rank	ARI	Combined
e_3	1.11×10^{-16}	1.000000	1.11×10^{-16}	4/5	No	Forward
e_4	0.00	1.000000	2.22×10^{-16}	4/5	No	Forward

3.4.2 Policy comparison

The combined policy forwards all 40 rank-deficient cases: eight in regression and 32 in parametric development. However, critical events without rank loss are still missed.

Table 3.2: Diagnostics at a CI threshold of 0.5; the denominator in each missed-event column is the number of events in the corresponding group.

Group	Policy	N	Forwarded	Rank events missed	Critical events missed
Canonical	ARI	64	5	6/8	14/17
Canonical	Guard	64	8	0/8	9/17
Canonical	Combined	64	11	0/8	8/17
Parametric	ARI	272	17	24/32	54/71
Parametric	Guard	272	32	0/32	39/71
Parametric	Combined	272	41	0/32	30/71

In regression, the combined policy forwards 11 of 64 disruptions and still misses 8 of 17 critical cases. In parametric development, it forwards 41 of 272 disruptions and still misses 30 of 71 critical cases. These values show improved mechanism detection alongside limitations for all diagnostic consequences. The guard alone also captures every rank-deficient case examined, so ARI's additional role must be assessed against other consequence types.

3.4.3 Automated checks and limits of interpretation

The package reports 1475 passing automated checks. Coverage includes analytical values, isostatic cases, unit and ordering invariance, guard independence from loads and capacities, near-singular cases, and forwarding when SVD fails or ARI abstains. This is a count of software and arithmetic checks, not independent models or a reliability level.

3.5 Discussion

The margin addresses constraint loss in the single-element update domain. It does not measure whether other bars receive forces exceeding capacity or whether displacement exceeds a threshold. Therefore, success in detecting mechanisms must not be reported as success in detecting all critical events.

Agreement between the guard and solver in rank checks is also not independent validation of ARI's ability to predict capacity. Both use related mechanical foundations. Main validation requires new models, a frozen configuration, simple comparators, and a comparable forwarding budget.

Initial factorization and cache construction have costs. The one-time cost per model must be separated from per-scenario margin updates and ARI costs. Local timing records do not establish savings in GNN integration. Forwarding mechanism cases also does not ensure that the downstream predictor supports those inputs.

3.6 Conclusion

The intact-model guard provides a separate kinematic check without changing the ARI score. In the diagnostics, the combined policy forwards all 40 rank-deficient cases, including two zero-initial-force bar removals not prioritized by ARI. Missed capacity and displacement events show that mechanism detection is not a substitute for assessing all consequences. The results remain limited development evidence; screening suitability and GNN benefits cannot yet be declared successful.

Data Source and Relationship to Other Manuscripts

Methods article draft with development results; no independent validation yet.

Results follow Section 4.2 and the archive `results/revision`. Tables are reproductions, not additional experiments. Regression reuses the four models from Article 2; the 16 additional development models are not the main validation dataset.

Provisional References

[1] Irawan, H. (2026). *Development and Validation of the Adjusted Redundancy Index for Mechanics-Based Screening of Disruption Scenarios Before Graph Neural Network Prediction.* Dissertation draft, Doctoral Program in Engineering, Graduate School, Yogyakarta

State University. Project source: `main.tex`, with the sections and result archives cited therein.

[2] Gade, J., Tkachuk, A., von Scheven, M., & Bischoff, M. (2021). A continuum-mechanical theory of redundancy in elastostatic structures. *International Journal of Solids and Structures, 226–227*, 110977. DOI: `10.1016/j.ijsolstr.2021.01.022`.

[3] von Scheven, M., Ramm, E., & Bischoff, M. (2021). Quantification of the redundancy distribution in truss and beam structures. *International Journal of Solids and Structures, 213*, 41–49. DOI: `10.1016/j.ijsolstr.2020.11.002`.

Axial-Demand and Displacement-Amplification Bounds from the Intact Model for Enriching Disruption Screening

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Abstract

The kinematic guard does not measure capacity exceedance or displacement amplification in structures that retain full rank. This article reorganizes a proposed pre-analysis bound from the intact model to complement the guarded ARI policy. For linear-elastic axial bars under fixed loads and with a positive disruption margin, a rank-one update identity is used to derive bounds on active-bar demand and nodal displacement norms. Operational information includes signed initial forces, stiffness contributions, capacity under the disruption rule, and local flexibility of the intact model. A bound that triggers forwarding does not prove that a consequence occurred, because the inequality may be conservative. Conversely, conditional coverage properties apply only to the stated domain, event definition, and arithmetic. The article reports no unmeasured recall, reduction, or savings. Its contribution is an analytical derivation, information contract, and verification design prior to implementation.

Keywords: rank-one update; demand bound; flexibility; displacement; pre-analysis enrichment.

4.1 Introduction

Guarded ARI diagnostics show that all tested rank-deficient cases can be forwarded, while critical capacity and displacement events are still missed [1]. This finding motivates separating consequence types into distinct channels rather than merging all information into one path weight.

The article asks whether intact-model information can provide useful consequence bounds before the full response to each disruption is computed. Pre-analysis means before full post-disruption evaluation, not without mechanical computation. The update identity and the relationship between redundancy and changes in flexibility have foundations discussed in source [2]; this article does not claim the identity as a new discovery.

4.2 Assumptions and Permitted Information

The model is limited to linear-elastic axial bars, positive-definite $\mathbf{K}_0$, fixed geometry and supports, and weakening of a single bar. Nodal load $\mathbf{f}$ is unchanged by the disruption. Element load release, initial strain, changes in load direction, plasticity, and multiple-bar disruptions are excluded.

The intact response and signed initial forces satisfy

$$\mathbf{K}_0\mathbf{u}_0 = \mathbf{f}, \qquad N_{j,0} = k_j\mathbf{a}_j\mathbf{u}_0, \qquad D_j = |N_{j,0}|. \tag{4.1}$$

Capacity $R_{j,s}$ is derived from initial attributes and the frozen disruption rule, not from reference-solver outputs. For area weakening in the current package, $R_{e,s} = \lambda R_e$ while the capacities of other bars remain unchanged. This rule does not automatically apply to all damage mechanisms.

The cache stores intact-model information and configuration identity. Post-disruption response features are unavailable at the decision stage. Learned normalization may use development data only; changes in loads or attributes invalidate dependent cache entries.

4.3 Derivation of the Axial-Demand Bound

4.3.1 Response-update identity

Write $\alpha_s = 1 - \lambda$ and $m_s = 1 - \alpha_s h_e$. Once the guard confirms that the margin is valid and sufficiently far from zero, the response change in the linear idealization is

$$\Delta \mathbf{u}_s = \frac{\alpha_s N_{e,0}}{m_s} \mathbf{K}_0^{-1} \mathbf{a}_e^{\mathsf{T}}. \tag{4.2}$$

This identity can be checked by substituting $\mathbf{u}_0 + \Delta \mathbf{u}_s$ into $\mathbf{K}_{e,\lambda} \mathbf{u}_s = \mathbf{f}$. This equation is not a solver-response column used as a feature; it is the basis for deriving the bound.

4.3.2 Unaffected and weakened bars

For $j \neq e$, Cauchy–Schwarz in the $\mathbf{K}_0^{-1}$ metric gives

$$\left| k_j \mathbf{a}_j \mathbf{K}_0^{-1} \mathbf{a}_e^{\mathsf{T}} \right| \leq \sqrt{\frac{k_j}{k_e} h_j h_e}. \tag{4.3}$$

Thus, the active-bar demand bound is

$$\overline{D}_{j,s} = \begin{cases} |N_{j,0}| + \dfrac{\alpha_s |N_{e,0}|}{m_s} \sqrt{\dfrac{k_j}{k_e} h_j h_e}, & j \neq e, \\ \dfrac{\lambda |N_{e,0}|}{m_s}, & j = e, \quad \lambda > 0. \end{cases} \tag{4.4}$$

The second row accounts for the stiffness change in the target bar. Using the unaffected-bar formula for a weakened bar would ignore that change. A removed element is no longer part of the active-bar set $\mathcal{E}_s$.

For positive capacities in all active bars,

$$\overline{U}_s = \max_{j \in \mathcal{E}_s} \frac{\overline{D}_{j,s}}{R_{j,s}}. \tag{4.5}$$

A value of $\overline{U}_s \geq 1$ triggers forwarding but does not prove that capacity was actually exceeded. A zero, unknown, or invalid active capacity results in abstention; regularization is not used to conceal this invalidity.

4.4 Derivation of the Displacement Bound

Let $\mathbf{P}_i$ extract the translational components of node i from the free degrees of freedom, with restrained components set to zero. Define

$$c_i = \lambda_{\max}\left(\mathbf{P}_i \mathbf{K}_0^{-1} \mathbf{P}_i^{\mathsf{T}}\right), \qquad d_0 = \max_i \|\mathbf{P}_i \mathbf{u}_0\|_2. \tag{4.6}$$

The operator norm gives

$$\|\mathbf{P}_i \Delta\mathbf{u}_s\|_2 \le \frac{\alpha_s |N_{e,0}|}{m_s} \sqrt{c_i \frac{h_e}{k_e}}. \tag{4.7}$$

The triangle inequality yields the maximum nodal displacement bound:

$$\overline{d}_s = \max_i \left\{ \|\mathbf{P}_i \mathbf{u}_0\|_2 + \frac{\alpha_s |N_{e,0}|}{m_s} \sqrt{c_i \frac{h_e}{k_e}} \right\}. \tag{4.8}$$

If d_0 is sufficiently far from zero, the amplification bound is $\overline{A}_s = \overline{d}_s / d_0$. The threshold $A_{\lim}$ must match the evaluation-event definition and be frozen before testing. If the denominator is too small, use an absolute threshold only if it was specified in the protocol; otherwise, forward by abstention.

One factorization of the intact model can provide the required quantities without forming an explicit inverse. However, constructing all local flexibility quantities still has time and memory costs. Approximations that may underestimate the bound must not be used without adequate error control.

4.5 Layered Policy and Conditional Properties

Let $F(s)$ denote forwarding by ARI, the kinematic guard, or initial abstention. The capacity channel has $g_C(s) = 1$ when $\overline{U}_s \ge 1$, and the displacement channel has $g_D(s) = 1$ when the bound exceeds its corresponding threshold. The proposed policy is

$$F^+(s) = \mathbf{1}\{F(s) = 1 \vee g_C(s) = 1 \vee g_D(s) = 1 \vee \text{bound unavailable}\}. \tag{4.9}$$

The kinematic guard must be checked first. For $N_{e,0} = 0$ and $m_s = 0$, one must not conclude that the response change is zero, because Equation (4.2) does not apply at the singularity. If $m_s > 0$, the load is fixed, and all linear assumptions hold, $N_{e,0} = 0$ does give zero displacement change for that load case; this conclusion does not extend to different loads.

In exact arithmetic and within the stated domain, the response does not exceed its upper bound. Therefore, capacity or displacement exceedance under the same definition triggers the corresponding channel. This is a conditional analytical property, not empirical recall or a safety guarantee. Numerical implementation must account for error or forward cases when the bound property cannot be confirmed.

4.6 Verification and Audit Design

The bound implementation is not yet available in the diagnostic package. The verification design separates three stages: preparation of the intact-model cache; operational decisions that read only the cache, disruption definition, and configuration; and the reference solver, which runs after

the decision is stored. Removing or randomizing labels must not change features or decisions. The screen must still run when access to the reference directory is unavailable.

Table 4.1: Planned verification cases; not results of checks already run.

Case	Required check
Control $\lambda = 1$	Zero response change and consistency with the initial bound
Stable disruption	Solver response does not exceed the bound at controlled tolerance
Margin near zero	Forward or bound error; do not evaluate an invalid ratio
Zero-initial-force bar	Distinguish positive-margin from singular cases
Element removed	Exclude the element from the active-capacity set
Load change	Invalidate the dependent cache and recompute initial quantities

Ablations should compare the guard alone, the guard with the capacity channel, the guard with the displacement channel, both channels without ARI, and the full policy. Recall for each event type and their intersections should be reported separately. Mandatory triggers must not be pruned to meet a quota, because doing so changes the OR policy and the basis for conditional coverage.

4.7 Discussion and Limitations

The absolute bound ignores force-sign cancellation and may therefore be loose. When m_s is small, the bound can grow and forward nearly all scenarios. High coverage alone therefore does not establish screening suitability. Evaluation must include bound tightness, reduction, and the total cost of computing it.

Computing the full vector in Equation (4.2) to obtain the complete response is a linear rank-one reanalysis. This procedure is a valid comparator, but must not be presented as a low-cost feature without further analysis. Preparation, storage, bound-evaluation, and forwarding costs must be compared with reanalysis on the same candidate set.

The enrichment also does not automatically extend the domain to buckling, plasticity, dynamics, or multi-element damage. The bound properties apply only when the model, capacities, loads, and event definition satisfy the derivation assumptions.

4.8 Conclusion

Axial-demand and displacement bounds can be derived from the intact model using a kinematic margin and a rank-one update in a restricted linear domain. This channel complements rank checks without changing CI. The results remain analytical: implementation, numerical robustness, recall–reduction, and computational savings have not been demonstrated. Verification, leakage audits, and cost evaluation are prerequisites for further development before application claims.

Evidence Status and Source

Analytical study draft; the bound channels have not been implemented or empirically tested.

The derivation follows Section 4.4. The bound channel has not been implemented in `results/revision`; the test-design table must not be read as checks that have passed. The article adds no experimental results and makes no universal novelty claim for the update formula.

Provisional References

[1] Irawan, H. (2026). *Development and Validation of the Adjusted Redundancy Index for Mechanics-Based Screening of Disruption Scenarios Before Graph Neural Network Prediction.* Dissertation draft, Doctoral Program in Engineering, Graduate School, Yogyakarta State University. Project source: `main.tex`, with the sections and result archives cited therein.

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[3] von Scheven, M., Ramm, E., & Bischoff, M. (2021). Quantification of the redundancy distribution in truss and beam structures. *International Journal of Solids and Structures, 213*, 41–49. DOI: `10.1016/j.ijsolstr.2020.11.002`.

A Protocol for Evaluating Coverage, Time, and Operational Energy of ARI Screening Before GNN Inference

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Abstract

Reducing the number of inference candidates does not establish computational benefit if screening costs and missed critical cases are not accounted for. This article develops a protocol for evaluating ARI before GNN inference, separating ranking quality, screening coverage, final coverage, operational time, electrical energy, and estimated emissions. Comparisons use a fixed-parameter GNN, the same initial candidate set, partitions by parent model, and paired measurements. Abstentions count as forwarded, whereas unselected scenarios receive neither zero predictions nor safe labels. Preparation, required intact-model analysis, ARI, the guard, inference, and failure handling are included in the system boundary. Benefit decisions combine quality criteria with uncertainty in time and energy differences. The article reports no unavailable inference results, power measurements, or savings. The draft's output is a traceable evaluation design to distinguish potential efficiency from measured benefit.

Keywords: ARI–GNN; evaluation protocol; recall; total time; computational energy; Green AI.

5.1 Introduction

Screening reduces the number of candidates to be predicted, but adds computation and can remove opportunities to identify critical cases. Therefore, reduction must be assessed together with output quality and system cost. In the source study [1], integration with a compatible trained GNN and energy measurements are not yet available. Local ARI or guard timing cannot replace an end-to-end experiment.

This article is a protocol, not a completed study. The main question is whether screening preserves the required final coverage while providing net savings in time and energy. Energy reporting follows the attention to system boundaries and traceability discussed in source [2]. Configuration selection is separated from final evaluation so that test results are not reused to optimize the method [3].

5.2 Planned Research Design

5.2.1 Unit of analysis and prerequisites

The independent unit is the parent model. All related variants, scenarios, and repeats from the same parent are assigned to the same partition. Normalization, ARI parameters, tolerances, thresholds, critical-event definitions, and GNN configuration are frozen before blind testing. If test results are used to improve the method, those data become development data and a new independent evaluation is required.

Main validation S1–S5, with an initial subtotal of 300 parent models, has not been carried out. That number remains a plan, not evidence of statistical adequacy. The four canonical models and 16 guard-development models do not replace it. The integration sample size must account for estimation precision and the number of parents with critical events.

Integration prerequisites include mathematical and algorithmic consistency, mechanical adequacy, stability under perturbation, and screening suitability in the supported domain. The GNN audit checks element types, units, features, supports, prediction targets, and mechanism handling. Forwarding by the guard does not guarantee that the GNN can process the forwarded state.

5.2.2 Treatments and comparators

The baseline comparisons include no screening, random screening, initial utilization, ARI, guard only, and ARI with the guard. The final list and rules for each are set before testing. All treatments use the same initial candidates, architecture, and trained GNN parameters. ARI is not added as a new GNN feature; this experiment does not assess label savings or changes to training.

Equal-budget experiments are compared separately from the operational policy with mandatory forwarding. Under equal budgets, the shared abstention set and its costs are counted, then the remaining quota is selected under each comparator's rule. If mandatory triggers exceed the quota, the excess is reported; cases are not silently pruned to force equal counts. Predictions for the same scenario across treatments must agree within a prespecified tolerance.

5.3 Output Definitions and Leakage Control

Let $\mathcal{S}$ be the candidate set, $\mathcal{C} \subseteq \mathcal{S}$ the critical cases according to the reference, and $\mathcal{F}$ the set of forwarded scenarios. Abstentions or invalid scores are included in $\mathcal{F}$. For sets with nonempty denominators,

$$\text{Recall}_{\text{screened}} = \frac{|\mathcal{F} \cap \mathcal{C}|}{|\mathcal{C}|}, \qquad \text{Reduction} = 1 - \frac{|\mathcal{F}|}{|\mathcal{S}|}. \tag{5.1}$$

Scenarios in $\mathcal{S} \setminus \mathcal{F}$ are not evaluated by the GNN. Let $\mathcal{H}_{\mathcal{F}}$ contain forwarded scenarios with successful inference and predictions meeting the frozen output threshold. Final coverage is

$$\text{Recall}_{\text{final}} = \frac{|\mathcal{H}_{\mathcal{F}} \cap \mathcal{C}|}{|\mathcal{C}|}. \tag{5.2}$$

Missed critical cases are separated into those not forwarded, those with failed or out-of-domain inference, and predictions below the threshold. All critical cases remain in the denominator. Handling by a conventional solver is reported separately and is not counted as a successful GNN prediction.

The decision process accesses only the intact model, scenario, and frozen configuration. Features, hashes, decisions, and abstention reasons are stored before reference outcomes are computed. Removing or randomizing labels must not change decisions. The audit also covers the origin of normalization, cache identity, and parent grouping, not only the final file split.

5.4 Measurement of Time, Energy, and Estimated Emissions

5.4.1 Operational time boundary

Total time includes input preparation, required intact-model costs, graph construction, ARI, the guard, inference, data transfer, and failure handling. One-time per-model costs are distinguished from per-scenario costs. For the same measurement boundary,

$$\Delta T = T_{\text{without}} - T_{\text{with}}. \tag{5.3}$$

Positive values indicate shorter time; negative values are still reported as additional cost. Inference time is not assumed to be linear in the number of candidates because batch size and data transfer can change its behavior.

Use paired measurements on comparable hardware, power boundaries, software versions, and configurations. The source protocol specifies ten repetitions, warm-up, device synchronization, and a balanced treatment order. Cold- and warm-cache conditions are distinguished; amortization uses only candidates actually processed. Instrument repeats are not independent structures.

5.4.2 Electrical energy

For power $P_{\text{el},v}(t)$ in watts and time in seconds, the energy for treatment v is

$$\mathcal{E}_v = \frac{1}{3.6 \times 10^6} \int_{t_{0,v}}^{t_{1,v}} P_{\text{el},v}(t)\,\mathrm{d}t \quad [\text{kWh}]. \tag{5.4}$$

The primary boundary includes energy for the entire machine during operation, including idle power over that interval. CPU- or GPU-only telemetry is reported as component energy, not whole-machine energy. Machine energy is not added to its component energies.

For concurrent stages, the same interval is not counted twice. Record instrument resolution, sampling interval, missing samples, background load, and monitoring overhead. Nominal power is not used as a substitute for consumption records. If adequate measurements are unavailable, energy remains unmeasured rather than being imputed from execution time.

For $\mathcal{E}_{\text{without}} > 0$, the energy difference and savings fraction are

$$\Delta\mathcal{E} = \mathcal{E}_{\text{without}} - \mathcal{E}_{\text{with}}, \qquad r_{\mathcal{E}} = \frac{\Delta\mathcal{E}}{\mathcal{E}_{\text{without}}}. \tag{5.5}$$

Energy per candidate uses the initial candidate count so that the denominator is comparable. Training and label generation are recorded separately as development costs, not claimed as costs eliminated by screening.

5.4.3 Estimated operational emissions

If a suitable electricity-intensity factor is available,

$$\widehat{M}_{\text{CO}_2\text{e},v} = \sum_j \Delta\mathcal{E}_{v,j}\,\iota_j, \tag{5.6}$$

where ι_j is in kg CO_2e/kWh. The factor's source, location, period, accounting method, and uncertainty are recorded. The main comparison uses the same factor to isolate computational differences. The result is an estimate, not a direct emissions measurement or evidence of avoided marginal emissions. If no suitable factor is available, reporting stops at energy. Structural material impacts and device life-cycle impacts are not inferred from these measurements.

5.5 Statistical Analysis and Decision Criteria

The primary estimand follows the source protocol: mean recall per parent model containing critical cases, not automatically the pooled proportion across all scenario rows. Related variants are summarized using frozen rules. Resampling is performed at the parent level while preserving paired treatments; the number of models without critical cases is reported, and no artificial recall is assigned to them. Small samples or degenerate intervals indicate insufficient evidence.

Initial targets in the source study include a one-sided 95% lower bound of at least 0.95 for mean recall, mean reduction of at least 0.30, and recall advantage over random screening at the same budget of at least 0.15. Comparisons against fixed comparators must also show interval-supported differences. These are research criteria to be frozen, not structural safety standards.

For integration, the source protocol requires a one-sided 95% lower bound of at least 0.95 for final recall, a one-sided 95% upper bound on recall loss relative to no screening no greater than 0.02, and a positive one-sided 95% lower bound for ΔT. The 0.02 margin is reviewed in a pilot and then frozen before testing. If the unscreened GNN does not meet the quality criterion, reducing candidates is not sufficient to claim a benefit.

An energy claim additionally requires a positive one-sided 95% lower bound for $\Delta\mathcal{E}$, accounting for instrument uncertainty. Savings smaller than instrument resolution are not reported as measured. Passing the energy criterion does not replace passing the quality, time, or representational-validity criteria; each is reported separately.

5.6 Planned Outputs and Discussion

Table 5.1: Outputs required by the protocol; not integration-experiment results.

Category	Required record	Current evidence status
Predictor	Architecture, trained parameters, features, domain, version	Not available for integration experiments
Decision	Candidates, forwarding, abstention, reason, hash	Local diagnostics available; integration untested
Quality	Screening and final recall, failures, valid prediction error	Not available for GNN workflow
Time	Paired duration and cost per stage	No measured integration benefit
Energy and emissions	Power and energy records, intensity factors	Not measured

The output table is not populated with assumptions presented as measurement results. Local screen timing does not prove a system-level benefit because overhead may exceed the inference avoided. Shorter runtime also does not automatically mean lower energy use. The evaluation is designed to determine whether a benefit exists, not to ensure that a benefit claim is accepted.

Zero or negative time and energy savings, failure to meet coverage criteria, and increased abstentions remain valid outcomes. Separating stages allows analysis of whether failure stems from representation, calibration, predictor compatibility, or computational cost. Sustainability claims must be limited to quantities and system boundaries actually measured.

5.7 Conclusion

ARI–GNN evaluation must distinguish forwarding, final prediction quality, and total operational cost. The protocol provides rules for parent-level partitioning, comparison, abstention, paired measurement, and statistical decisions for time and energy. All targets remain untested plans. A benefit can be claimed only after quality prerequisites are met and net savings are supported by traceable data.

Protocol Status and Source

Research protocol draft; not executed or claimed to be preregistered.

The article follows the validation design and energy protocol in Section 3.18.7 and the dissertation's methods section. This manuscript has not been preregistered and does not claim that experiments have taken place. Finalizing parameters, a trained GNN, sample size, measurement instruments, and complete references remain tasks to be completed before execution or submission.

Provisional References

[1] Irawan, H. (2026). *Development and Validation of the Adjusted Redundancy Index for Mechanics-Based Screening of Disruption Scenarios Before Graph Neural Network Prediction.* Dissertation draft, Doctoral Program in Engineering, Graduate School, Yogyakarta State University. Project source: `main.tex`, with the sections and result archives cited therein.

[2] Henderson, P., Hu, J., Romoff, J., Brunskill, E., Jurafsky, D., & Pineau, J. (2020). Towards the systematic reporting of the energy and carbon footprints of machine learning. *Journal of Machine Learning Research, 21*(248), 1–43.

[3] Cawley, G. C., & Talbot, N. L. C. (2010). On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research, 11*, 2079–2107.